A finite construction of matrix multiplication algorithms with exponent approaching 9/4

Abstract

The exponent ω of matrix multiplication is the least real number such that, for every $\varepsilon > 0$, two $n \times n$ matrices can be multiplied with $O(n^{\omega+\varepsilon})$ arithmetic operations. A preprint posted by OpenAI in October 2026 announced the bound $\omega \le 9/4$. Its proof concludes through an existence argument for tensor characters and does not specify an algorithm. Building on the convolution relations and the shared-leg separator of that preprint, we give a self-contained construction by explicit tensor maps: for every rational $\varepsilon > 0$, a terminating procedure outputs an exact bilinear algorithm with rational coefficients whose exponent is below $9/4 + \varepsilon$. The construction uses two kinds of tensor relations among products of polynomial multiplication tensors (splits and triplings). How often each relation is applied is given by the expected visit counts of a random walk whose transition probabilities come from the least solution of the inequalities that these relations impose. These frequencies are rounded to integers without breaking the balance of any tensor factor. For every integer $a \ge 2^{64}$ we obtain a scheme that computes several products of $B \times B$ matrices from products of smaller $A \times A$ matrices; its recursive use has exponent below $9/4 + 900/\log a$ (natural logarithm), and $\log A \le 181 a^5 \lceil \log_2 a \rceil^2$. Within the construction, the least A that gives exponent at most $9/4 + \varepsilon$ is $\exp(\exp(\Theta(1/\varepsilon)))$, and this family attains that order. Replacing roots of unity by rational interpolation nodes, we also give the seed side by a formula and bound the bit lengths of the coefficients and the bit cost of computing them; these bounds are computable from ε and astronomically large. For a fixed rational seed, we bound the bit cost of multiplying integer matrices exactly and the precision that suffices for a prescribed normwise forward error. Counting the computation of the seed as Turing-machine steps and the multiplication as field operations, one algorithm multiplies $n \times n$ matrices over any field of characteristic zero in $n^{9/4+o(1)}$ steps for every n. A Lean formalization of a simpler rational variant of the construction proves $\omega \le 9/4$ over every field of characteristic zero, without explicit constants.

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1 Introduction

The exponent ω of matrix multiplication is the least real number such that, for every $\varepsilon > 0$, two $n \times n$ complex matrices can be multiplied with $O(n^{\omega+\varepsilon})$ arithmetic operations. In a preprint dated 2 October 2026 and released on 6 October [22], OpenAI announced $\omega \leq 9/4$, well below the bounds near 2.371 obtained by the laser method [2, 11, 25, 34]. Its proof works with tensor characters: functions on tensors that add over direct sums, multiply over tensor products, do not increase under restriction, and take the value one on the unit tensor. These are the points of Strassen's asymptotic spectrum [31]. The preprint derives inequalities for the values of characters on tensors that encode polynomial multiplication, and deduces a bound for their values on matrix multiplication tensors. An existence theorem for characters that detect the rank of matrix multiplication then turns this into a bound on ω. The proof therefore shows that for every $\varepsilon > 0$ some size d admits a bilinear algorithm for $d \times d$ matrices with fewer than $d^{9/4+\varepsilon}$ multiplications, but it specifies neither d nor the algorithm.

This paper constructs such algorithms. We realize the inequalities that the preprint's diagonal bound uses by explicit tensor maps, decide how often to apply each map, and bound the exponent of the resulting finite construction. Theorem 1.1 below gives, for each rational $\varepsilon > 0$, a terminating procedure that outputs an exact rational bilinear algorithm, and Theorem 1.2 gives a family of relative schemes (defined below) whose exponents converge to $9/4$ at a proved rate.

1.1 The results

Arithmetic operations are additions, subtractions and multiplications of scalars, including multiplications by fixed constants (Section 2). A *rational bilinear algorithm* for $d_0 \times d_0$ matrix multiplication with R multiplications consists of R pairs of linear forms with rational coefficients, one form in the entries of each input matrix, together with rational linear combinations of the R products of these pairs that give the entries of the product matrix. Equivalently, it is a rank decomposition of the matrix multiplication tensor with rational coefficients.

Theorem 1.1. *For every positive rational ε, a terminating procedure produces an integer $d_0 \geq 2$ and a rational bilinear algorithm for $d_0 \times d_0$ matrix multiplication with fewer than $d_0^{9/4+\varepsilon}$ multiplications. Recursive use of this algorithm multiplies $n \times n$ matrices over every field of characteristic zero with $O_\varepsilon(n^{9/4+\varepsilon})$ arithmetic operations.*

We call such an algorithm a *rational seed* with *seed side* d_0. In particular $\omega \le 9/4$: for a real $\varepsilon > 0$, apply the theorem to a rational number in $(0, \varepsilon)$. Theorem 1.1 is stated in characteristic zero, but a rational seed is also a bilinear algorithm, with the same number of multiplications, over every field whose characteristic divides none of the denominators of its coefficients. Other work proves $\omega \le 9/4$ over every field [12, 20, 35] and the corresponding operation count over every associative unital ring [21] (Section 1.4).

Termination alone would not require a construction: by Navot [21], for every $\varepsilon > 0$ some d admits a bilinear algorithm for $d \times d$ matrices with integer coefficients and fewer than $d^{9/4+\varepsilon}$ multiplications, so an exhaustive search would also terminate (OpenAI [22, Remark 5.2] observes this for algebraic coefficients). What the construction adds is that prescribed tensor maps produce the seed and that its finite resources are bounded: the source side in Theorem 1.2, and the seed side, the coefficients and the preprocessing cost in Theorems 9.2 and 9.3.

The quantitative result is stated for relative schemes. Write M_d for the tensor of $d \times d$ matrix multiplication, $m\mathsf{M}_d$ for the direct sum of m copies of it with independent variables, and $T \succeq T'$ when T' is a restriction of T, that is, when linear maps on the three legs of T produce T' (Section 2). A *relative scheme* is a restriction $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ with positive integers r_*, m_*, A, B and $A < B$. It computes m_* independent products of $B \times B$ matrices from r_* products of $A \times A$ matrices, with all input transformations and output recombinations given by specified linear maps. We call A the *source side* and B the *target side*. Applying the scheme recursively, and pooling all subproblems of the same depth into batches of m_*, multiplies $n \times n$ matrices with $O(n^p)$ arithmetic operations whenever $p > 2$, where

$$p = \frac{\log(r_*/m_*)}{\log(B/A)}$$

is the *pooled exponent* (Definition 2.3 and Lemma 2.4). All logarithms are natural. The construction is indexed by an integer $a \ge 2$, the *target index*: it converts copies of the tensor $S^+_{a,a}$ of Section 1.2 into matrix multiplication tensors, and larger a gives exponents closer to $9/4$.

Theorem 1.2 (Quantitative relative construction)**.** *For every integer $a \ge 2^{64}$, the construction of Section 7, with the parameters of Theorem 8.1, gives a relative scheme $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ whose coefficients lie in a cyclotomic field $\mathbb{Q}(\zeta_\Lambda)$, with Λ computed by the construction. With $\ell_a = \lceil \log_2 a \rceil$, its pooled exponent and source side satisfy*

$$\frac{9}{4} < p < \frac{9}{4} + \frac{900}{\log a}, \qquad \log A \le 181 a^5 \ell_a^2.$$

Theorem 8.1 proves the two bounds, and Remark 8.5 identifies the field. The guaranteed bound is below 3 only when $\log a > 1200$ (Remark 8.3), whereas Section 10 evaluates the construction at $a = 12$ with other parameters and obtains $2.479 < p < 2.676$. For a rational $\varepsilon > 0$, the index $a = 2^\ell$ with $\ell = \max\{64, \lceil 1800/\varepsilon \rceil\}$ gives $p < 9/4 + \varepsilon$ (Corollary 8.2). Theorem 1.1 follows from one scheme of this family: Theorem 8.4 turns it into an algorithm over $\mathbb{Q}(\zeta_\Lambda)$ with exponent below $9/4 + \varepsilon/2$, and Corollary 8.7 unrolls the recursion, restricts the coefficients to $\mathbb{Q}$ (Lemma 8.6), and stops at the first side d_0 with fewer than $d_0^{9/4+\varepsilon}$ multiplications, which exists because of the gap $\varepsilon/2$.

Replacing the roots of unity in the rank decompositions and in exactification by rational interpolation nodes, and the separator by the polynomial separator of Lü [18], gives a variant of the construction with rational coefficients and the same counts, hence the same bounds, as in Theorem 1.2 (Section 9.1). It gives a second proof of Theorem 1.1, in which the seed side d_0 is given by an explicit formula in the parameters of the construction instead of a search (Theorem 9.2), and in which the bit lengths of the coefficients and the bit cost of computing them are bounded explicitly (Theorem 9.3). For a fixed rational seed, Section 9.4 bounds the bit cost of exact integer

multiplication and the precision that suffices for a prescribed normwise forward error. Counting the preprocessing, one algorithm multiplies $n \times n$ matrices in $n^{9/4+o(1)}$ steps for every n (Corollary 9.7); Remark 9.8 sketches a proof that, for a suitable implementation of the preprocessing, the $o(1)$ term is $O(1/\log\log n)$.

Within this construction the source side must be large (Section 11.2): every scheme it produces with pooled exponent below 3 has $A \geq 12^{18}$, and the least source side of a scheme with pooled exponent at most $9/4+\varepsilon$ is $\exp(\exp(\Theta(1/\varepsilon)))$, an order that the family of Theorem 1.2 attains. Section 11.3 describes a formalization in Lean 4 with Mathlib [9, 32] of a simpler rational variant of the construction, which clears denominators instead of rounding. It formalizes the tensor maps for splits and triplings, the occupations and the relative scheme of Sections 4, 5 and 7, and derives $\omega \leq 9/4$ over every field of characteristic zero, without explicit constants.

1.2 Where the exponent $9/4$ comes from

The following heuristic explains the exponent. No proof depends on it; Sections 4–7 make each step rigorous and bound its cost. Say that a tensor T is *worth* a matrix side V if, for large n, explicit maps convert $T^{\otimes n}$, with the help of a reusable auxiliary tensor and with losses subexponential in n, into a matrix multiplication tensor of side $V^{n-o(n)}$. Worth is multiplicative under tensor products, M_d is worth d, a tensor is worth at least as much as every tensor into which it can be converted, and a bounded factor in the number of copies does not change worth. If T has rank R and is worth V, recursion gives $\omega \leq (\log R)/\log V$, because $T^{\otimes n}$ has rank at most R^n, while M_d has rank at least d^ω.

The construction works with the *convolution tensor* $C(r,b)$, which encodes the multiplication of a polynomial with r coefficients by a polynomial with b coefficients, and with the dot product $B_X(d) = x\sum_{\gamma=0}^{d-1} y_\gamma z_\gamma$, whose first leg has a single variable. The *symmetrized convolution tensor* $S^\sigma_{r,b}$ is the tensor product of the three copies $C(r,b)^\pi$ whose legs are permuted by the even permutations π (sign $\sigma = +$) or by the odd ones ($\sigma = -$); see Section 2. Let $V(r,b)$ be the supremum of the sides that $S^\sigma_{r,b}$ is worth. It does not depend on σ, because an odd permutation of all three legs exchanges $S^+_{r,b}$ and $S^-_{r,b}$ and maps M_d to an isomorphic tensor. It is symmetric, $V(r,b) = V(b,r)$, because $S^\sigma_{r,b} \cong S^{-\sigma}_{b,r}$. Since $C(1,b) = B_X(b)$ and $S^\sigma_{1,b} \cong \mathsf{M}_b$, the boundary values are $V(1,b) = b$.

Two kinds of tensor relations, *splits* and *triplings*, constrain the other values. Each is built in three steps (Section 4). First, a restriction of a tensor power leaves M *sectors*: blocks with their own variables on the second and third legs but a common first leg. Second, the *separator* (Proposition 4.6) converts $2M-1$ copies of this tensor into M sectors that are independent on all three legs, and it attaches to each a dot product of dimension M. Third, the relation is applied to all three factors $C(r,b)^\pi$ of $S^\sigma_{r,b}$, one *orientation* π each. The three oriented copies of a dot product $B_X(d)$ have their single-variable legs in the three different positions, and their product is M_d by (2.2). So $(2M-1)^3$ source copies yield M^3 output copies, a ratio below 8, each carrying the factor M_M that the relation gains.

Split. Let $1 \leq h \leq b-1$. In bases adapted to the determinant filtration of binary forms (Section 4.1), the tensor $C(r,b)B_X(h+1)$ is block triangular with diagonal blocks $C(r,b+h-2j)$, $0 \leq j \leq h$. Fix nonnegative integers $n_0,\dots,n_h$ with sum n. In the n-th tensor power, keeping only the coordinates whose words of block indices have type $(n_0,\dots,n_h)$ removes the off-diagonal blocks and leaves $M = n!/(n_0!\cdots n_h!)$ sectors. The three orientations of the auxiliary dot product $B_X(h+1)$ form the factor M_{h+1} that the split consumes. Altogether (Corollary 4.8), $(2M-1)^3$

copies of $(S^{\sigma}_{r,b}\mathsf{M}_{h+1})^{\otimes n}$ are converted into M^3 independent copies of

$$\prod_{j=0}^{h}\left(S^{\sigma}_{r,b+h-2j}\right)^{\otimes n_j}\otimes \mathsf{M}_M.$$

Here $M = e^{nH(\theta)+O(\log n)}$ for the probability vector $\theta_j = n_j/n$, where H is the Shannon entropy. In terms of worth, therefore, $(h+1)V(r,b) \geq e^{H(\theta)}\prod_j V(r,b+h-2j)^{\theta_j}$. The right side is largest when θ_j is proportional to $V(r,b+h-2j)$, and its maximum is the sum of these worths. Hence

$$(h+1)V(r,b) \geq \sum_{j=0}^{h} V(r,b+h-2j). \tag{1.1}$$

For $h=1$ this is the discrete concavity $2V(r,b) \geq V(r,b-1)+V(r,b+1)$. For larger h it follows from concavity (Remark 4.4), but wider splits lower the finite costs.

Tripling. The second-leg indices of $C(r,4r-1)$ fall into three intervals, of lengths r, $2r-1$ and r, whose blocks are $C(r,r)$, $C(r,r)^{(23)}$ and $C(r,r)$ up to coordinate permutations (Section 4.2). In the n-th tensor power, fixing the number of middle blocks leaves $M \geq 3^n/(n+1)$ sectors, and $(2M-1)^3$ copies of $(S^{\sigma}_{r,4r-1})^{\otimes n}$ are converted into M^3 copies of $(S^{\sigma}_{r,r})^{\otimes n}\otimes \mathsf{M}_M$; the middle blocks contribute $S^{-\sigma}_{r,r} \cong S^{\sigma}_{r,r}$. Hence $V(r,4r-1) \geq 3V(r,r)$.

Normalization $V(1,b)=b$, symmetry, concavity and tripling are the axioms of a *profile* (Definition 3.1), except that a profile satisfies the tripling axiom $P(r,3b+r-1) \geq 3P(r,b)$ at every column b, whereas the tensor maps supply only the case $b=r$. Theorem 3.2 shows that every profile P satisfies $P \geq F$ for an explicit *least profile* F, with $F(a,a) = s_a(3a-1)/2$, $s_a = \prod_{h=1}^{a-1}(1+\frac{1}{3h})$, and $\log F(a,a) = \frac{4}{3}\log a + O(1)$. The diagonal part $P(a,a) \geq F(a,a)$ uses only normalization, symmetry, concavity and the tripling axiom with $b=r$. The worth V satisfies all of these, so $V(a,a) \geq F(a,a)$. Since $\mathrm{R}(S^{+}_{a,a}) \leq (2a-1)^3$, the heuristic gives

$$\omega \leq \tau_a = \frac{3\log(2a-1)}{\log F(a,a)} \longrightarrow \frac{3}{4/3} = \frac{9}{4}.$$

We call τ_a the *profile exponent*; it exceeds $9/4$ because $F(a,a) < (2a-1)^{4/3}$ for $a \geq 2$. Since F is itself a profile, these inequalities cannot show that $S^{+}_{a,a}$ is worth more than $F(a,a)$, so $9/4$ is the limit of this approach (Remark 3.3).

1.3 From the heuristic to a finite algorithm

A finite construction must decide how many times to apply each relation, so that every intermediate convolution factor produced by one relation is consumed by another. The only net effect on these factors should be to consume copies of $S^{+}_{a,a}$ and to produce copies of $\mathsf{M}_2 \cong S^{\sigma}_{1,2}$; the auxiliary matrix tensors are accounted for separately. We obtain the frequencies from a random walk in each row, the *row walk* of Section 5. For $r=2,\dots,a$, the *states* $x \in \{0,\dots,3r\}$ of row r stand for the factors $S^{\sigma_r}_{r,r-1+x}$, with column $b = r-1+x$ and sign $\sigma_r = (-1)^{a-r}$. At an interior state x, a split of width $h_{r,x} = \min(x, 3r-x)$ moves to a *child* $y \in \{x-h_{r,x}, x-h_{r,x}+2, \dots, x+h_{r,x}\}$ with probability proportional to $F(r,r-1+y)$. Since F is affine on the columns $r-1,\dots,4r-1$ of row r, which contain all children, it satisfies (1.1) with equality there, and these probabilities attain the maximum in the split step with V replaced by F. At state $3r$ (column $4r-1$) a tripling returns to state 1 (column r). State 0 is column $r-1$; by the symmetry $S^{\sigma}_{r,r-1} \cong S^{-\sigma}_{r-1,r}$, it is state 2 of row $r-1$, the *injection state* at which the walk of that row starts. This identification is why the sign alternates

between rows (Section 4.4). The walk of row a starts at state 1, the factor $S^{+}_{a,a}$, and the walk of row 2 ends at column 1, the factor $S^{\sigma_2}_{2,1} \cong \mathsf{M}_2$.

The expected numbers of visits, called *occupations*, are frequencies at which every intermediate factor is produced exactly as often as it is consumed (Proposition 5.3). Since the probabilities are tight for F, the potential $\log F$ telescopes along the walk. Per injected copy of $S^{+}_{a,a}$, the expected total gain is exactly $\log F(a,a)$, where a split gains the entropy of its child distribution minus the logarithm of its auxiliary dimension $h_{r,x}+1$, a tripling gains $\log 3$, and the final M_2 contributes $\log 2$. Because every split has the largest width that keeps its children in the row, the expected number of splits in a row and the sum of the logarithms of their auxiliary dimensions are bounded independently of the row, and both totals over all rows are at most $\Gamma_a = 105 + 180(a-2)$ (Proposition 5.3). These totals control the finite costs: the auxiliary dimensions multiply into the source side, and the number of splits sets the size of the catalyst described below.

The occupations are rationals whose denominators can be large; for the exact binary certificate of Appendix A, which uses splits of width one only, the scale needed to clear them grows at least exponentially in a. Instead, we multiply the occupations by a *rounding scale* K and round the resulting transition counts to integers so that every balance stays exact (Section 6). The telescoping identity then holds up to a *rounding loss* $\mathcal{L} \le 4J_a$, where $J_a = O(a^3)$, the number of transition coordinates, counts the possible split transitions $x \to y$ from interior states in all rows. This bound holds for every K, even when some occupations are tiny or zero (Proposition 6.3).

Section 7 applies every relation its rounded number of times, inside tensor powers of fixed type at a *type scale* N, separates the sectors and takes the three orientations. The factors common to both sides form a *catalyst* $\mathcal{C}_N$, and the result is one degeneration, the *catalytic degeneration* (Proposition 7.1): it turns L copies of $\mathcal{C}_N(S^{+}_{a,a})^{\otimes NK}\mathsf{M}_{A_1}$ into M^3 copies of $\mathcal{C}_N\mathsf{M}_{B_1}$. Here M_{A_1} collects the auxiliary factors of the splits, M is the product of the sector counts, $B_1 = 2^{NK}M$ collects the NK final factors M_2 and the separator auxiliaries, and L counts the separator source copies. Interpolation at k-th tensor powers with E points turns this degeneration into a restriction (Lemma 2.2). Using the restriction j times with one copy of the catalyst, supplied once from a rank decomposition of $\mathcal{C}_N^{\otimes k}$ with κ_N^k terms, gives a restriction $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ with $A = A_1^{kj}$ and $B = B_1^{kj}$ (Proposition 7.3). With $\Delta_N = \log(B_1/A_1)$, it is a relative scheme exactly when $\Delta_N > 0$, which is not automatic for small K. In that case Lemma 2.4 turns it into an algorithm, and its pooled exponent (7.9) is

$$p = \frac{3NK\log(2a-1) + \log(L/M^3) + (\log E)/k + (\log \kappa_N)/j}{\Delta_N}. \tag{1.2}$$

The main term $3NK\log(2a-1)$ pays for the rank of the NK source copies of $S^{+}_{a,a}$. Without the overheads of Table 1, Δ_N would equal $NK\log F(a,a)$ and p would equal τ_a; all overheads are nonnegative, so $p \ge \tau_a$. The family of Theorem 1.2 uses $K = a^3\ell_a$, $N = k = 1$ and $j = a\ell_a$, for which every overhead is at most a constant multiple of K while the main terms are of order $K\log a$; this gives the rate $900/\log a$ (Section 8).

1.4 Relation to the OpenAI preprint and later work

The bound $\omega \le 9/4$ and the scalar framework behind it are due to OpenAI [22]. That preprint introduced the profile axioms (its equations (4.5), (4.6) and (4.9)) and proved the diagonal growth bound $P(a,a) \ge a^{4/3}$ (its Lemma 5.1); the diagonal part of Theorem 3.2, $P(a,a) \ge F(a,a)$, follows from the same argument. It also gave the determinant filtration step for width one (its Lemma 4.1), the partition into three intervals behind our tripling (its Lemma 4.2), and the shared-leg separator

Overhead	Size	Made negligible by	Bounded in
rounding loss	$0 \le \mathcal{L} \le 4J_a$	rounding scale K	Proposition 6.3
type penalty	$\mathcal{T}_N = O(\log(NK))$	N or K	(7.10), (7.11)
copy overhead	$0 \le \log(L/M^3) < 3I \log 2$	N or K	(7.12)
interpolation overhead	$(\log E)/k$	interpolation power k	Lemma 2.2, (7.12)
catalyst overhead	$(\log \kappa_N)/j$	reuse count j	(7.5), (7.12)

Table 1: The overheads of the finite construction. Here $\mathcal{L}$ is the rounding loss, $L = \prod_i (2M_i - 1)^3$ is the number of separator source copies, I is the number of relations used, and the O-bound for the type penalty holds for fixed a. The rounding loss, multiplied by N, and the type penalty reduce the denominator Δ_N of (1.2); the other three overheads enter its numerator.

with $5M$ source copies (its Proposition 3.1). It applied these constructions to tensor characters and concluded through the existence of detecting characters (its Lemma 2.2).

The present paper uses none of the character theory and reproves the constructions it needs in the following forms. The local relations become type restrictions instead of degenerations that scale coordinates by powers of t (weight degenerations), so that the separator is the only degeneration. The tripling restriction fixes only the number of middle blocks, so it works for every tripling count without a divisibility condition. The splits have arbitrary width h, with auxiliary dimension $h + 1$. The separator uses $2M - 1$ source copies instead of $5M$, as did the first version of this paper (7 October 2026); Lü [18, Proposition 2.1] obtained this count independently with a polynomial separator of smaller degree (Remark 4.7). On the scalar side, we determine the least profile off the diagonal as well, show that F satisfies the axioms, so that the bound $P \ge F$ is attained, and prove the upper bound $F(m, n) \le (m + n - 1)^{4/3}$, which is strict for $m = n > 1$. Filmus [12, Section 4.3], who cites an earlier version of this paper, obtains $F/2$ on rows $2, \dots, a$ as an optimal solution of a finite linear program; the factor $1/2$ comes from his normalization. The finite analysis is new: the occupations of the row walk and their uniform bounds, the integer rounding that keeps every balance exact and its uniform bound on the rounding loss, the bounds on all overheads (Table 1), the rate of Theorem 1.2, the lower bounds of Section 11.2, and a procedure that constructs rational seeds, with the bounds of Theorems 9.2 and 9.3.

Several works that appeared after the OpenAI preprint extend its results. Navot [20] and Zuiddam [35] proved in Lean that $\omega \le 9/4$ over every field, choosing a Fourier period that is invertible in the field and descending from the algebraic closure; Navot [21] extended the operation count to every associative unital ring through decompositions with integer coefficients. Lü [18] used his separator to extend the OpenAI argument to algebraically closed fields of every characteristic, checking its other steps in outline. Filmus [12] proved a criterion that turns a finite family of shared-leg block identities with a nonnegative certificate into an exponent bound, using tensor value and asymptotic cancellation instead of characters, and reproved $\omega \le 9/4$ over every field, also in Lean. Uznański [33] extended the analysis to rectangular products in characteristic zero, and Filmus [12] extended these rectangular bounds to every field. These works prove that algorithms exist; none of them specifies a seed or bounds its size.

1.5 Related work

Strassen's algorithm showed that recursion on a fixed bilinear identity gives an exponent below three [28]. Approximate bilinear algorithms and interpolation give the standard passage from degenerations to exact algorithms [3], and Schönhage's asymptotic sum inequality bounds ω through

simultaneous independent matrix products [26]. Stothers [27, p. 24 and Theorem 7] gives an accessible account of the passage from approximate to exact algorithms by coefficient extraction and of the sum inequality. Lemma 2.2 is a root-of-unity form of interpolation, and Lemma 2.4 handles simultaneous products directly for our relative schemes.

Strassen's laser method [29, 30] and the construction of Coppersmith and Winograd [8] extract independent matrix products from tensor powers, with restrictions to words of a fixed type, entropy estimates for their number, and products over permuted legs as standard tools; Le Gall [16, Appendix A of the full version] gives a detailed example. Alman and Vassilevska Williams [1] refine the extraction of independent components, Alman et al. [2] analyze powers of the Coppersmith–Winograd tensor asymmetrically and prove $\omega < 2.371339$, Dupont et al. [11] optimize that analysis numerically and report $\omega < 2.371177$, Zadeh [34] reports $\omega < 2.37115924$ in the same framework, and an OpenAI preprint dated 24 September 2026 and released with OpenAI [22] proves $\omega < 2.371054887$ over every field [25]. A second OpenAI preprint of the same date, released at the same time, adjoins auxiliary group tensors to the extraction and proves $\omega < 2.258$ over every field of characteristic zero [23].

Strassen's asymptotic spectrum compares tensor powers through characters [7, 31], and relative bilinear complexity [30] is the setting of our relative schemes. Catalytic comparisons, which keep an auxiliary tensor on both sides of a reduction, are discussed in Christandl et al. [6, Section 3.2], which also defines the cyclic symmetrization $T \otimes T^{(123)} \otimes T^{(132)}$ that underlies $S^{+}_{r,b}$.

The determinant filtration is a case of the universal Clebsch–Gordan filtration of Maakestad [19, Lemma 2.5, Theorem 2.8, Corollary 2.9 and Definition 2.10]; we identify the multiplication induced on its quotients and the restrictions obtained by fixing types. Integer circulations between integral bounds exist by total unimodularity [14, Theorem 2], and rounding through a return edge appears in Kang and Payor [15, Sections 2.2–3.1]. Böcherer and Geiger [5, Section IV.D, Eq. (42)] bound the relative entropy of the best approximation of a distribution by a type with a fixed total; our rounding may change these totals, and Proposition 6.3 proves the bound that we need. The row walk uses the fundamental matrix of an absorbing chain [13, Theorem 11.4] and a Doob transform [17, Section 17.6.1].

1.6 Scope and organization

All tensor identities are exact, and Theorem 1.1 counts arithmetic operations on exact field elements for a fixed allowance ε: its algorithm depends on ε but not on the input size. All sizes are astronomically large: even the example with $a = 12$ in Section 10 has a source side A with about 2.3×10^{11} decimal digits, and the explicit bounds of Section 9 are far larger. The lower bounds of Section 11.2 concern only this construction; they are not lower bounds for matrix multiplication or for other constructions. Every pooled exponent of the family is strictly above $9/4$, and for a seed we prove only an exponent below $9/4 + \varepsilon$; so neither a fixed seed nor the algorithm of Corollary 9.7, whose allowance shrinks with n, is shown to use $O(n^{9/4})$ operations. That algorithm uses a seed only once n exceeds the cost of computing one, so below an astronomically large size it multiplies classically. The precision bound of Section 9.4 is a sufficient condition for a normwise forward error; we make no claim about backward stability or about accuracy in a fixed format such as binary64.

Sections 2–8 contain the proofs: general tools (Section 2), the least profile (Section 3), the splits, triplings and separator (Section 4), the row walk (Section 5), the rounding (Section 6), the finite construction and its overheads (Section 7), and the rate and rational seeds (Section 8). Section 9 gives the rational variant, the explicit seed and its preprocessing cost, the bit complexity and precision of recursion with a fixed seed, and the algorithm for all sizes. Section 10 evaluates finite examples. Section 11 discusses possible improvements, proves lower bounds on the source side

within the construction (Section 11.2), describes the Lean formalization (Section 11.3), lists the supporting files and states open questions. Appendices A–D treat the exact binary certificate, the composed relation maps on their full variable spaces, the separator with general parameters, and the notation.

2 Tensors, restrictions, and arithmetic algorithms

This section fixes conventions and proves, in the form used later, the standard tools that turn tensor identities into algorithms: identities among matrix multiplication, dot-product, and convolution tensors; composition and exactification of degenerations; and the pooled recursion, which turns a relative scheme $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ into an algorithm with exponent $\log(r_*/m_*)/\log(B/A)$ (Lemma 2.4).

2.1 Tensors, restrictions, and leg permutations

All logarithms are natural, and all vector spaces are finite dimensional. We work over $\mathbb{C}$; the coefficients of the main construction lie in a cyclotomic field (Theorem 8.4), and those of its rational variant in $\mathbb{Q}$ (Section 9.1). For a positive integer m, ζ_m denotes a primitive m-th root of unity.

A *tensor* is a trilinear form in three disjoint groups of variables, called its *legs*. We name the variables of legs 1, 2, and 3 with the letters x, y, and z, respectively. The rank $\mathrm{R}(T)$ of a tensor T is the least number of products of three linear forms, one in the variables of each leg, whose sum is T. The *flattening* of T along a leg is its coefficient matrix with rows indexed by the variables of that leg and columns indexed by pairs of variables of the other two legs; its rank is at most $\mathrm{R}(T)$.

We write $T \succeq T'$, and call T' a *restriction* of T, if linear substitutions on each leg transform T into T': every variable of T is replaced by a linear form in the variables of the same leg of T'. We read a tensor as a bilinear map: its two arguments are the values of the variables of legs 1 and 2, and its result is the vector of coefficients of the variables of leg 3. In this reading, $T \succeq T'$ says that T' can be evaluated by applying linear maps to its two arguments, evaluating T, and applying a linear map to the result. In particular $\mathrm{R}(T') \le \mathrm{R}(T)$, and no flattening rank increases under restriction. We write $T \cong T'$, and call the tensors *isomorphic*, if T' arises from T by a bijection from the variables of each leg of T to the variables of the same leg of T'; isomorphic tensors are restrictions of each other.

For a positive integer m, the tensor mT is the full direct sum of m copies of T, with independent variables on all three legs; $\oplus$ denotes full direct sums in general. Juxtaposition denotes the tensor product, whose leg-i variables are the products of the leg-i variables of the factors, and $T^{\otimes n}$ is a tensor power. Tensor products distribute over direct sums, so that $(mT)T' \cong m(TT')$, and reordering the factors of a tensor product is an isomorphism.

For $\pi \in S_3$, the tensor T^π places leg i of T at position $\pi(i)$, so that $(T^{\pi'})^\pi = T^{\pi\pi'}$. In a product such as $T^\pi T'$, leg i of T lies in leg $\pi(i)$ of the product; we call the legs of the product its *global* legs. Let $\mathcal{A}_+$ and $\mathcal{A}_-$ be the sets of even and odd permutations, and write (23) for the transposition of legs 2 and 3. Every identity of this paper also holds under the opposite convention ($T^\pi \mapsto T^{\pi^{-1}}$, $\pi\pi' \mapsto \pi'\pi$): permutations enter only through the sets $\mathcal{A}_\pm$, which are closed under inversion, and $\pi\pi'$ and $\pi'\pi$ lie in the same set because $\mathcal{A}_+$ is normal in S_3.

2.2 Matrix multiplication, dot-product, and convolution tensors

The matrix multiplication tensor of side d is $\mathsf{M}_d = \sum_{\alpha,\beta,\gamma=0}^{d-1} x_{\alpha\beta}\, y_{\beta\gamma}\, z_{\gamma\alpha}$. As a bilinear map it multiplies the matrices $(x_{\alpha\beta})$ and $(y_{\beta\gamma})$: the coefficient of $z_{\gamma\alpha}$ is the entry (α,γ) of their product.

The *dot-product tensors* $B_X(d) = x\sum_{\gamma=0}^{d-1} y_\gamma z_\gamma$, $B_Y(d) = y\sum_{\gamma=0}^{d-1} x_\gamma z_\gamma$, and $B_Z(d) = z\sum_{\gamma=0}^{d-1} x_\gamma y_\gamma$ each have one leg with a single variable. Relabeling variables gives

$$\mathsf{M}_d\mathsf{M}_{d'} \cong \mathsf{M}_{dd'}, \qquad B_X(d)B_Y(d)B_Z(d) \cong \mathsf{M}_d. \tag{2.1}$$

For the first, merge corresponding indices of the two factors. For the second, call the summation indices of $B_X(d)$, $B_Y(d)$, and $B_Z(d)$ γ, α, and β; then the leg-1, leg-2, and leg-3 variables of the product are determined by (α,β), (β,γ), and (γ,α), and renaming them gives M_d.

The tensor $B_X(d)^\pi$ has its single-variable leg at position $\pi(1)$, and $B_X(d)^{(23)} \cong B_X(d)$; hence $B_X(d)^\pi$ is isomorphic to $B_X(d)$, $B_Y(d)$, or $B_Z(d)$ according as $\pi(1)$ is 1, 2, or 3. In each of $\mathcal{A}_+ = \{\mathrm{id}, (123), (132)\}$ and $\mathcal{A}_- = \{(12), (13), (23)\}$, the value $\pi(1)$ runs through $1, 2, 3$ once. By (2.1), therefore,

$$\bigotimes_{\pi\in\mathcal{A}_\sigma} B_X(d)^\pi \cong \mathsf{M}_d \quad (\sigma \in \{+,-\}), \qquad \bigotimes_{\pi\in S_3} B_X(d)^\pi \cong \mathsf{M}_{d^2}. \tag{2.2}$$

The construction uses the first identity, for the three orientations of one set $\mathcal{A}_\sigma$; the second, for all six orientations, serves only the exact binary certificate of Appendix A.

For positive integers r and b, the *convolution tensor* is

$$C(r,b) = \sum_{\alpha=0}^{r-1}\sum_{\beta=0}^{b-1} x_\alpha\, y_\beta\, z_{\alpha+\beta}. \tag{2.3}$$

Its legs have r, b, and $r+b-1$ variables. If x_α and y_β are the coefficients of two polynomials with r and b coefficients, the bilinear map $C(r,b)$ returns the $r+b-1$ coefficients of their product. For $r=1$, the tensor $C(1,b)$ is $B_X(b)$.

The rank of $C(r,b)$ is $r+b-1$. To see this, put $\zeta = \zeta_{r+b-1}$. Evaluation at the points ζ^ℓ, followed by interpolation, gives the rank decomposition

$$C(r,b) = \frac{1}{r+b-1}\sum_{\ell=0}^{r+b-2}\Big(\sum_{\alpha=0}^{r-1}\zeta^{\ell\alpha}x_\alpha\Big)\Big(\sum_{\beta=0}^{b-1}\zeta^{\ell\beta}y_\beta\Big)\Big(\sum_{\gamma=0}^{r+b-2}\zeta^{-\ell\gamma}z_\gamma\Big). \tag{2.4}$$

Indeed, the coefficient of $x_\alpha y_\beta z_\gamma$ on the right is 1 if $\alpha+\beta \equiv \gamma \pmod{r+b-1}$ and 0 otherwise, and since $\alpha+\beta$ and γ lie in $\{0,\dots,r+b-2\}$, the congruence forces $\gamma = \alpha+\beta$. Conversely, the flattening of $C(r,b)$ along leg 3 has rank $r+b-1$, because the rows of the variables z_γ are nonzero and have disjoint supports. The tensors $B_X(d)$ and $C(r,b)$, the second identity in (2.1), and the rank of $C(r,b)$ also appear in OpenAI [22, Sections 2 and 4].

Since Section 4 applies each relation simultaneously to the three oriented copies $C(r,b)^\pi$, $\pi\in\mathcal{A}_\sigma$, the construction tracks the *symmetrized convolution tensors*

$$S^\sigma_{r,b} = \bigotimes_{\pi\in\mathcal{A}_\sigma} C(r,b)^\pi, \qquad \sigma\in\{+,-\}. \tag{2.5}$$

We call σ the *sign* of $S^\sigma_{r,b}$, and we regard $S^\sigma_{r,b}$ as a product of three factors indexed by $\pi\in\mathcal{A}_\sigma$, also when some of these factors are isomorphic. These tensors satisfy

$$S^\sigma_{r,b} \cong S^{-\sigma}_{b,r}, \qquad S^+_{r,r} \cong S^-_{r,r}, \qquad S^\sigma_{1,b} \cong \mathsf{M}_b, \qquad \mathrm{R}(S^\sigma_{r,b}) \le (r+b-1)^3. \tag{2.6}$$

For the first identity, exchanging the first two legs of $C(r,b)$ gives $C(b,r)$, that is, $C(b,r) = C(r,b)^{(12)}$. Hence $S^{-\sigma}_{b,r} = \bigotimes_{\pi\in\mathcal{A}_{-\sigma}}(C(r,b)^{(12)})^\pi = \bigotimes_{\pi\in\mathcal{A}_{-\sigma}} C(r,b)^{\pi(12)}$. As π runs through $\mathcal{A}_{-\sigma}$, the product

$\pi(12)$, which has the opposite parity, runs through $\mathcal{A}_\sigma$, so both sides have the same three factors in a different order. Taking $b = r$ gives the second identity. For the third, $C(1,b) = B_X(b)$, so $S^{\sigma}_{1,b} \cong \mathsf{M}_b$ by (2.2). The rank bound follows by tensoring three copies of (2.4), since permuting legs does not change rank. For $r, b \geq 2$ with $r \neq b$, these identities do not relate $S^{+}_{r,b}$ to $S^{-}_{r,b}$; Section 4.4 explains why the construction needs no such relation.

Appendix A, which compares the construction with the exact binary certificate, uses the sixfold product $Q_{r,b}$ below; its properties follow from (2.6) and (2.1):

$$Q_{r,b} = S^{+}_{r,b} S^{-}_{r,b} = \bigotimes_{\pi \in S_3} C(r,b)^{\pi}, \tag{2.7}$$

$$Q_{r,b} \cong Q_{b,r}, \qquad \mathrm{R}(Q_{r,b}) \leq (r+b-1)^6, \qquad Q_{1,2} \cong \mathsf{M}_2\mathsf{M}_2 \cong \mathsf{M}_4. \tag{2.8}$$

2.3 Degenerations and exactification

The local relations of Section 4 are degenerations: their maps depend on a parameter t, and they produce the desired tensor only as the constant term of a polynomial in t. The letter t denotes this parameter; as a subscript, as in M_{t} and $\mathcal{I}_t$, it abbreviates "tripling".

Let T and T' be tensors and let $D \geq 0$ be an integer. We write $T \trianglerighteq T'$, and call this a *degeneration of degree at most D*, if a triple $\Psi(t)$ of linear maps, one on each leg, whose matrix entries are Laurent polynomials in t, transforms T into a polynomial in t of degree at most D with tensor coefficients and constant term T'; the entries of $\Psi(t)$ may contain negative powers of t, but the transformed tensor may not. Degenerations enter the construction only through the next two lemmas. Each local degeneration (Corollary 4.8) is a type restriction, then a separator, then a reindexing, and Section 7 tensors all of them with one parameter; Lemma 2.1 shows that maps independent of t, applied before or after a degeneration, keep its degree bound, and that degree bounds add under tensor products.

Lemma 2.1 (Composition)**.** *For each i in a finite index set, let $\mathcal{U}_i$ be a tensor, let T_i and R_i be triples of linear maps that do not depend on t, and let $\mathsf{S}_i(t)$ be a triple of linear maps with Laurent-polynomial entries. Suppose that $\mathsf{S}_i(t)$ transforms $\mathsf{T}_i(\mathcal{U}_i)$ into a polynomial $\widetilde{T}_i(t)$ in t of degree at most D_i with constant term $\widetilde{T}_{i,0}$, and that R_i is defined on all target variables of $\mathsf{S}_i(t)$. Then the composite $\mathsf{R}_i \circ \mathsf{S}_i(t) \circ \mathsf{T}_i$ transforms $\mathcal{U}_i$ into the polynomial obtained by applying R_i to each coefficient of $\widetilde{T}_i(t)$, so $\mathcal{U}_i \trianglerighteq \mathsf{R}_i(\widetilde{T}_{i,0})$ with degree at most D_i. Applying these composites to the factors of $\bigotimes_i \mathcal{U}_i$, with one common parameter t, gives a polynomial in t of degree at most $\sum_i D_i$ with constant term $\bigotimes_i \mathsf{R}_i(\widetilde{T}_{i,0})$.*

For $\pi \in S_3$, let T_i^{π}, $\mathsf{S}_i(t)^{\pi}$, and R_i^{π} assign to leg $\pi(l)$ the maps that T_i, $\mathsf{S}_i(t)$, and R_i assign to leg l. Then $\mathsf{S}_i(t)^{\pi} \circ \mathsf{T}_i^{\pi}$ transforms $\mathcal{U}_i^{\pi}$ into $\widetilde{T}_i(t)^{\pi}$, on which R_i^{π} acts coefficient by coefficient as R_i acts on $\widetilde{T}_i(t)$. Hence both statements also hold with each $\mathcal{U}_i$ replaced by $\mathcal{U}_i^{\pi_i}$, for every choice of $\pi_i \in S_3$.

Proof. Since R_i is linear and independent of t, it acts coefficient by coefficient. In the expanded tensor product the exponents of t add and are nonnegative, so the degree is at most $\sum_i D_i$, and a term has exponent zero exactly when every factor contributes its constant term. Permuting legs reassigns the maps of each triple to other legs and changes no coefficient and no power of t. □

Lemma 2.2 (Exactification [3])**.** *Suppose that a triple $\Psi(t)$ of linear maps with Laurent-polynomial entries transforms a tensor T into a polynomial $\widetilde{T}(t)$ in t of degree at most D with constant term $\widetilde{T}_0$. Then, for every positive integer k,*

$$(kD+1)\, T^{\otimes k} \trianglerighteq \widetilde{T}_0^{\otimes k}.$$

With $E = kD+1$, this restriction uses the maps $\Psi(\zeta_E^{\ell})^{\otimes k}$, $0 \leq \ell < E$, and the scalar $1/E$.

Proof. Put $E = kD + 1$ and $\zeta = \zeta_E$. Since $\zeta^\ell \neq 0$, the Laurent entries of Ψ can be evaluated at ζ^ℓ, and $\Psi(\zeta^\ell)^{\otimes k}$ transforms $T^{\otimes k}$ into $\widetilde{T}(\zeta^\ell)^{\otimes k}$. As $\widetilde{T}(t)^{\otimes k}$ is a polynomial in t of degree at most $kD = E - 1$ with constant term $\widetilde{T}_0^{\otimes k}$, and $\sum_{\ell=0}^{E-1} \zeta^{\ell\eta} = 0$ for $0 < \eta < E$,

$$\frac{1}{E} \sum_{\ell=0}^{E-1} \widetilde{T}(\zeta^\ell)^{\otimes k} = \widetilde{T}_0^{\otimes k}.$$

Apply the ℓ-th triple to the ℓ-th summand of $E\,T^{\otimes k}$, send the copy of each target variable in every summand to that variable, which turns the direct sum of the E images into their sum, and multiply the map on one leg by $1/E$. All these maps are linear on each leg, so $E\,T^{\otimes k} \succeq \widetilde{T}_0^{\otimes k}$. □

2.4 From relative schemes to algorithms

We measure algorithms by their arithmetic cost: the number of additions, subtractions, and multiplications of field elements, including multiplications by fixed coefficients. An algorithm may depend on a fixed exponent allowance ε but not on the input dimension n; its fixed coefficients are computed once, in a preprocessing step whose cost depends on ε only.

Definition 2.3 (Relative scheme)**.** A *relative scheme* is a restriction $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$, given by its three linear maps, where r_*, m_*, A, and B are positive integers with $B > A$. We call A its *source side*, B its *target side*, and

$$p = \frac{\log(r_*/m_*)}{\log(B/A)}$$

its *pooled exponent.*

In the bilinear reading of Section 2.1, a relative scheme computes m_* independent products of $B \times B$ matrices from r_* products of $A \times A$ matrices whose inputs are formed, and whose results are combined, by linear maps. Grouping the r_* products created by one batch only among themselves would give the exponent $(\log\lceil r_*/m_*\rceil)/\log(B/A)$; pooling all products of one recursion depth removes the ceiling.

Lemma 2.4 (Pooled recursion)**.** *Let $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ be a relative scheme with pooled exponent p. Then $p \geq 2$. The scheme yields a fixed arithmetic algorithm that multiplies $n \times n$ matrices with $O(n^p)$ operations if $p > 2$, and with $O(n^2 \log n)$ operations if $p = 2$; the implied constants depend only on the scheme. Apart from multiplications by fixed coefficients, every multiplication in this algorithm multiplies a linear form in the entries of the left input matrix by a linear form in the entries of the right input matrix.*

Proof. The idea is to pool all products of one recursion depth, which have the same side, and to cut them into batches of m_* regardless of which batch created them; their number then grows by the factor r_*/m_* per depth, up to one partly filled batch, while the side shrinks by the factor A/B.

Step 1: $p \geq 2$. The flattening of M_A along leg 1 has rank A^2, because the rows of the variables $x_{\alpha\beta}$ are nonzero and have disjoint supports. Flattening ranks add over direct sums and do not increase under restriction, so $r_*A^2 \geq m_*B^2$. Thus $r_*/m_* \geq (B/A)^2 > 1$, which gives $p \geq 2$ and $r_* > m_*$.

Step 2: the recursion. Put $\rho = A/B < 1$ and $n_\infty = A/(1-\rho)$, and call $n_* = \lceil n_\infty \rceil + 1$ the *leaf threshold.* At depth ℓ the pooled recursion holds ν_ℓ products of $n_\ell \times n_\ell$ matrices, starting from $\nu_0 = 1$ and $n_0 = n$. If $n_\ell \leq n_*$, then ℓ is the last depth, denoted $\ell_{\max}$, and its products are computed by the classical algorithm. If $n_\ell > n_*$, the products are cut into $\lceil \nu_\ell/m_* \rceil$ batches of m_*, the last one filled

up with products of zero matrices, and each input matrix is padded with zeros to side $B\lceil n_\ell/B\rceil$ and cut into a $B\times B$ array of blocks of side $\lceil n_\ell/B\rceil$. A batch is then an instance of $m_*\mathsf{M}_B$ whose entries are blocks; the maps of the scheme on legs 1 and 2 form from it the inputs of r_* products of side $A\lceil n_\ell/B\rceil$, and the map on leg 3 forms the m_* outputs from their results. This is valid for blocks: the scheme is an identity of trilinear forms with scalar coefficients in which each product has a linear form in the left input before one in the right input, so it holds over the noncommutative ring of block matrices. Therefore

$$\nu_{\ell+1} = r_*\lceil \nu_\ell/m_*\rceil \le \frac{r_*}{m_*}\nu_\ell + r_*, \qquad n_{\ell+1} = A\lceil n_\ell/B\rceil. \tag{2.9}$$

Apart from the products, the work at a depth $\ell < \ell_{\max}$ (padding, cutting into blocks, the fixed linear maps on at most ν_ℓ batches of blocks of side $\lceil n_\ell/B\rceil \le n_\ell$, and removing the padding) takes $O(\nu_\ell n_\ell^2)$ operations, since $B\lceil n_\ell/B\rceil < n_\ell + B$.

The computation runs in two passes: a downward pass forms the inputs and records the batches depth by depth, the products at depth $\ell_{\max}$ are computed classically, and an upward pass applies the output maps depth by depth, discarding the padding and the zero products. Inputs at depth $\ell+1$ depend only on inputs at depth ℓ, and outputs at depth ℓ only on outputs at depth $\ell+1$, so products created by different batches may share a batch.

Step 3: counts and depths. Since $r_* > m_*$, induction on (2.9) with the geometric sum $\sum_{\ell'<\ell}(r_*/m_*)^{\ell'} \le (r_*/m_*)^\ell m_*/(r_*-m_*)$ gives $\nu_\ell \le (1+\frac{r_*m_*}{r_*-m_*})(r_*/m_*)^\ell$; as $r_*/m_* = \rho^{-p}$ by the definition of p, this is $\nu_\ell = O(\rho^{-p\ell})$. For the sides, $n_\ell/B \le \lceil n_\ell/B\rceil < n_\ell/B + 1$ gives $\rho n_\ell \le n_{\ell+1} < \rho n_\ell + A$, and since $A = (1-\rho)n_\infty$, the upper bound reads $n_{\ell+1} - n_\infty < \rho(n_\ell - n_\infty)$. Hence

$$\rho^\ell n \le n_\ell < n_\infty + \rho^\ell n.$$

If depth ℓ is not the last one, then $n_\ell > n_* \ge n_\infty + 1$, so $\rho^\ell n > n_\ell - n_\infty > 1$; as $\rho^\ell n \to 0$, the recursion stops. Taking $\ell = \ell_{\max} - 1$ gives $\rho^{\ell_{\max}} n > \rho$, which also holds for $\ell_{\max} = 0$ because $n \ge 1 > \rho$; and $\rho^{\ell_{\max}} n \le n_{\ell_{\max}} \le n_*$. Thus

$$\rho < \rho^{\ell_{\max}} n \le n_*, \qquad \text{so} \qquad \ell_{\max} = \log_{1/\rho} n + O(1).$$

Step 4: total cost. Each classical product at the last depth costs at most $2n_*^3$ operations, so the leaves cost $O(\nu_{\ell_{\max}}) = O(\rho^{-p\ell_{\max}})$. By Steps 2 and 3, and since $n_\ell^2 \le 2n_\infty^2 + 2\rho^{2\ell}n^2$, the work at a depth $\ell < \ell_{\max}$ is $O(\rho^{-p\ell} + \rho^{(2-p)\ell}n^2)$. Including the last depth, the total cost is therefore

$$O\Big(\sum_{\ell=0}^{\ell_{\max}} \big(\rho^{-p\ell} + \rho^{(2-p)\ell}n^2\big)\Big).$$

The first sum is geometric with ratio $\rho^{-p} > 1$, hence $O(\rho^{-p\ell_{\max}}) = O\big(n^p(\rho^{\ell_{\max}}n)^{-p}\big)$. If $p > 2$, the second sum is geometric with ratio $\rho^{2-p} > 1$, hence $O(\rho^{(2-p)\ell_{\max}}n^2) = O\big(n^p(\rho^{\ell_{\max}}n)^{2-p}\big)$. Both are $O(n^p)$ because $\rho^{\ell_{\max}}n > \rho$. If $p = 2$, each of the $\ell_{\max}+1 = O(\log n)$ terms of the second sum equals n^2.

Step 5: bilinearity. Padding, cutting into blocks, batching, and the maps of the scheme on legs 1 and 2 act linearly and separately on left and right inputs, so by induction on the depth every entry of a left (right) input at every depth is a linear form in the entries of the original left (right) input. The classical algorithm at the last depth multiplies an entry of a left input by an entry of a right input, and all other multiplications are by fixed coefficients. □

3 The least profile

This section is purely scalar. A *profile* is a positive symmetric function of two positive integers that satisfies three conditions, the profile axioms of Definition 3.1. We prove that the function F in (3.4) is the least profile and that $\log F(a,a) = \frac{4}{3}\log a + O(1)$ (Theorem 3.2). The construction of Sections 4–7 uses F as a potential (Remark 3.3), and the diagonal values of F determine the profile exponent $\tau_a = 3\log(2a-1)/\log F(a,a)$.

Definition 3.1. A *profile* is a positive symmetric function $P : \mathbb{N}_{>0}^2 \to \mathbb{R}$ such that, for all positive integers r and b,

$$P(1,b) = b, \tag{3.1}$$

$$2P(r,b) \geq P(r,b-1) + P(r,b+1) \quad (b \geq 2), \tag{3.2}$$

$$P(r,3b+r-1) \geq 3P(r,b). \tag{3.3}$$

We call these the normalization, concavity and tripling axioms.

In the heuristic of Section 1.2, $P(r,b)$ is the matrix side that $S^{\sigma}_{r,b}$ is worth. Normalization and symmetry then come from the isomorphisms $S^{\sigma}_{1,b} \cong \mathsf{M}_b$ and $S^{\sigma}_{r,b} \cong S^{-\sigma}_{b,r}$ of (2.6), concavity from the split of width one (Lemma 4.3 with $h = 1$), and the tripling axiom at column $b = r$, that is, $P(r,4r-1) \geq 3P(r,r)$, from Lemma 4.5. Wider splits give no further inequality (Remark 4.4).

For a positive integer m put

$$s_m = \prod_{h=1}^{m-1}\left(1 + \frac{1}{3h}\right), \qquad F(m,n) = s_{\min(m,n)}\left(\max(m,n) + \frac{\min(m,n)-1}{2}\right). \tag{3.4}$$

The empty product gives $s_1 = 1$, and $s_{m+1}/s_m = (3m+1)/(3m)$. Along each row r, F is affine from column $r-1$ on:

$$F(r,b) = s_r\,\frac{2b+r-1}{2} \qquad (r \geq 1,\ b \geq \max(1, r-1)). \tag{3.5}$$

For $b \geq r$ this is (3.4). For $b = r-1 \geq 1$, symmetry gives $F(r,r-1) = s_{r-1}(3r-2)/2$, and $s_{r-1} = s_r(3r-3)/(3r-2)$ turns this into $s_r(3r-3)/2$, which is (3.5) at $b = r-1$. Thus every increment $F(r,b+1) - F(r,b)$ with $b \geq r-1$ equals s_r, and (3.5) gives $F(r,4r-1) = s_r(9r-3)/2 = 3s_r(3r-1)/2 = 3F(r,r)$.

Theorem 3.2 (Least profile)**.** *The function F is a profile, and every profile P satisfies $P(m,n) \geq F(m,n)$ for all positive integers m and n. In addition,*

$$F(m,n) \leq (m+n-1)^{4/3}, \qquad \log F(a,a) = \frac{4}{3}\log a + O(1). \tag{3.6}$$

The first inequality is strict when $m = n > 1$.

The three axioms are the relations derived by OpenAI [22, eqs. (4.5), (4.6) and (4.9)] for a symmetrized convolution profile, with tripling proved there at every column $b \geq 1$ [22, Lemma 4.2]; our tensor maps realize only $b = r$. The diagonal bound $P(a,a) \geq F(a,a)$ is also due to OpenAI [22]: the proof of its Lemma 5.1 (Diagonal growth) establishes it as eq. (5.2), in the form $2P(a,a)/(3a-1) \geq s_a$, on the way to the bound $P(a,a) \geq a^{4/3}$ stated there, and Step 2 below follows that proof. Steps 1, 3 and 4 are not in OpenAI [22]. Filmus [12, Section 4.3] considers the finite linear program formed by concavity and the tripling axiom at $b = r$ on rows $2, \ldots, a$. His normalization gives the

value 1 at $(2,1)$, where $F(2,1)=2$, and his optimal profile [12, eq. (4.6)] is $F/2$ on the columns $r-1\le b\le 4r-1$ of these rows.

Besides the axioms, the proof uses two facts about a *concave sequence*, that is, a sequence $g(1),g(2),\dots$ whose increments $g(b+1)-g(b)$ are nonincreasing. Let $b'<b''$. On $[b',b'']$ the sequence lies on or above the chord through $(b',g(b'))$ and $(b'',g(b''))$, and outside $[b',b'']$ it lies on or below the extension of that chord. Both facts follow by comparing average increments: for $b'<b<b''$, the average increment over $[b',b]$ is at least that over $[b,b'']$; for $b>b''$, the average over $[b',b'']$ is at least that over $[b'',b]$; the case $b<b'$ is symmetric. By (3.2), every row $b\mapsto P(r,b)$ of a profile is a concave sequence.

Proof of Theorem 3.2. Step 1: F is a profile. Positivity and symmetry are clear from (3.4), and $F(1,b)=s_1b=b$ gives normalization. Fix a row r. For every column $b\le r+1$ we have $F(r,b)=s_b(2r+b-1)/2$: for $b\le r$ this is (3.4), and for $b=r+1$ both sides equal $s_r(3r+1)/2$ because $s_{r+1}=s_r(3r+1)/(3r)$. Let $2\le b\le r$. The columns $b-1,b,b+1$ are at most $r+1$, so substituting $s_{b-1}=s_b(3b-3)/(3b-2)$ and $s_{b+1}=s_b(3b+1)/(3b)$ gives

$$2F(r,b)-F(r,b-1)-F(r,b+1)=\frac{s_b}{2}\left(2(2r+b-1)-\frac{(3b-3)(2r+b-2)}{3b-2}-\frac{(3b+1)(2r+b)}{3b}\right).$$

Over the common denominator $3b(3b-2)$, the numerator of the bracket is

$$6b(3b-2)(2r+b-1)-3b(3b-3)(2r+b-2)-(3b-2)(3b+1)(2r+b)=4(r-b),$$

because its coefficient of r is $12b(3b-2)-6b(3b-3)-2(3b-2)(3b+1)=4$ and its remaining terms add up to $-4b$. So the second difference equals $2(r-b)s_b/\big(3b(3b-2)\big)\ge 0$. For $b>r$, the columns $b-1,b,b+1$ are at least r, where the row is affine by (3.5), so the second difference vanishes. Hence every row of F is concave.

For tripling, recall that the increments of row r equal s_r from column $r-1$ on. The extension of the chord through columns r and $r+1$ is therefore the line $b\mapsto s_r(2b+r-1)/2$, and the second fact about concave sequences gives $F(r,b)\le s_r(2b+r-1)/2$ for every $b\ge 1$. Since $3b+r-1\ge r$, (3.5) and this bound give

$$F(r,3b+r-1)=s_r\,\frac{6b+3r-3}{2}=3s_r\,\frac{2b+r-1}{2}\ge 3F(r,b),$$

which is (3.3) for F, with equality when $b\ge r-1$.

Step 2: the diagonal bound. Let P be a profile, and for $a\ge 1$ put $\chi_a=2P(a,a)/(3a-1)$, so that $P(a,a)=(3a-1)\chi_a/2$. Since $F(a,a)=s_a(3a-1)/2$, the claim $P(a,a)\ge F(a,a)$ is the inequality $\chi_a\ge s_a$. The idea is the following. Tripling forces row a to rise by at least $2P(a,a)$ over the $3a-1$ columns after the diagonal, and by concavity the first of these increments is the largest, so it is at least χ_a. Symmetry then turns the increments next to the diagonal into the step from $P(a-1,a-1)$ to $P(a,a)$, which yields a recursion for χ_a with the same factors as s_a.

Fix $a\ge 1$. The tripling axiom in row a at column a gives $P(a,4a-1)\ge 3P(a,a)$, so

$$2P(a,a)\le P(a,4a-1)-P(a,a)=\sum_{b=a}^{4a-2}\big(P(a,b+1)-P(a,b)\big)\le(3a-1)\big(P(a,a+1)-P(a,a)\big),$$

because the sum has $3a-1$ increments and none exceeds the first. Dividing by $3a-1$,

$$P(a,a+1)-P(a,a)\ge\chi_a. \tag{3.7}$$

Now let $a \geq 2$. Symmetry gives $P(a, a-1) = P(a-1, a)$, so

$$P(a,a) - P(a-1,a-1) = \big(P(a,a) - P(a,a-1)\big) + \big(P(a-1,a) - P(a-1,a-1)\big).$$

By concavity of row a, the first difference is at least $P(a,a+1) - P(a,a)$, hence at least χ_a by (3.7). The second difference is at least χ_{a-1}, by (3.7) for $a-1$. Thus $P(a,a) - P(a-1,a-1) \geq \chi_a + \chi_{a-1}$. Substituting $P(a,a) = (3a-1)\chi_a/2$ and $P(a-1,a-1) = (3a-4)\chi_{a-1}/2$, and multiplying by 2, we obtain

$$(3a-1)\chi_a - (3a-4)\chi_{a-1} \geq 2\chi_a + 2\chi_{a-1}, \qquad \text{that is,} \qquad (3a-3)\chi_a \geq (3a-2)\chi_{a-1}.$$

The product s_a obeys the same recursion with equality: $s_a/s_{a-1} = 1+1/(3(a-1)) = (3a-2)/(3a-3)$. Normalization gives $\chi_1 = P(1,1) = 1 = s_1$, and if $\chi_{a-1} \geq s_{a-1}$, then $\chi_a \geq (3a-2)\chi_{a-1}/(3a-3) \geq (3a-2)s_{a-1}/(3a-3) = s_a$. By induction $\chi_a \geq s_a$, that is, $P(a,a) \geq F(a,a)$, for every $a \geq 1$. This step uses tripling only at the column equal to the row index, the instances realized by Lemma 4.5.

Step 3: the off-diagonal bound. By symmetry it suffices to treat $m < n$. The idea is to reach column n of row m from the diagonal by repeated tripling and to interpolate by concavity. Define columns $b_0 = m$ and $b_{\nu+1} = 3b_\nu + m - 1$ for $\nu \geq 0$. Since $b_0 - m = 0$ and $b_{\nu+1} - m = 3(b_\nu - m) + (3m-1)$, we have $b_\nu - m = (3^\nu - 1)(3m-1)/2$ for $\nu \geq 0$. Tripling in row m at column b_ν gives $P(m, b_{\nu+1}) \geq 3P(m, b_\nu)$, so $P(m, b_\nu) \geq 3^\nu P(m,m)$ for every $\nu \geq 0$. The columns b_ν increase without bound, so we may fix $\nu \geq 1$ with $b_\nu \geq n$. Column n then lies between m and b_ν, and the chord fact for the concave row $b \mapsto P(m,b)$ gives

$$\begin{aligned} P(m,n) &\geq P(m,m) + \frac{n-m}{b_\nu - m}\big(P(m,b_\nu) - P(m,m)\big) \geq P(m,m)\left(1 + \frac{2(n-m)}{3m-1}\right) \\ &= \frac{2P(m,m)}{3m-1}\left(n + \frac{m-1}{2}\right). \end{aligned}$$

By Step 2, $2P(m,m)/(3m-1) = \chi_m \geq s_m$, so $P(m,n) \geq s_m(n + (m-1)/2) = F(m,n)$. This step uses tripling at columns b_ν beyond the row index, and for each pair (m,n) only finitely many instances of the axioms.

Step 4: the upper bounds. First, $s_a^3 \leq (3a-1)/2$ for every $a \geq 1$, with strict inequality for $a > 1$. Equality holds at $a = 1$. If $s_a^3 \leq (3a-1)/2$, then $s_{a+1}^3 = s_a^3(3a+1)^3/(3a)^3 \leq (3a-1)(3a+1)^3/(2(3a)^3) < (3a+2)/2$, where the last inequality is equivalent to $(3a+2)(3a)^3 - (3a-1)(3a+1)^3 = 6a+1 > 0$. Now let $m \leq n$. Both $(3m-1)/2$ and $n + (m-1)/2$ are at most $m+n-1$. Cubing (3.4) therefore gives $F(m,n)^3 = s_m^3\big(n+(m-1)/2\big)^3 \leq \big((3m-1)/2\big)\big(n+(m-1)/2\big)^3 \leq (m+n-1)^4$, which proves the first part of (3.6). For $m = n > 1$ the middle inequality is strict. Finally, take logarithms in the product for s_a. Since $\log(1 + 1/(3h)) = 1/(3h) + O(h^{-2})$ and the series $\sum_h h^{-2}$ converges, $\log s_a = \sum_{h=1}^{a-1} 1/(3h) + O(1) = \frac{1}{3}\log a + O(1)$. Together with $\log((3a-1)/2) = \log a + O(1)$, this gives $\log F(a,a) = \frac{4}{3}\log a + O(1)$. □

For $a \geq 2$ we have $F(a,a) \geq (3a-1)/2 > 1$, since $s_a \geq 1$, and the strict case of (3.6) gives $0 < \log F(a,a) < \frac{4}{3}\log(2a-1)$. Hence the profile exponent satisfies $\tau_a > 9/4$, and $\tau_a \to 9/4$ by the asymptotic formula in (3.6).

Remark 3.3 (How F is used, and what the axioms can certify). The construction of Sections 4–7 uses F only as a potential. The transition probabilities of Section 5 are ratios of values of F within one row, which sum to one because the row is affine (3.5); the identity $F(r, 4r-1) = 3F(r,r)$ makes each tripling contribute exactly $\log 3$; and the changes of $\log F$ along the row walk telescope, so only $\log F(a,a)$ enters the exponent (Proposition 6.3). The lower bound $P \geq F$ is not used. Together with

the fact that F is itself a profile (Step 1), however, it shows that $F(a,a)$ is the largest lower bound on $P(a,a)$ that holds for every profile. Through the heuristic of Section 1.2, the axioms therefore give the exponent τ_a at target index a and nothing better, so $9/4$ is the limit of this approach. This remains true when the axioms are combined with the upper bound $P(m,n) \le (m+n-1)^{4/3}$, which F satisfies by (3.6). That bound is equivalent to $P(m,n)^{9/4} \le (m+n-1)^3$, whose right side bounds the rank of $S^{\sigma}_{m,n}$ by (2.6); OpenAI [22, eq. (4.5)] uses a bound of this kind, from the rank of $C(m,n)$.

4 Tensor maps for splits and triplings

This section constructs the tensor maps behind the profile inequalities $(h+1)P(r,b) \ge \sum_{j=0}^{h} P(r,b+h-2j)$, a consequence of the concavity axiom (3.2) (Remark 4.4), and $P(r,4r-1) \ge 3P(r,r)$, the tripling axiom (3.3) at $b=r$. A *split* of width h converts the convolution tensor $C(r,b)$, together with a dot product of dimension $h+1$, into the convolution tensors $C(r,b+h-2j)$, $0 \le j \le h$. A *tripling* converts $C(r,4r-1)$ into three copies of $C(r,r)$, one of them with its second and third legs exchanged. Corollary 4.8 states both as relations among the symmetrized convolution tensors $S^{\sigma}_{r,b}$ that also produce an auxiliary matrix multiplication tensor; Section 7 applies it to every relation of the construction.

Each relation is built in three steps. First, a restriction of a tensor power (Sections 4.1 and 4.2) produces a sum $T = \sum_{g=0}^{M-1} T_g$ of M *sectors sharing the first leg*: $T_g \in X \otimes Y_g \otimes Z_g$ with $Y = \bigoplus_g Y_g$ and $Z = \bigoplus_g Z_g$. This is not a direct sum, because all sectors use the same first-leg variables, so a coordinate change that depends on the sector can be applied on the second and third legs but not yet on the first. Second, the separator (Section 4.3) gives each sector its own first leg. Third, the relation is applied in three orientations (Section 4.4). The split, the tripling and the separator each adapt a construction of OpenAI [22]; we state the correspondence after each result.

4.1 Splits from the determinant filtration

We obtain a split from a filtration, preserved by multiplication, of the input and output spaces of polynomial multiplication. Its successive quotients carry the tensors $C(r,b+h-2j)$, so in adapted bases $C(r,b)B_X(h+1)$ is block triangular with these blocks on the diagonal. For $r \ge 2$ the off-diagonal blocks are not zero (Example 4.2), but in a tensor power they disappear when we keep only the coordinates with a prescribed number of factors from each quotient (Lemma 4.3).

Binary forms. For $n \ge 1$ let $V_n = \mathbb{Q}[u,v]_{n-1}$ be the space of binary forms with n coefficients (indexed by dimension, not degree), with monomial basis $u^{n-1-i}v^i$, $0 \le i < n$. The convolution tensor (2.3) is then the tensor of multiplication $V_r \times V_b \to V_{r+b-1}$: the monomials with indices α and β multiply to the monomial with index $\alpha+\beta$, and z_γ reads the coefficient of the γ-th monomial of V_{r+b-1}. Let $V'_n = \mathbb{Q}[u',v']_{n-1}$ be a copy of V_n in new variables. Multiplication by $f \in V_r$ on the first factor maps $V_b \otimes V'_{h+1}$ to $V_{r+b-1} \otimes V'_{h+1}$, and in monomial bases its tensor is $C(r,b)B_X(h+1)$: the factor V'_{h+1} passes through unchanged, which is the dot product $B_X(h+1)$ with its one-dimensional first leg.

The filtration. Put $\Delta = uv' - vu'$. For integers $h \ge 0$ and $c \ge h+1$, filter $V_c \otimes V'_{h+1}$ by

$$\mathcal{F}^j(V_c \otimes V'_{h+1}) = \Delta^j\,(V_{c-j} \otimes V'_{h+1-j}) \quad (0 \le j \le h), \qquad \mathcal{F}^{h+1}(V_c \otimes V'_{h+1}) = 0,$$

where Δ^j multiplies in $\mathbb{Q}[u,v,u',v']$ and $V'_1=\mathbb{Q}$. The letter c stands for a number of coefficients; we use $c=b$ for the second leg and $c=r+b-1$ for the output. Lemma 4.1 below shows that dividing by Δ^j and substituting $(u',v')=(u,v)$ gives an isomorphism

$$\mathcal{F}^j/\mathcal{F}^{j+1}\cong V_{c+h-2j}. \tag{4.1}$$

This is the universal Clebsch–Gordan filtration of Maakestad [19, Lemma 2.5, Theorem 2.8, Corollary 2.9 and Definition 2.10] for the symmetric powers of degrees h and $c-1$ of a rank-two module; we construct the integer bases we need directly.

For $0\le i<c+h$, the *lift* of the i-th monomial of V_{c+h} is

$$L_{c,h}(i)=\begin{cases} u^{c-1-i}v^iu'^h, & 0\le i\le c-1,\\ v^{c-1}u'^{c+h-1-i}v'^{i-c+1}, & c-1<i<c+h.\end{cases}$$

It lies in $V_c\otimes V'_{h+1}$, its total exponent of v and v' is i, and the substitution $(u',v')=(u,v)$ maps it to $u^{c+h-1-i}v^i$. The *determinant basis* of $V_c\otimes V'_{h+1}$ consists of the vectors

$$\mathbf{b}_{j,i}=\Delta^j L_{c-j,h-j}(i),\qquad 0\le j\le h,\quad 0\le i<c+h-2j. \tag{4.2}$$

We call j the *filtration index* of $\mathbf{b}_{j,i}$, and we write $\mathbf{b}^{(c)}_{j,i}$ when the space must be indicated.

Lemma 4.1 (Determinant basis)**.** *Let $h\ge 0$ and $c\ge h+1$.*

(a) The vectors (4.2) form a basis of $V_c\otimes V'_{h+1}$ in which every monomial has integer coordinates. Those with filtration index at least j form a basis of $\mathcal{F}^j(V_c\otimes V'_{h+1})$.

(b) Dividing by Δ^j and substituting $(u',v')=(u,v)$ gives the isomorphism (4.1), which maps the class of $\mathbf{b}_{j,i}$ to the i-th monomial $u^{c+h-2j-1-i}v^i$ of V_{c+h-2j}.

(c) For $r\ge 1$ and $f\in V_r$, multiplication by f maps $\mathcal{F}^j(V_c\otimes V'_{h+1})$ into $\mathcal{F}^j(V_{r+c-1}\otimes V'_{h+1})$. On the j-th quotients it induces multiplication by f from V_{c+h-2j} to $V_{r+c+h-2j-1}$.

Proof. Step 1: part (a). We rewrite monomials with the relation $uv'=vu'+\Delta$ until they become lifts. Let ψ be a monomial in $V_c\otimes V'_{h+1}$ with total exponent i of v and v'. If uv' divides ψ, write $\psi=\psi_1uv'=\psi_1vu'+\Delta\psi_1$ with $\psi_1\in V_{c-1}\otimes V'_h$; the monomial ψ_1vu' has the same total exponent i and smaller exponents of u and v'. Repeating this, we reach a monomial not divisible by uv', namely $u^{c-1-i}v^iu'^h$ or $v^{c-1}u'^{c+h-1-i}v'^{i-c+1}$, which is the lift $L_{c,h}(i)$. Hence $\psi=L_{c,h}(i)+\Delta\rho$, where ρ is an integer combination of monomials of $V_{c-1}\otimes V'_h$; when $h=0$, every monomial is already a lift and $\rho=0$. Since $c-1\ge h$, the monomials of ρ can be treated in the same way, and after at most h rounds every monomial of $V_c\otimes V'_{h+1}$ is an integer combination of the vectors $\mathbf{b}_{j,i}$. Their number is $\sum_{j=0}^{h}(c+h-2j)=c(h+1)=\dim(V_c\otimes V'_{h+1})$, so they form a basis in which every monomial has integer coordinates. Applying this to $V_{c-j}\otimes V'_{h+1-j}$ and multiplying by Δ^j, which is injective, shows that the $\mathbf{b}_{j',i}$ with $j'\ge j$ form a basis of $\mathcal{F}^j$.

Step 2: part (b). The map $\mathcal{F}^j\to V_{c+h-2j}$ that sends $\Delta^j\rho$ to $\rho(u,v,u,v)$ is well defined, because $\Delta^j\rho$ determines ρ. It sends $\mathbf{b}_{j,i}$ to the i-th monomial, and $\mathbf{b}_{j',i}$ with $j'>j$ to zero, because Δ vanishes when $(u',v')=(u,v)$. By (a), it is surjective with kernel $\mathcal{F}^{j+1}$.

Step 3: part (c). For $\rho\in V_{c-j}\otimes V'_{h+1-j}$ we have $f\Delta^j\rho=\Delta^j(f\rho)$ with $f\rho\in V_{r+c-j-1}\otimes V'_{h+1-j}$, so multiplication by f preserves the filtration. Since $(f\rho)(u,v,u,v)=f\cdot\rho(u,v,u,v)$, the induced map on quotients is multiplication by f. □

The tensor in the determinant bases. Let Π_c be the matrix whose columns are the vectors $\mathbf{b}^{(c)}_{j,i}$ in monomial coordinates; by Lemma 4.1(a), Π_c and Π_c^{-1} have integer entries. We keep the monomial basis $u^{r-1-\alpha}v^\alpha$ of the first leg V_r, with variables x_α, and change the bases of the other two legs. On the second leg, the input space $V_b \otimes V'_{h+1}$, using the basis $\mathbf{b}^{(b)}_{j,i}$ means substituting $y = \Pi_b y'$ for the vector y of second-leg variables. The third leg is the dual of the output space $V_{r+b-1} \otimes V'_{h+1}$, since its variables read off the coordinates of an output, so changing the output basis to $\mathbf{b}^{(r+b-1)}_{j,i}$ changes the third-leg variables contragrediently, by $z = \Pi_{r+b-1}^{-\top} z'$. After these substitutions, the coefficient of $x_\alpha y'_{(j,i)} z'_{(j',i')}$ is the coordinate along $\mathbf{b}^{(r+b-1)}_{j',i'}$ of the product $u^{r-1-\alpha}v^\alpha\,\mathbf{b}^{(b)}_{j,i}$. By Lemma 4.1, this coefficient is zero when $j' < j$, and for $j' = j$ it is one if $i' = i + \alpha$ and zero otherwise. Thus multiplication never decreases the filtration index, and the diagonal blocks of the tensor are exactly $C(r, b+h)$, $C(r, b+h-2)$, $\ldots$, $C(r, b-h)$.

Example 4.2. Take $r = 2$, $b = 3$ and $h = 1$, so that $C(2,3)B_X(2)$ is multiplication by $f \in V_2$ from $V_3 \otimes V'_2$ to $V_4 \otimes V'_2$. The determinant bases of these two spaces are as follows, with $i = 0, 1, 2, \ldots$ from left to right in each row:

	$\mathbf{b}^{(3)}_{j,i} \in V_3 \otimes V'_2$	$\mathbf{b}^{(4)}_{j,i} \in V_4 \otimes V'_2$
$j = 0$	$u^2u',\ uvu',\ v^2u',\ v^2v'$	$u^3u',\ u^2vu',\ uv^2u',\ v^3u',\ v^3v'$
$j = 1$	$u\Delta,\ v\Delta$	$u^2\Delta,\ uv\Delta,\ v^2\Delta$

The diagonal blocks are $C(2,4)$ for filtration index 0 and $C(2,2)$ for index 1. Most products of a basis vector with u or v are again basis vectors; for instance $v\,\mathbf{b}^{(3)}_{0,3} = v^3v' = \mathbf{b}^{(4)}_{0,4}$ and $u\,\mathbf{b}^{(3)}_{1,0} = u^2\Delta = \mathbf{b}^{(4)}_{1,0}$. The only exception is $u \cdot v^2v' = v^3u' + v^2\Delta = \mathbf{b}^{(4)}_{0,3} + \mathbf{b}^{(4)}_{1,2}$. The first term is the diagonal entry $x_0 y'_{(0,3)} z'_{(0,3)}$, since u is the monomial with $\alpha = 0$ and $i' = i + \alpha = 3$. The second gives the single off-diagonal coefficient, at $x_0 y'_{(0,3)} z'_{(1,2)}$. In a tensor power, a term that uses this coefficient in some position has a larger total filtration index on the third leg than on the second, so the type restriction of Lemma 4.3 removes it.

Words, types and sectors. In the n-th tensor power we use on the second leg the product basis vectors $\mathbf{b}_{j_1,i_1} \otimes \cdots \otimes \mathbf{b}_{j_n,i_n}$, and on the third leg the coordinates dual to the corresponding product basis of the output. The *word* of such a basis vector, and of its coordinate, is the sequence of filtration indices $\mathbf{j} = (j_1, \ldots, j_n) \in \{0, \ldots, h\}^n$. The *type* of a word is its count vector $(n_0, \ldots, n_h)$, where n_j is the number of positions θ with $j_\theta = j$. A type restriction keeps, on the second and third legs, only the coordinates whose words have a prescribed type. If every surviving term has the same word on both legs, the surviving terms with word $\mathbf{j}$ form the *sector* of $\mathbf{j}$, a tensor $T_\mathbf{j} \in X \otimes Y_\mathbf{j} \otimes Z_\mathbf{j}$, where $Y_\mathbf{j}$ and $Z_\mathbf{j}$ are spanned by the coordinates with word $\mathbf{j}$ and X is the whole first leg; we use the word, its coordinates and the tensor $T_\mathbf{j}$ interchangeably. The sectors thus share the first leg.

Lemma 4.3 (Type restriction for a split)**.** *Let $r \geq 1$, $h \geq 1$ and $b \geq h+1$, and let $n_0, \ldots, n_h$ be nonnegative integers with positive sum n. In the determinant bases on the second and third legs, keeping only the coordinates whose words have type $(n_0, \ldots, n_h)$ is a restriction of $(C(r,b)B_X(h+1))^{\otimes n}$ to a tensor with $M = n!/\prod_{j=0}^h n_j!$ sectors sharing the first leg $X = V_r^{\otimes n}$, one for each word of this type. After its tensor positions are reordered according to its word, each sector is $\prod_{j=0}^h C(r, b+h-2j)^{\otimes n_j}$. On the shared first leg, this word-dependent reordering is made only after separation.*

Proof. Apply the coordinate change above in every tensor position. In product bases the coefficient of a term is the product of the coefficients in the n positions, so a term with second-leg word $\mathbf{j}$

and third-leg word $\mathbf{j}'$ can be nonzero only if $j'_\theta \geq j_\theta$ for every position θ. Setting to zero every second- and third-leg coordinate whose word does not have type $(n_0, \dots, n_h)$ is a restriction. For a surviving term both words have this type, so $\sum_\theta j'_\theta = \sum_j j n_j = \sum_\theta j_\theta$, and with the positionwise inequalities this forces $j'_\theta = j_\theta$ for every θ. Hence the restricted tensor is the sum of the sectors $T_{\mathbf{j}}$ over the M words of the given type. In $T_{\mathbf{j}}$ position θ uses the diagonal block of index j_θ, so $T_{\mathbf{j}} = \bigotimes_{\theta=1}^{n} C(r, b+h-2j_\theta)$, with the factors in the order of the positions. Reordering the positions so that their indices j_θ increase gives the stated product. □

The restriction in Lemma 4.3 involves no degeneration parameter, so it adds nothing to the interpolation degree. For $h = 1$, the same filtration and the same two diagonal blocks $C(r, b+1)$ and $C(r, b-1)$ appear in the proof of OpenAI [22, Lemma 4.1], which removes the off-diagonal block by a degeneration that scales coordinates by powers of t; here the type restriction removes it.

Remark 4.4 (Wider splits). A split of width h uses an auxiliary dot product of dimension $h+1$, whereas producing the same lengths $b+h-2j$ by h successive splits of width one would use dimension 2^h. Wider splits therefore lower the finite cost; the auxiliary dimension $d_{r,x} = h_{r,x} + 1$ enters through the occupation bounds of Section 5. They give no new scalar inequality: since $b+h-2j$ and $b-h+2j$ lie symmetrically about b, the concavity inequality (3.2) along the row gives $P(r, b+h-2j) + P(r, b-h+2j) \leq 2P(r,b)$ for every profile P, and summing over $0 \leq j \leq h$ gives $(h+1)P(r,b) \geq \sum_{j=0}^{h} P(r, b+h-2j)$.

4.2 Tripling with only the middle count fixed

The coordinates of $C(r, 4r-1)$ fall into three blocks, two copies of $C(r,r)$ and one of $C(r,r)^{(23)}$. As in Section 4.1, a restriction of a tensor power removes the terms that join the middle block to an outer one; it suffices to fix the number of middle labels in a word, which leaves at least $3^n/(n+1)$ sectors in the n-th tensor power.

Write the terms of $C(r, 4r-1)$ as $x_\alpha y_\beta z_\gamma$ with $0 \leq \alpha < r$, $0 \leq \beta < 4r-1$ and $\gamma = \alpha + \beta$. Label each second-leg index β and each third-leg index γ *left*, *middle* or *right* according to the following intervals, with endpoints included:

	left	middle	right
β	$[0, r-1]$	$[r, 3r-2]$	$[3r-1, 4r-2]$
γ	$[0, 2r-2]$	$[2r-1, 3r-2]$	$[3r-1, 5r-3]$

.

Figure 1 shows the case $r = 2$. A left label on the second leg forces a left label on the third, since $\gamma \leq 2r-2$ when $\beta \leq r-1$. A right label on the second leg forces a right label on the third, since $\gamma \geq \beta$. A middle label on the third leg forces a middle label on the second, since $2r-1 \leq \gamma \leq 3r-2$ and $0 \leq \alpha \leq r-1$ give $r \leq \beta = \gamma - \alpha \leq 3r-2$. Hence every term whose two labels differ has a middle label on the second leg and an outer label on the third. In the n-th tensor power, the word of a second- or third-leg coordinate is its sequence of n labels.

Lemma 4.5 (Type restriction for a tripling). *Let $r \geq 2$, $n \geq 1$ and $0 \leq m \leq n$. Keeping on the second and third legs of $C(r, 4r-1)^{\otimes n}$ only the coordinates whose words have exactly m middle labels is a restriction to a tensor with $M_{\mathrm{t}}(n,m) = \binom{n}{m} 2^{n-m}$ sectors sharing the first leg, one for each such word. Each sector is a tensor product of $n-m$ factors isomorphic to $C(r,r)$ and m factors isomorphic to $C(r,r)^{(23)}$. The choice $m = \lfloor (n+1)/3 \rfloor$ maximizes $M_{\mathrm{t}}(n,m)$ and gives*

$$M_{\mathrm{t}}(n,m) \geq \frac{3^n}{n+1}. \tag{4.3}$$

No divisibility condition on n is required.

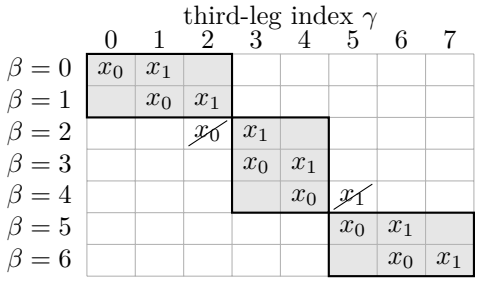


Figure 1: The terms of $C(2,7)$ (Lemma 4.5 with $r=2$): the cell (β,γ) contains x_α when $\gamma=\alpha+\beta$. Labels: left for $\beta\le 1$, $\gamma\le 2$; middle for $2\le\beta\le 4$, $3\le\gamma\le 4$; right for $\beta\ge 5$, $\gamma\ge 5$. The shaded blocks are $C(2,2)$, $C(2,2)^{(23)}$ after reversal of the first index, and a translate of $C(2,2)$. The two crossed terms have a middle second-leg label and an outer third-leg label; the restriction removes them.

Proof. In each position of a surviving term, a middle label on the third leg forces a middle label on the second leg. Hence the set of middle positions on the third leg is contained in that on the second leg; both sets have m elements, so they coincide. At the other $n-m$ positions both labels are outer, and the second-leg label determines the third-leg label. Every surviving term therefore carries the same word on both legs, and the sectors are indexed by the words with m middle labels: there are $\binom{n}{m}$ choices of the middle positions and 2^{n-m} choices of the outer labels.

A sector is the tensor product, over the n positions, of the blocks selected by the labels. The left block is $C(r,r)$ itself, and the right block becomes $C(r,r)$ after subtracting $3r-1$ from β and γ. In the middle block put $\alpha'=r-1-\alpha$ and $\gamma'=\gamma-(2r-1)$; its terms are then $x_{r-1-\alpha'}\,y_{r+\alpha'+\gamma'}\,z_{2r-1+\gamma'}$ with $0\le\alpha',\gamma'<r$. Reversing the first index and shifting the second and third indices by r and $2r-1$ turns them into $\sum_{\alpha',\gamma'} x_{\alpha'}y_{\alpha'+\gamma'}z_{\gamma'}$, which is $C(r,r)^{(23)}$. These identifications are coordinate permutations within each leg. On the shared first leg, the reversal at the middle positions depends on the word, so, like the reordering of positions, it is applied only after separation.

Finally, $M_{\mathrm{t}}(n,m+1)/M_{\mathrm{t}}(n,m)=(n-m)/(2(m+1))$, which is at least one exactly when $m\le(n-2)/3$. Hence the counts do not decrease up to $m=\lfloor(n+1)/3\rfloor$ and strictly decrease afterwards, so this m is a maximizer (tied with $m-1$ when $n\equiv 2 \pmod 3$). The $n+1$ counts sum to $(1+2)^n=3^n$, which proves (4.3). □

The interval partition and the identification of the middle block are those of OpenAI [22, Lemma 4.2] with both of its parameters equal to r. There the terms between blocks are removed by a degeneration that scales coordinates by powers of t; here the type restriction removes them.

4.3 The separator

The separator makes the M sectors of Lemmas 4.3 and 4.5 independent on all three legs, so that each sector can be reindexed on its own. The idea is to average over copies with Fourier phases. On the shared leg each variable receives a tentative sector label g', and on the other two legs the variables receive auxiliary indices e and e'. The phases are chosen so that averaging over the $2M-1$ copies keeps exactly the terms with $e-e'+g'-g=0$, where g is the true sector label. Weights in the parameter t then multiply each such term by $t^{(g'-g)^2}$. At $t=0$ only $g'=g$ survives, which forces $e=e'$; the free index $e=e'$ is the dot product.

Proposition 4.6 (Separator). *Let $T = \sum_{g=0}^{M-1} T_g$ with $T_g \in X \otimes Y_g \otimes Z_g$, where $Y = \bigoplus_g Y_g$ and $Z = \bigoplus_g Z_g$. The substitutions in the proof give a degeneration*

$$(2M-1)\,T \trianglerighteq \bigoplus_{g=0}^{M-1} T_g\, B_X(M), \tag{4.4}$$

in which each summand on the right has its own copy of X. Its polynomial has degree at most $(M-1)^2$, and its exponents lie in $\{\delta^2 : 0 \le \delta \le M-1\}$. Permuting the three legs gives the same statement with any chosen shared leg.

Proof. Choose bases and write $T_g = \sum_{\alpha,\beta,\gamma} T_g[\alpha,\beta,\gamma]\, x_\alpha y_{g,\beta} z_{g,\gamma}$, where the x_α are the coordinates of X and the $y_{g,\beta}$ and $z_{g,\gamma}$ those of Y_g and Z_g; the ranges of β and γ may depend on g. Put $\zeta = \zeta_{2M-1}$, and index the source copies by $\ell = 0, \dots, 2M-2$. The target variables are $X_{\alpha,g'}$, $Y_{g,\beta,e}$ and $Z_{g,\gamma,e'}$, with $g', e, e' \in \{0, \dots, M-1\}$.

Step 1: Fourier averaging. In copy ℓ substitute

$$x_\alpha \longmapsto \sum_{g'=0}^{M-1} \zeta^{\ell g'} X_{\alpha,g'}, \qquad y_{g,\beta} \longmapsto \sum_{e=0}^{M-1} \zeta^{\ell(e-g)} Y_{g,\beta,e},$$
$$z_{g,\gamma} \longmapsto (2M-1)^{-1} \sum_{e'=0}^{M-1} \zeta^{-\ell e'} Z_{g,\gamma,e'}.$$

Each phase uses only indices visible on its own leg, since the second- and third-leg coordinates carry their sector label g. All copies map into the same target variables, so the substitutions are linear on each leg of $(2M-1)T$. Summing over the copies, the term $x_\alpha y_{g,\beta} z_{g,\gamma}$ becomes

$$\sum_{g',e,e'} \left(\frac{1}{2M-1} \sum_{\ell=0}^{2M-2} \zeta^{\ell(e-e'+g'-g)} \right) X_{\alpha,g'} Y_{g,\beta,e} Z_{g,\gamma,e'}.$$

The bracket is one if $e-e'+g'-g \equiv 0 \pmod{2M-1}$ and zero otherwise. Since $|e-e'+g'-g| \le 2M-2$, the congruence is the equality

$$e - e' + g' - g = 0. \tag{4.5}$$

Step 2: weights. Multiply $X_{\alpha,g'}$ by $t^{g'^2}$, $Y_{g,\beta,e}$ by t^{2ge-g^2}, and $Z_{g,\gamma,e'}$ by $t^{-2ge'}$. On a term satisfying (4.5) we have $e - e' = g - g'$, so the total exponent is $g'^2 + 2g(e-e') - g^2 = g'^2 + 2g(g-g') - g^2 = (g'-g)^2$. The weights complete the square: the cross term $-2gg'$ mixes the shared-leg label g' with the sector label g, and no single leg sees both, but on (4.5) it equals $2g(e-e') - 2g^2$, which splits between the second and third legs. Individual weights may be negative, but every surviving exponent is $(g'-g)^2$, so the transformed tensor is a polynomial in t of degree at most $(M-1)^2$ with exponents in $\{\delta^2 : 0 \le \delta \le M-1\}$.

Step 3: the constant term. The terms of exponent zero have $g' = g$, and then (4.5) gives $e = e'$. The constant term is thus

$$\sum_{g=0}^{M-1} \sum_{\alpha,\beta,\gamma} T_g[\alpha,\beta,\gamma] \sum_{e=0}^{M-1} X_{\alpha,g} Y_{g,\beta,e} Z_{g,\gamma,e}.$$

For each g this is $T_g B_X(M)$, with first-leg variables $X_{\alpha,g}$, second-leg variables $Y_{g,\beta,e}$ and third-leg variables $Z_{g,\gamma,e}$. Different g use disjoint variables on all three legs, so the constant term is the direct sum (4.4). □

The output of the separator has two multiplicities: the label g counts M direct summands, for which the source pays $2M-1$ copies of T, and the index $e=e'$ is the dot product $B_X(M)$ inside each summand, whose first leg is one-dimensional. This dot product carries the gain of a relation: in the three orientations of Section 4.4 it becomes the auxiliary tensor M_M, whose side M is the number of sectors, namely the multinomial coefficient $n!/\prod_j n_j!$ for a split and at least $3^n/(n+1)$ for a tripling with the middle count of Lemma 4.5. For a split, this gain is set against the auxiliary $\mathsf{M}_{(h+1)^n}$ that the split consumes. The copies cost only a constant: the three orientations turn $(2M-1)^3$ source copies into M^3 output summands (Corollary 4.8), a ratio below 8 per relation that does not grow with n.

Proposition 4.6 is a variant of OpenAI [22, Proposition 3.1], which uses $5M$ source copies; with the phases above, the quantity in the Fourier condition has absolute value at most $2M-2$, so $2M-1$ copies suffice. Proposition C.1 in Appendix C allows a dot product of another dimension and a general phase slope; the main construction uses only Proposition 4.6.

Remark 4.7 (A polynomial separator). Independently of this work, Lü [18, Proposition 2.1] constructed a separator by polynomial evaluation and interpolation instead of Fourier phases. For M sectors and a dot product of dimension K_0, it uses $M+K_0-1$ source copies, as Proposition C.1 does with $c=1$; its polynomial has degree at most $\min(M-1,K_0-1)$, as the end of Lü's proof shows, instead of $\min(M-1,K_0-1)^2$; and its maps are defined over every field with at least $M+K_0-1$ elements. For $K_0=M$ it gives (4.4) over $\mathbb{Q}$, with the target variables and the constant term of Proposition 4.6. Lemma 9.1(b) writes it out in our coordinates and proves the degree bound, and the rational variant of Section 9.1 uses it. Filmus [12, Lemma 2.5] gives another interpolation separator, with $6M-5$ source copies, in every characteristic. We keep Proposition 4.6 in the main construction because Appendix B writes out its composed maps, Appendix C generalizes it using its exponents δ^2, and Section 10 evaluates its degree bound; the smaller degree would lower only the interpolation overhead (Section 11.1).

4.4 Three orientations

We apply each relation in the three orientations $\pi\in\mathcal{A}_\sigma$ of one parity class and tensor the results, as in the cyclic symmetrization of Christandl et al. [6, Section 3.2], with the class alternating between rows. In orientation π we apply Lemma 4.3 or 4.5 to the permuted tensor and the separator with shared leg $\pi(1)$. We call the legs of a tensor product of oriented factors its *global legs*; leg l of a factor T^π lies in global leg $\pi(l)$. The orientations serve two purposes. First, they put the single-variable leg of a dot product $B_X(d)$ on each global leg once, so that, up to the order of the factors within each global leg, the three oriented copies combine into M_d by (2.2); this applies both to the auxiliary dot product of a split and to the dot product produced by the separator. Second, the symmetrized tensors satisfy $S^\sigma_{r,b}\cong S^{-\sigma}_{b,r}$ by (2.6), which links adjacent rows (see the end of this section).

After separation, each sector receives its own coordinate permutation, which identifies it with a fixed *canonical sector tensor* $\mathcal{V}$, the same for all sectors of the relation. For a split, the positions are sorted by increasing filtration index, and $\mathcal{V}=\prod_j C(r,b+h-2j)^{\otimes n_j}$. For a tripling, the $n-m$ outer positions come first and the m middle positions follow, each group in its original order, and $\mathcal{V}=C(r,r)^{\otimes(n-m)}(C(r,r)^{(23)})^{\otimes m}$; since the left and right blocks have the same dimensions, $\mathcal{V}$ does not depend on the outer labels. Independent choices of sectors in the three orientations therefore all give the same tensor. For a tripling, the convolution factors of the three oriented copies of $\mathcal{V}$ regroup as

$$\bigotimes_{\pi\in\mathcal{A}_\sigma}\Big(C(r,r)^{\otimes(n-m)}\big(C(r,r)^{(23)}\big)^{\otimes m}\Big)^{\pi}\cong(S^\sigma_{r,r})^{\otimes(n-m)}(S^{-\sigma}_{r,r})^{\otimes m}\cong(S^\sigma_{r,r})^{\otimes n}. \tag{4.6}$$

The first isomorphism reorders tensor factors within each global leg and uses $(C(r,r)^{(23)})^{\pi} = C(r,r)^{\pi(23)}$, where $\pi(23)$ runs over $\mathcal{A}_{-\sigma}$ as π runs over $\mathcal{A}_{\sigma}$. The second applies the diagonal isomorphism $S^{+}_{r,r} \cong S^{-}_{r,r}$ of (2.6) to the last m factors. Both are coordinate identifications within each global leg.

Corollary 4.8 (Oriented relations)**.** *Let $\sigma \in \{+,-\}$.*

(a) Split. *Let $r \geq 1$, $h \geq 1$ and $b \geq h+1$, let $(n_0,\dots,n_h)$ be a type with positive sum n, and put $M = n!/\prod_j n_j!$. Then*

$$(2M-1)^3\,(S^{\sigma}_{r,b})^{\otimes n}\,\mathsf{M}_{(h+1)^n} \trianglerighteq M^3\Big(\prod_{j=0}^{h}(S^{\sigma}_{r,b+h-2j})^{\otimes n_j}\Big)\mathsf{M}_M. \tag{4.7}$$

(b) Tripling. *Let $r \geq 2$, $n \geq 1$ and $0 \leq m \leq n$, and put $M = M_{\mathrm{t}}(n,m)$. Then*

$$(2M-1)^3\,(S^{\sigma}_{r,4r-1})^{\otimes n} \trianglerighteq M^3\,(S^{\sigma}_{r,r})^{\otimes n}\,\mathsf{M}_M. \tag{4.8}$$

Both degenerations are composed of the maps of this section, and their polynomials have degree at most $3(M-1)^2$.

Proof. Let $\mathcal{U}$ be $(C(r,b)B_X(h+1))^{\otimes n}$ or $C(r,4r-1)^{\otimes n}$, and let $\mathcal{V}$ be the canonical sector tensor. In each orientation $\pi \in \mathcal{A}_{\sigma}$, the type restriction of Lemma 4.3 or 4.5, the separator of Proposition 4.6 with shared leg $\pi(1)$, and the sector-specific coordinate permutations give $(2M-1)\,\mathcal{U}^{\pi} \trianglerighteq M\,(\mathcal{V}B_X(M))^{\pi}$. The coordinate permutations act after separation and do not involve t. On the first leg they reindex the copy with tentative label g' according to the word of sector g', so they are defined on all target variables of the separator, including those of its terms of positive degree (see (B.6) in Appendix B), as Lemma 2.1 requires. Tensoring the three orientations with one common parameter t gives, by Lemma 2.1, a degeneration

$$(2M-1)^3 \bigotimes_{\pi\in\mathcal{A}_{\sigma}} \mathcal{U}^{\pi} \trianglerighteq M^3 \bigotimes_{\pi\in\mathcal{A}_{\sigma}} (\mathcal{V}B_X(M))^{\pi}$$

of degree at most $3(M-1)^2$. It remains to identify both sides by coordinate permutations within each global leg; such fixed maps change neither the degree nor the constant term. For a split, $B_X(h+1)^{\otimes n} \cong B_X((h+1)^n)$, so the source is $(S^{\sigma}_{r,b})^{\otimes n}\mathsf{M}_{(h+1)^n}$ and each summand of the target is $\prod_j (S^{\sigma}_{r,b+h-2j})^{\otimes n_j}\mathsf{M}_M$. For a tripling, the source is $(S^{\sigma}_{r,4r-1})^{\otimes n}$, and (4.6) identifies each summand of the target with $(S^{\sigma}_{r,r})^{\otimes n}\mathsf{M}_M$. □

Alternating signs between rows. Row r of the construction uses the class $\mathcal{A}_{\sigma_r}$, where $\sigma_r = (-1)^{a-r}$ is the row sign of Section 5, so adjacent rows use opposite classes. The reason is the boundary between rows: the walk of row r ends at column $r-1$, the factor $S^{\sigma_r}_{r,r-1}$, and the walk of row $r-1$ starts at column r, the factor $S^{\sigma_{r-1}}_{r-1,r}$. By (2.6) the first is isomorphic to $S^{-\sigma_r}_{r-1,r}$, which is the second exactly when $\sigma_{r-1} = -\sigma_r$. With one sign σ for all rows, the boundary factors would be $S^{\sigma}_{r,r-1}$ and $S^{\sigma}_{r-1,r}$; for $r \geq 3$ the identities (2.6), the only ones the ledger of Proposition 6.2 uses, do not relate them, and the ledger would not close. Alternation makes an isomorphism between $S^{+}_{r,b}$ and $S^{-}_{r,b}$ for $r \neq b$ unnecessary, and the construction uses none. Only the alternation matters: the choice $\sigma_a = +$ is only a normalization, since $S^{+}_{a,a} \cong S^{-}_{a,a}$. Applying every relation in all six orientations, with the products $Q_{r,b} = S^{+}_{r,b}S^{-}_{r,b} \cong Q_{b,r}$, would need no signs but squares all dimensions; Section A.4 compares this sixfold construction with the threefold one used here.

5 The row walk and its occupations

This section decides how often to apply each split and each tripling of Section 4. Apart from the copies of $S^{+}_{a,a}$ that are fed in and the copies of M_2 that come out, every tensor factor that one relation produces must be consumed by another relation, and conversely; in addition, the logarithms of the auxiliary dimensions, weighted by how often each split is used, must have a small sum, because the auxiliary dimensions multiply into the source side of the final scheme. We make these choices with a random walk in each row, the *row walk*, whose expected numbers of visits, the *occupations*, balance every factor exactly. Proposition 5.3 proves this and bounds both the total occupation and the weighted sum of logarithms by $\Gamma_a = 105 + 180(a-2)$.

5.1 States, splits and triplings

Fix the target index $a \geq 2$ and a row r with $2 \leq r \leq a$, and put $\sigma_r = (-1)^{a-r}$. Row r consists of the tensors $S^{\sigma_r}_{r,b}$ with columns $r-1 \leq b \leq 4r-1$. We index them by the *state* $x = b-r+1 \in \{0,\dots,3r\}$, so that state x is column $b = r-1+x$. With $f_r(x) = \frac{3(r-1)}{2} + x$, the row-affine form (3.5) of the least profile reads, in the state variable,

$$F(r, r-1+x) = s_r f_r(x) \qquad (0 \leq x \leq 3r). \tag{5.1}$$

The states $1 \leq x \leq 3r-1$ are *interior*. For an interior state x we define its *width*, its auxiliary dimension, its *children* and its transition probabilities by

$$\begin{aligned} h_{r,x} &= \min(x, 3r-x), & d_{r,x} &= h_{r,x}+1, \\ \mathcal{Y}_{r,x} &= \{x-h_{r,x}, x-h_{r,x}+2, \dots, x+h_{r,x}\}, & q_{r,x,y} &= \frac{f_r(y)}{d_{r,x} f_r(x)} \quad (y \in \mathcal{Y}_{r,x}). \end{aligned} \tag{5.2}$$

The $d_{r,x}$ children are placed symmetrically about x, so their average is x. Since f_r is affine, it follows that $\sum_{y\in\mathcal{Y}_{r,x}} f_r(y) = d_{r,x} f_r(x)$; hence the $q_{r,x,y}$ are positive rationals with sum one. By (5.1), a child is chosen with probability proportional to the profile value $F(r, r-1+y)$ at its column; in other words, $(h+1)F(r,b) = \sum_{j=0}^{h} F(r, b+h-2j)$ for $b = r-1+x$ and $h = h_{r,x}$. This makes $\log F$ a potential that telescopes along the walk (Proposition 6.3).

Since $h_{r,x} = \min(x, 3r-x)$ is the largest h with $0 \leq x-h$ and $x+h \leq 3r$, every child satisfies $0 \leq y \leq 3r$, so its column $r-1+y$ lies in $[r-1, 4r-1]$, and no wider split keeps all its children in the row. This maximal width makes the walk cross a row quickly, which keeps the cost of a row bounded (Proposition 5.3). Since $r \geq 2$ and $h_{r,x} \leq x$, the column of x satisfies $b = r-1+x \geq h_{r,x}+1$, which is the hypothesis of Lemma 4.3.

The walk keeps track of tensor factors. State x of row r stands for the factor $S^{\sigma_r}_{r,r-1+x}$. A step from an interior state x stands for one *split* at (r,x): the restriction of Lemma 4.3 with $b = r-1+x$, $h = h_{r,x}$ and auxiliary $B_X(d_{r,x})$, used in the three orientations of $\mathcal{A}_{\sigma_r}$ as in Corollary 4.8, so that the three auxiliaries combine into $\mathsf{M}_{d_{r,x}}$ by (2.2). Its block with filtration index $j = (x + h_{r,x} - y)/2$ is $S^{\sigma_r}_{r,r-1+y}$, the factor of the child y. In a tensor power, each copy of the factor at x becomes one factor at some child, and the type of the restriction fixes how many copies go to each child; the walk moves from x to y with the frequency $q_{r,x,y}$ that this type approximates (Section 6).

The two end states have special roles. State $3r$ is column $4r-1$, and the step from it to state 1, column r, stands for one *tripling*: the restriction of Lemma 4.5 in the three orientations of $\mathcal{A}_{\sigma_r}$, which turns copies of $S^{\sigma_r}_{r,4r-1}$ into sectors that each contain the same number of copies of $S^{\sigma_r}_{r,r}$, by (4.6). State 0 is column $r-1$. By (2.6) and $\sigma_{r-1} = -\sigma_r$, its factor $S^{\sigma_r}_{r,r-1} \cong S^{\sigma_{r-1}}_{r-1,r}$ is the factor at state 2 of row $r-1$ (Section 4.4). A walk that reaches state 0 therefore leaves row r and continues

in row $r-1$ at state 2. Hence the walk of a lower row starts at state 2, whereas the walk of the top row starts at state 1, the diagonal factor $S^{\sigma_a}_{a,a} = S^{+}_{a,a}$ whose powers are the input of the construction; accordingly, the *injection state* of row r is $x^{\mathrm{in}}_r = 1$ if $r = a$ and $x^{\mathrm{in}}_r = 2$ if $r < a$. In row 2, state 0 is $S^{\sigma_2}_{2,1} \cong S^{-\sigma_2}_{1,2} \cong \mathsf{M}_2$, the output of the construction.

Definition 5.1 (Row walk and occupations)**.** The *row walk* of row r is the Markov chain $X_0, X_1, X_2, \dots$ on the states $\{0, \dots, 3r\}$ with $X_0 = x^{\mathrm{in}}_r$ and the following transitions. From an interior state x, it moves to the child $y \in \mathcal{Y}_{r,x}$ with probability $q_{r,x,y}$; from state $3r$, it moves to state 1; and state 0 is absorbing. The *occupation* $\lambda_{r,x}$ of an interior state x is the expected number of times $\nu \geq 0$ with $X_\nu = x$, that is, the expected number of splits at (r, x). The expected number of times ν with $X_\nu = 3r$ is denoted by μ_r; it is the expected number of triplings in row r.

A priori these expectations lie in $[0, \infty]$; Proposition 5.3 shows that they are finite and rational. Figure 2 and Example 5.2 show the row walk of row 2.

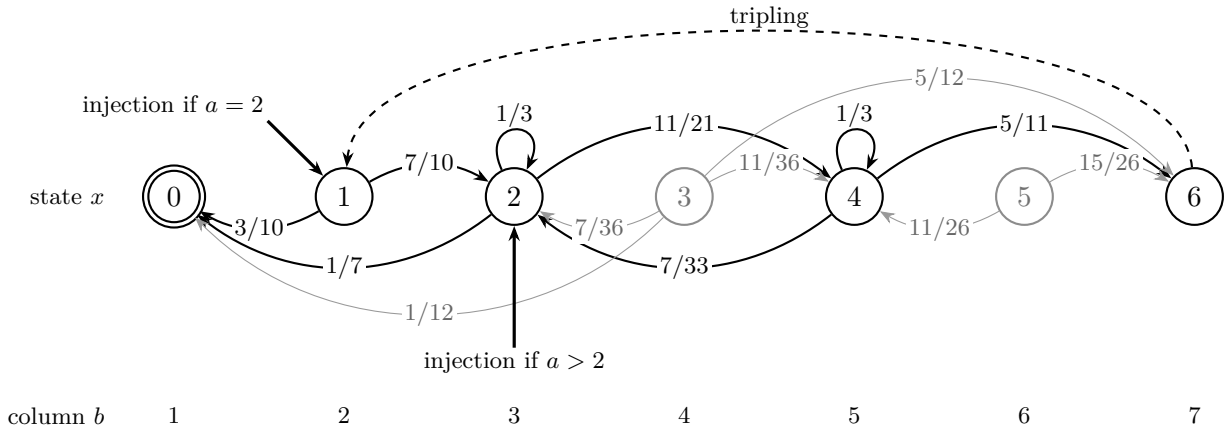


Figure 2: The row walk of row $r = 2$ (Example 5.2). Arcs above the states move right, arcs below move left, and loops stay; labels are the probabilities $q_{2,x,y}$. Dashed: the tripling $6 \to 1$. Double circle: the absorbing state 0, the factor $S^{\sigma_2}_{2,1} \cong \mathsf{M}_2$. Gray: states not reachable from the injection states.

Example 5.2 (Row $r = 2$)**.** Here $s_2 = 4/3$ and $f_2(x) = x + 3/2$, so $F(2, b) = \frac{4}{3}(b + \frac{1}{2})$ for $1 \leq b \leq 7$, and the states $0, \dots, 6$ are the columns $1, \dots, 7$. The widths of the interior states $1, \dots, 5$ are $1, 2, 3, 2, 1$. For instance, the children of state 2 are $0, 2, 4$; their values of f_2 are $3/2, 7/2, 11/2$, whose sum is $3f_2(2) = 21/2$, so $q_{2,2,0} = 1/7$, $q_{2,2,2} = 1/3$ and $q_{2,2,4} = 11/21$. Figure 2 shows all transition probabilities. Every child in this row is even: a state $x \leq 3$ has the children $0, 2, \dots, 2x$, and a state $x \geq 3$ has the children $2x - 6, \dots, 6$. Consequently a walk that starts at state 1 or 2 never visits states 3 and 5, and $\lambda_{2,3} = \lambda_{2,5} = 0$. For a state other than 0, the expected number of visits equals the number of starts there plus the expected number of arrivals. For the states $1, 2, 4, 6$ this gives

$$\lambda_{2,1} = \delta_{1,x^{\mathrm{in}}_2} + \mu_2, \qquad \lambda_{2,2} = \delta_{2,x^{\mathrm{in}}_2} + \tfrac{7}{10}\lambda_{2,1} + \tfrac{1}{3}\lambda_{2,2} + \tfrac{7}{33}\lambda_{2,4},$$
$$\lambda_{2,4} = \tfrac{11}{21}\lambda_{2,2} + \tfrac{1}{3}\lambda_{2,4}, \qquad \mu_2 = \tfrac{5}{11}\lambda_{2,4},$$

where δ is the Kronecker delta. These are the coordinates $1, 2, 4, 6$ of the row identity (5.3) below.

If $a = 2$, then $x^{\mathrm{in}}_2 = 1$, and the solution is $\lambda_{2,1} = 2$, $\lambda_{2,2} = 14/5$, $\lambda_{2,4} = 11/5$ and $\mu_2 = 1$: the walk makes 7 splits on average, and $\sum_x \lambda_{2,x} \log d_{2,x} = 2 \log 2 + 5 \log 3 \approx 6.88$. If $a > 2$, then $x^{\mathrm{in}}_2 = 2$, and the solution is $\lambda_{2,1} = 10/7$, $\lambda_{2,2} = 4$, $\lambda_{2,4} = 22/7$ and $\mu_2 = 10/7$: the walk makes $60/7 \approx 8.57$ splits on average, and $\sum_x \lambda_{2,x} \log d_{2,x} = \frac{10}{7} \log 2 + \frac{50}{7} \log 3 \approx 8.84$. In both cases μ_2 agrees with (5.4).

5.2 Occupations

Let $\mathbf{e}_0, \dots, \mathbf{e}_{3r}$ be the standard basis of $\mathbb{Q}^{3r+1}$, indexed by the states of row r.

Proposition 5.3 (Occupations)**.** *Let $a \geq 2$ and $2 \leq r \leq a$. The row walk of row r, started at any state, is absorbed at state 0 in finite expected time. Its occupations $\lambda_{r,x}$, for $1 \leq x \leq 3r-1$, and μ_r are nonnegative rationals. They are the unique solution of a nonsingular rational linear system, and they satisfy the row identity*

$$\sum_{x=1}^{3r-1} \lambda_{r,x}\Big(\mathbf{e}_x - \sum_{y \in \mathcal{Y}_{r,x}} q_{r,x,y}\mathbf{e}_y\Big) + \mu_r(\mathbf{e}_{3r} - \mathbf{e}_1) = \mathbf{e}_{x_r^{\mathrm{in}}} - \mathbf{e}_0. \tag{5.3}$$

The expected numbers of triplings are

$$\mu_a = \frac{1}{a-1}, \qquad \mu_r = \frac{2(3r-1)}{(r-1)(3r+1)} \quad (r < a); \tag{5.4}$$

in particular, $\mu_r \leq 2$. Each of the two row sums $\sum_x \lambda_{r,x}$ and $\sum_x \lambda_{r,x} \log d_{r,x}$ is less than 105 if $r = a$ and less than 180 if $r < a$. Consequently, each $\lambda_{r,x}$ and each $\lambda_{r,x} \log d_{r,x}$ is less than 180, and, with $\Gamma_a = 105 + 180(a-2)$,

$$\sum_{r=2}^{a} \sum_{x=1}^{3r-1} \lambda_{r,x} \leq \Gamma_a, \qquad \sum_{r=2}^{a} \sum_{x=1}^{3r-1} \lambda_{r,x} \log d_{r,x} \leq \Gamma_a. \tag{5.5}$$

On the left side of (5.3), the coefficient of $\mathbf{e}_y$ is the expected number of steps out of y minus the expected number of steps into y; by the right side, it is 1 at x_r^{in}, -1 at state 0 and 0 elsewhere, so one unit enters the row at x_r^{in} and leaves it at state 0. In terms of tensor factors, the splits and triplings of row r, used with these frequencies, consume one more copy of $S^{\sigma_r}_{r,r-1+x_r^{\mathrm{in}}}$ than they produce, produce one more copy of $S^{\sigma_r}_{r,r-1}$ than they consume, and balance every other factor of the row.

Proof. The idea is to compare each excursion of the row walk between visits to the end states 0 and $3r$ with the *uniform walk*, which picks every child with equal probability. The uniform walk is a martingale, and the function g below decreases in expectation by at least $\gamma\sqrt{h_{r,x}}$ at each step from x; this bounds the expected number of splits in an excursion and the expected sum of $\log d_{r,x}$ over them. Between visits to the end states, the row walk is the Doob transform of the uniform walk by the affine function f_r [17, Section 17.6.1]; this changes these expectations by at most the factor $f_r(3r)/f_r(1) = 3$. Finally, triplings chain the excursions together, and their number is geometrically distributed.

Notation. Fix the row and drop the index r from f_r, $h_{r,x}$, $d_{r,x}$, $\mathcal{Y}_{r,x}$, $q_{r,x,y}$, $\lambda_{r,x}$, μ_r and x_r^{in}. Let $\theta_{\mathrm{ex}} = \min\{\nu \geq 0 : X_\nu \in \{0, 3r\}\}$ be the exit time from the interior ($\theta_{\mathrm{ex}} = \infty$ if there is no such ν). We write $\mathbb{P}^q_x$ and $\mathbb{E}^q_x$ for the row walk started at x, and $\mathbb{P}^{\mathrm{u}}_x$ and $\mathbb{E}^{\mathrm{u}}_x$ for the uniform walk started at x, which moves from an interior state x to each child $y \in \mathcal{Y}_x$ with probability $1/d_x$ and stops at 0 and at $3r$. An *excursion* of the row walk runs from an *entry* state (the starting state, or state 1 right after a tripling) up to the next visit to 0 or $3r$. A *charge* is a function $\varphi \geq 0$ on the interior states, and the charge of a path is $\sum_{\nu < \theta_{\mathrm{ex}}} \varphi(X_\nu)$. We use $\varphi = 1$, which counts splits, and $\varphi(x) = \log d_x$. Both satisfy $\varphi(x) \leq \sqrt{h_x}$ at interior states, the first because $h_x \geq 1$, and the second because $\log(1+h) \leq \sqrt{h}$ for $h \geq 1$: the difference $\sqrt{h} - \log(1+h)$ is positive at $h = 1$, and its derivative $(\sqrt{h}-1)^2/(2\sqrt{h}\,(1+h))$ is nonnegative.

Step 1: a square-root drift. Put $g(x) = \sqrt{x} + \sqrt{3r-x} - \sqrt{3r}$ and $\gamma = 1 - 2\sqrt{2}/3 > 0$. Since the square root is subadditive, $g \geq 0$ on $[0, 3r]$. We claim that

$$g(x) - \mathbb{E}^{\mathrm{u}}_x\, g(X_1) \geq \gamma\sqrt{h_x} \qquad (1 \leq x \leq 3r-1). \tag{5.6}$$

First let $x \leq 3r/2$. Then $h_x = x$, and the children are $0, 2, \ldots, 2x$, each with probability $1/(x+1)$. On each interval $[\ell-1, \ell]$, the concave square root lies above its chord, so

$$\sum_{\ell=0}^{x} \sqrt{\ell} - \frac{\sqrt{x}}{2} = \sum_{\ell=1}^{x} \frac{\sqrt{\ell-1} + \sqrt{\ell}}{2} \leq \int_0^x \sqrt{\xi}\, d\xi = \frac{2}{3}x^{3/2}.$$

Dividing by $x+1$ and using $\frac{2}{3}x + \frac{1}{2} \leq \frac{2}{3}(x+1)$, we obtain

$$\mathbb{E}^{\mathrm{u}}_x \sqrt{X_1} = \frac{\sqrt{2}}{x+1} \sum_{\ell=0}^{x} \sqrt{\ell} \leq \sqrt{2}\,\sqrt{x}\, \frac{\frac{2}{3}x + \frac{1}{2}}{x+1} \leq \frac{2\sqrt{2}}{3}\sqrt{x}.$$

The children average to x, so Jensen's inequality for the concave function $\xi \mapsto \sqrt{3r-\xi}$ gives $\mathbb{E}^{\mathrm{u}}_x \sqrt{3r - X_1} \leq \sqrt{3r-x}$. Together,

$$g(x) - \mathbb{E}^{\mathrm{u}}_x\, g(X_1) = \big(\sqrt{x} - \mathbb{E}^{\mathrm{u}}_x \sqrt{X_1}\big) + \big(\sqrt{3r-x} - \mathbb{E}^{\mathrm{u}}_x \sqrt{3r - X_1}\big) \geq \Big(1 - \frac{2\sqrt{2}}{3}\Big)\sqrt{x} = \gamma\sqrt{h_x}.$$

For $x \geq 3r/2$, the reflection $\xi \mapsto 3r - \xi$ maps $\mathcal{Y}_x$ onto $\mathcal{Y}_{3r-x}$, preserves the width, and leaves g unchanged. Hence the first case, applied to the state $3r - x \leq 3r/2$, proves (5.6) at x.

Step 2: excursions of the uniform walk. We show that

$$\mathbb{E}^{\mathrm{u}}_x \sum_{\nu < \theta_{\mathrm{ex}}} \sqrt{h_{X_\nu}} \leq \frac{g(x)}{\gamma} \leq \frac{\sqrt{x}}{\gamma} \qquad (1 \leq x \leq 3r-1). \tag{5.7}$$

Fix an integer $\bar{\nu} \geq 1$. On the event $\{\nu < \theta_{\mathrm{ex}}\}$ the state X_ν is interior, so the Markov property and (5.6) give $\mathbb{E}^{\mathrm{u}}_x[g(X_\nu) - g(X_{\nu+1}) \mid X_0, \ldots, X_\nu] \geq \gamma\sqrt{h_{X_\nu}}$ there. The event $\{\nu < \theta_{\mathrm{ex}} \wedge \bar{\nu}\}$ is determined by $X_0, \ldots, X_\nu$. Multiplying by its indicator, taking expectations and summing over $\nu < \bar{\nu}$, we obtain

$$\gamma\, \mathbb{E}^{\mathrm{u}}_x \sum_{\nu < \theta_{\mathrm{ex}} \wedge \bar{\nu}} \sqrt{h_{X_\nu}} \leq \mathbb{E}^{\mathrm{u}}_x \sum_{\nu < \theta_{\mathrm{ex}} \wedge \bar{\nu}} (g(X_\nu) - g(X_{\nu+1})) = g(x) - \mathbb{E}^{\mathrm{u}}_x\, g(X_{\theta_{\mathrm{ex}} \wedge \bar{\nu}}) \leq g(x),$$

where the last step uses $g \geq 0$. Letting $\bar{\nu} \to \infty$, monotone convergence gives the first inequality in (5.7), and $\sqrt{3r-x} \leq \sqrt{3r}$ gives the second. This argument does not assume that θ_{ex} is finite; since $1 \leq \sqrt{h_x}$ and $\varphi(x) \leq \sqrt{h_x}$ at interior states, it gives $\mathbb{E}^{\mathrm{u}}_x \theta_{\mathrm{ex}} \leq \sqrt{x}/\gamma < \infty$ and $\mathbb{E}^{\mathrm{u}}_x \sum_{\nu < \theta_{\mathrm{ex}}} \varphi(X_\nu) \leq \sqrt{x}/\gamma$ for both charges.

Step 3: excursions of the row walk. The function f is positive, since $f(0) = 3(r-1)/2 > 0$, and harmonic for the uniform walk at every interior state, because the children average to x and f is affine. By (5.2), $q_{x,y} = d_x^{-1} f(y)/f(x)$ is the Doob transform of the uniform probabilities by f. The tripling step $3r \to 1$ is not of this form, since $f(1) \neq f(3r)$, so we transform only excursions: for every path $x_0, x_1, \ldots, x_\nu$ whose states $x_0, \ldots, x_{\nu-1}$ are interior, whether or not x_ν is interior, the ratio of the path probabilities telescopes:

$$\mathbb{P}^{q}_{x_0}(X_1 = x_1, \ldots, X_\nu = x_\nu) = \prod_{\ell=1}^{\nu} \frac{f(x_\ell)}{d_{x_{\ell-1}} f(x_{\ell-1})} = \frac{f(x_\nu)}{f(x_0)}\, \mathbb{P}^{\mathrm{u}}_{x_0}(X_1 = x_1, \ldots, X_\nu = x_\nu).$$

Since f is increasing, every entry state $x_0 \geq 1$ and every state $x_\nu \leq 3r$ satisfy $f(x_\nu)/f(x_0) \leq f(3r)/f(1) = 3$, because $f(1) = (3r-1)/2$ and $f(3r) = 3(3r-1)/2$. For each ν, the event $\{\nu < \theta_{\mathrm{ex}}\}$ consists of the paths whose states $X_0, \dots, X_\nu$ are all interior. Summing the path identity over these paths with $x_0 = x$ and applying the last bound gives $\mathbb{E}_x^q[\varphi(X_\nu)\mathbf{1}_{\{\nu<\theta_{\mathrm{ex}}\}}] \leq 3\,\mathbb{E}_x^{\mathrm{u}}[\varphi(X_\nu)\mathbf{1}_{\{\nu<\theta_{\mathrm{ex}}\}}]$. We sum over ν and use Step 2. The expected charge $\psi_x = \mathbb{E}_x^q \sum_{\nu<\theta_{\mathrm{ex}}} \varphi(X_\nu)$ of an excursion of the row walk from the entry state x therefore satisfies

$$\psi_x \leq \frac{3\sqrt{x}}{\gamma} \qquad (1 \leq x \leq 3r-1) \tag{5.8}$$

for both charges. With $\varphi = 1$, this gives $\mathbb{E}_x^q \theta_{\mathrm{ex}} \leq 3\sqrt{x}/\gamma$, so every excursion of the row walk ends in finite expected time.

Step 4: exit probabilities. Under $\mathbb{P}_x^{\mathrm{u}}$, the stopped process $X_{\theta_{\mathrm{ex}}\wedge\nu}$ is a martingale with values in $[0, 3r]$, since the children of an interior state average to that state. By Step 2, $\theta_{\mathrm{ex}} < \infty$ almost surely, so bounded convergence gives $\mathbb{E}_x^{\mathrm{u}} X_{\theta_{\mathrm{ex}}} = x$. As $X_{\theta_{\mathrm{ex}}} \in \{0, 3r\}$, the uniform walk leaves the interior at $3r$ with probability $x/(3r)$. Summing the path identity of Step 3 over the paths that first leave the interior at $3r$, we find that the row walk started at x leaves the interior at $3r$ with probability

$$\rho_x = \mathbb{P}_x^q(X_{\theta_{\mathrm{ex}}} = 3r) = \frac{x f(3r)}{3r f(x)}, \qquad \rho_1 = \frac{1}{r}, \qquad \rho_2 = \frac{2(3r-1)}{r(3r+1)} < \frac{2}{r}, \tag{5.9}$$

where we used $f(2) = (3r+1)/2$ and the values of $f(1)$ and $f(3r)$ from Step 3.

Step 5: chaining the excursions. Start the row walk at an interior state x. Each excursion ends either at 0, where the walk is absorbed, or at $3r$, after which a tripling moves the walk to state 1 and a new excursion starts there. By the strong Markov property at the exit times, each new excursion is independent of the past and has the law of an excursion under $\mathbb{P}_1^q$; in particular, it ends at $3r$ with probability ρ_1. Hence the walk makes at least ℓ triplings with probability $\rho_x \rho_1^{\ell-1}$ for $\ell \geq 1$, so the expected number of triplings is $\rho_x/(1-\rho_1)$. The excursion that follows the ℓ-th tripling takes place exactly when at least ℓ triplings occur, and its expected charge is then ψ_1. Summing over ℓ, we find that the expected total charge of the row walk started at x is $\psi_x + \rho_x\psi_1/(1-\rho_1)$.

Now start at $x = x^{\mathrm{in}}$, so that the expected number of triplings is μ and the expected total charge is $\sum_{y=1}^{3r-1} \lambda_y \varphi(y)$. Since $1 - \rho_1 = (r-1)/r$, (5.9) gives $\mu = \rho_1/(1-\rho_1) = 1/(r-1)$ if $r = a$, and $\mu = \rho_2/(1-\rho_1) = 2(3r-1)/\big((r-1)(3r+1)\big)$ if $r < a$. This proves (5.4), and both values are at most $2/(r-1) \leq 2$. By (5.8), $\psi_1 \leq 3/\gamma$ and $\psi_2 \leq 3\sqrt{2}/\gamma$. Using $r \geq 2$, through $r/(r-1) \leq 2$ and $2/(r-1) \leq 2$, and $1/\gamma = 3/(3-2\sqrt{2}) = 9 + 6\sqrt{2}$, we find that the expected total charge is

$$\begin{cases} \psi_1 + \dfrac{\rho_1\psi_1}{1-\rho_1} = \dfrac{r}{r-1}\,\psi_1 \leq \dfrac{6}{\gamma} = 54 + 36\sqrt{2} < 105, & r = a, \\[2ex] \psi_2 + \dfrac{\rho_2\psi_1}{1-\rho_1} \leq \psi_2 + \dfrac{2}{r-1}\,\psi_1 \leq \dfrac{3\sqrt{2}+6}{\gamma} = 90 + 63\sqrt{2} < 180, & r < a. \end{cases}$$

Applied to the charges $\varphi = 1$ and $\varphi = \log d$, these estimates prove the bounds on the two row sums.

Step 6: absorption and the row identity. Every visit to a state other than 0 is followed by exactly one step. Hence the absorption time $\theta_{\mathrm{abs}} = \min\{\nu : X_\nu = 0\}$ of the row walk started at an interior state x is the number of visits to interior states plus the number of triplings, and by Step 5 with $\varphi = 1$ its expectation is $\psi_x + (1+\psi_1)\rho_x/(1-\rho_1) < \infty$. From state $3r$ the walk first moves to state 1, so absorption has finite expected time from every state. Now start the walk at x^{in}. The telescoping sum $\sum_{\nu<\theta_{\mathrm{abs}}} (\mathbf{e}_{X_\nu} - \mathbf{e}_{X_{\nu+1}}) = \mathbf{e}_{x^{\mathrm{in}}} - \mathbf{e}_0$ has terms of ℓ^1-norm at most 2, so the norms of its terms add up to at most $2\theta_{\mathrm{abs}}$, which has finite expectation. Hence we may take expectations term by term:

$\mathbf{e}_{x^{\mathrm{in}}} - \mathbf{e}_0 = \sum_{\nu \geq 0} \mathbb{E}^q_{x^{\mathrm{in}}}\big[\mathbf{1}_{\{\nu < \theta_{\mathrm{abs}}\}}(\mathbf{e}_{X_\nu} - \mathbf{e}_{X_{\nu+1}})\big]$. Given $X_0, \dots, X_\nu$ with $X_\nu = x \neq 0$, the conditional expectation of $\mathbf{e}_{X_{\nu+1}}$ is $\sum_{y \in \mathcal{Y}_x} q_{x,y}\mathbf{e}_y$ if x is interior and $\mathbf{e}_1$ if $x = 3r$. The ν-th term is therefore

$$\sum_{x=1}^{3r-1} \mathbb{P}^q_{x^{\mathrm{in}}}(X_\nu = x)\Big(\mathbf{e}_x - \sum_{y \in \mathcal{Y}_x} q_{x,y}\mathbf{e}_y\Big) + \mathbb{P}^q_{x^{\mathrm{in}}}(X_\nu = 3r)(\mathbf{e}_{3r} - \mathbf{e}_1).$$

Summing over ν and using $\sum_\nu \mathbb{P}^q_{x^{\mathrm{in}}}(X_\nu = x) = \lambda_x$ and $\sum_\nu \mathbb{P}^q_{x^{\mathrm{in}}}(X_\nu = 3r) = \mu$, we obtain (5.3).

Step 7: the linear system. Let Π be the matrix indexed by the states $1, \dots, 3r$ with $\Pi_{x,y} = q_{x,y}$ for interior x and $y \in \mathcal{Y}_x \setminus \{0\}$, with $\Pi_{3r,1} = 1$, and with all other entries 0. Since state 0 is absorbing, $(\Pi^\nu)_{x,y} = \mathbb{P}^q_x(X_\nu = y)$ for $x, y \geq 1$, so the row sums of $\sum_{\nu \geq 0} \Pi^\nu$ are the expected absorption times, which are finite by Step 6. The series therefore converges, so $\Pi^\nu \to 0$, and letting $\bar{\nu} \to \infty$ in $\sum_{\nu=0}^{\bar{\nu}} \Pi^\nu(\mathrm{Id} - \Pi) = \mathrm{Id} - \Pi^{\bar{\nu}+1}$ shows that $\mathrm{Id} - \Pi$ is nonsingular, with inverse the fundamental matrix $\sum_{\nu \geq 0} \Pi^\nu$ [13, Theorem 11.4]. The coordinates $1, \dots, 3r$ of (5.3) state that $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_{3r-1}, \mu)$ satisfies $(\mathrm{Id} - \Pi^{\mathsf{T}})\boldsymbol{\lambda} = \mathbf{e}'_{x^{\mathrm{in}}}$, where $\mathbf{e}'_{x^{\mathrm{in}}}$ is the standard basis vector of $\mathbb{Q}^{3r}$ at x^{in}. This nonsingular system has rational coefficients, so $\boldsymbol{\lambda}$ is its unique solution and is rational, and Gaussian elimination in exact arithmetic computes it. The occupations are nonnegative as expected counts.

Step 8: the totals. The final row contributes less than 105 to each sum in (5.5), and each of the $a - 2$ lower rows contributes less than 180. Both sums are therefore at most $105 + 180(a - 2) = \Gamma_a$. All terms of the row sums are nonnegative, since $d_{r,x} \geq 2$. Hence each $\lambda_{r,x}$ and each $\lambda_{r,x} \log d_{r,x}$ is at most its row sum, which is less than 180. □

Remark 5.4 (The constants are conservative). The constants 105 and 180 are far above the actual row sums. A floating-point evaluation of the occupations for $2 \leq r \leq 120$ gives $\sum_x \lambda_{r,x} \leq 7$ and $\sum_x \lambda_{r,x} \log d_{r,x} < 6.88$ in the final row, and $\sum_x \lambda_{r,x} \leq 60/7 < 8.58$ and $\sum_x \lambda_{r,x} \log d_{r,x} < 9.98$ in a lower row. The first three maxima are attained at $r = 2$ (Example 5.2), and the last at $r = 18$.

6 Rounding the occupations to integer counts

The tensor maps of Section 4 apply each split and each tripling a whole number of times, whereas the occupations $\lambda_{r,x}$ and μ_r of Proposition 5.3 are rational numbers whose denominators may be large. For every integer rounding scale $K \geq 1$, this section rounds K times the expected transition counts of each row to integers that keep every state balanced exactly, as the row identities (5.3) do. Then every tensor factor that one relation produces is consumed by another, apart from K copies of $S^+_{a,a}$ that are only consumed and K copies of M_2 that are only produced (Proposition 6.2). The price, at every scale, is a rounding loss of at most $4J_a$, where $J_a = O(a^3)$ is the number of transition coordinates (Proposition 6.3). Clearing the denominators of the exact certificate of Appendix A would instead require a scale that grows at least exponentially.

6.1 Rounding a circulation

A *circulation* on a finite directed multigraph assigns a nonnegative value to each edge so that at every vertex the values of the entering edges and of the leaving edges have the same sum; a self-loop both enters and leaves its vertex. The following lemma is classical: it follows from Hoffman and Kruskal [14, Theorem 2], since the incidence matrix of a directed graph is totally unimodular, and rounding through a return edge is described by Kang and Payor [15, Sections 2.2–3.1]. We include a short proof because it fixes the counts used in the tensor maps and allows self-loops and opposing edges.

Lemma 6.1 (Rounding a circulation)**.** *Let φ be a circulation with rational values on a finite directed multigraph, which may have self-loops and opposing edges. There is an integer circulation n on the same multigraph with*

$$\lfloor \varphi_e \rfloor \le n_e \le \lceil \varphi_e \rceil \quad \text{for every edge } e,$$

and it can be computed with exact rational and integer arithmetic. In particular, $|n_e - \varphi_e| < 1$ for every edge, and $n_e = \varphi_e$ whenever φ_e is an integer.

Proof. Put $\eta_e = \varphi_e - \lfloor \varphi_e \rfloor \in [0, 1)$. Since φ is a circulation, the out-minus-in balance β_v of the η_e at a vertex v, to which self-loops contribute nothing, equals the in-minus-out balance of the integer parts $\lfloor \varphi_e \rfloor$, so β_v is an integer. Form a network with a new source v_+ and a new sink v_-: each nonintegral edge with distinct endpoints becomes its own arc of capacity one, and we add an arc $v_+ \to v$ of capacity β_v when $\beta_v > 0$ and an arc $v \to v_-$ of capacity $-\beta_v$ when $\beta_v < 0$. The values η_e, with the added arcs saturated, form a flow of value $\beta_{\mathrm{tot}} = \sum_{v:\, \beta_v > 0} \beta_v$, because at each vertex v the arc from v_+ or to v_- supplies or absorbs the balance β_v; since $\sum_v \beta_v = 0$, β_{tot} is also the total capacity of the arcs at v_+, and of those at v_-. Since all capacities are integers, an augmenting-path algorithm started at the zero flow raises the value by a positive integer at every step and stops after at most β_{tot} steps with an integral flow g. When it stops, the vertices reachable from v_+ in the residual network form a cut of capacity equal to the value of g; this capacity is at least the value β_{tot} of the fractional flow, so g saturates all arcs at v_+ and v_-. Set $n_e = \lfloor \varphi_e \rfloor + g_e$, where $g_e \in \{0, 1\}$ is the flow of g on the arc of e, and $g_e = 0$ for integral edges and self-loops. At every vertex the g_e have out-minus-in balance β_v and the integer parts have balance $-\beta_v$, so n is an integer circulation with the required bounds. Every step uses exact rational and integer arithmetic. □

6.2 The rounded counts

Fix $a \ge 2$ and $K \ge 1$, and use the notation of Section 5; sums $\sum_{r,x}$ run over $2 \le r \le a$ and $1 \le x \le 3r - 1$. We round one graph for each row. The *graph of row r* has the states $0, \dots, 3r$ as vertices and three kinds of edges, each with a *scaled value*. For every interior state x and every child $y \in \mathcal{Y}_{r,x}$ there is a *transition edge* $x \to y$ with scaled value $K\lambda_{r,x}q_{r,x,y}$; it records the splits at x whose child is y. The *tripling edge* $3r \to 1$ has scaled value $K\mu_r$ and records the triplings of row r. The *return edge* $0 \to x_r^{\mathrm{in}}$ has scaled value K; it returns the K units absorbed at state 0 to the injection state. These scaled values form a circulation: an edge from x to y with value φ contributes $\varphi(\mathbf{e}_x - \mathbf{e}_y)$ to the out-minus-in balances, so, since $\sum_y q_{r,x,y} = 1$, the transition and tripling edges together contribute K times the left side of (5.3), which is $K(\mathbf{e}_{x_r^{\mathrm{in}}} - \mathbf{e}_0)$, and the return edge contributes $K(\mathbf{e}_0 - \mathbf{e}_{x_r^{\mathrm{in}}})$. The graph has self-loops, at the states x with $h_{r,x}$ even, and opposing edges, such as $0 \to x_r^{\mathrm{in}}$ and $x_r^{\mathrm{in}} \to 0$. A loop does not affect any balance, but its count is part of the number of splits at x and of their type, so it must be rounded too; for example, in row 2 with $a = 2$ the loop at state 2 has scaled value $14K/15$.

Apply Lemma 6.1 to the graph of each row separately. The return edge has the integer value K, so it keeps this value; we remove it after rounding. Let $n_{r,x,y}$ be the rounded count on the transition edge $x \to y$, which is the number of splits at state x of row r with child y, and let z_r be the rounded count on the tripling edge, which is the number of triplings in row r. Put $w_{r,x} = \sum_{y \in \mathcal{Y}_{r,x}} n_{r,x,y}$, the number of splits at (r, x), and $\bar{w} = \sum_{r,x} w_{r,x}$ and $\bar{z} = \sum_{r=2}^{a} z_r$. Together with the return edge, the rounded counts form a circulation, so they satisfy the rounded row identities

$$\sum_{x=1}^{3r-1} \Big(w_{r,x}\mathbf{e}_x - \sum_{y \in \mathcal{Y}_{r,x}} n_{r,x,y}\mathbf{e}_y \Big) + z_r(\mathbf{e}_{3r} - \mathbf{e}_1) = K(\mathbf{e}_{x_r^{\mathrm{in}}} - \mathbf{e}_0) \qquad (2 \le r \le a). \tag{6.1}$$

The lemma gives the first two of the strict error bounds

$$|n_{r,x,y} - K\lambda_{r,x}q_{r,x,y}| < 1, \qquad |z_r - K\mu_r| < 1, \qquad |w_{r,x} - K\lambda_{r,x}| < d_{r,x}, \tag{6.2}$$

and the third follows by adding the first over the $d_{r,x}$ children of x, since $\sum_y q_{r,x,y} = 1$.

An edge with scaled value zero keeps the count zero, so $w_{r,x} = 0$ whenever $\lambda_{r,x} = 0$. The split at (r, x) is *active* if $w_{r,x} > 0$; only active splits are used in the tensor maps. The count z_r may be any nonnegative integer, since Lemma 4.5 imposes no divisibility condition. Every rule for choosing augmenting paths gives counts with these properties; Theorem 8.4 fixes one.

6.3 The ledger

To see that the relations of all rows fit together, we record in one formal sum, the *ledger*, every tensor factor that a relation consumes, with a plus sign, and every factor that it produces, with a minus sign. It shows that all intermediate factors cancel, and it determines the catalyst: the factors that are both consumed and produced. Recall the sign $\sigma_r = (-1)^{a-r}$ of row r (Section 5), and write $b_r(x) = r - 1 + x$ for the column of state x; thus state x of row r stands for the tensor $S^{\sigma_r}_{r,b_r(x)}$. We record tensor factors by formal symbols $[\sigma; r, b]$, one for each tensor $S^{\sigma}_{r,b}$, subject to the identifications

$$[\sigma; r, b] = [-\sigma; b, r], \qquad [+; r, r] = [-; r, r], \qquad [\sigma; 1, b] = [\mathsf{M}_b], \tag{6.3}$$

which follow the isomorphisms $S^{\sigma}_{r,b} \cong S^{-\sigma}_{b,r}$, $S^{+}_{r,r} \cong S^{-}_{r,r}$ and $S^{\sigma}_{1,b} \cong \mathsf{M}_b$ of (2.6); the second is the case $b = r$ of the first. We work in the free abelian group generated by the resulting equivalence classes, which we still call symbols. The classes $[\mathsf{M}_b]$ are the *boundary symbols*.

In row r, a split at state x consumes the symbol $[\sigma_r; r, b_r(x)]$ and produces $[\sigma_r; r, b_r(y)]$ for its child y (Lemma 4.3); a tripling consumes $[\sigma_r; r, 4r-1]$ and produces $[\sigma_r; r, r]$ (Lemma 4.5 and (4.6)). The *source multiplicity* $\mathrm{src}[\sigma; r, b]$ and the *output multiplicity* $\mathrm{out}[\sigma; r, b]$ of a symbol are the numbers of times it is consumed and produced: the $w_{r,x}$ splits at (r, x) add $w_{r,x}$ to $\mathrm{src}[\sigma_r; r, b_r(x)]$, and those among them with child y add $n_{r,x,y}$ to $\mathrm{out}[\sigma_r; r, b_r(y)]$; the z_r triplings of row r add z_r to $\mathrm{src}[\sigma_r; r, 4r-1]$ and to $\mathrm{out}[\sigma_r; r, r]$. Summed over all symbols, both multiplicities give $\bar{w} + \bar{z}$. The *catalyst multiplicities* and the *catalyst* are

$$c_{\sigma,r,b} = \min\bigl(\mathrm{src}[\sigma; r, b], \mathrm{out}[\sigma; r, b]\bigr), \qquad \mathcal{C} = \bigotimes_{\sigma,r,b} (S^{\sigma}_{r,b})^{\otimes c_{\sigma,r,b}}, \qquad \mathcal{C}_N = \mathcal{C}^{\otimes N}. \tag{6.4}$$

Here the product, like every sum over (σ, r, b) below, runs over the symbols, with one representative (σ, r, b) chosen in each class; by (2.6), another choice changes $\mathcal{C}$ only by an isomorphism.

Proposition 6.2 (Rounded ledger)**.** *The rounded counts satisfy*

$$\sum_{r=2}^{a} \sum_{x=1}^{3r-1} \Bigl(w_{r,x}[\sigma_r; r, b_r(x)] - \sum_{y \in \mathcal{Y}_{r,x}} n_{r,x,y}[\sigma_r; r, b_r(y)] \Bigr) + \sum_{r=2}^{a} z_r \bigl([\sigma_r; r, 4r-1] - [\sigma_r; r, r] \bigr) = K[+; a, a] - K[\mathsf{M}_2]. \tag{6.5}$$

Consequently, the source and output multiplicities agree for every symbol except two: the source multiplicity of $[+; a, a]$ exceeds its output multiplicity by K, and $[\mathsf{M}_2]$, the only boundary symbol

that occurs, is never consumed and is produced K times. Hence its catalyst multiplicity is zero, the catalyst $\mathcal{C}$ is a tensor product of factors $S^{\sigma}_{r,b}$ with $r, b \geq 2$, and

$$\sum_{\sigma,r,b} c_{\sigma,r,b} = \bar{w} + \bar{z} - K. \tag{6.6}$$

In tensor terms, the relations together consume $\mathcal{C} \otimes (S^{+}_{a,a})^{\otimes K}$ and produce $\mathcal{C} \otimes (S^{\sigma_2}_{2,1})^{\otimes K} \cong \mathcal{C} \otimes \mathsf{M}_{2^K}$.

Proof. Step 1: the ledger identity. In row r, map each basis vector $\mathbf{e}_x$ to the symbol $[\sigma_r; r, b_r(x)]$ and extend linearly. Since $b_r(3r) = 4r - 1$ and $b_r(1) = r$, this map sends the left side of (6.1) to the terms of row r on the left side of (6.5), and it sends the right side to $K([\sigma_r; r, b_r(x^{\mathrm{in}}_r)] - [\sigma_r; r, r-1])$. In the final row, $x^{\mathrm{in}}_a = 1$ and $\sigma_a = +$, so this is $K([+; a, a] - [+; a, a-1])$. A lower row has $x^{\mathrm{in}}_r = 2$ and gives $K([\sigma_r; r, r+1] - [\sigma_r; r, r-1])$. Because $\sigma_{r+1} = -\sigma_r$, the first identification in (6.3) gives $[\sigma_r; r, r+1] = [\sigma_{r+1}; r+1, r]$; thus, for $2 \leq r < a$, the positive term of row r cancels the negative term of row $r+1$. Apart from naming $[+; a, a] = [-; a, a]$, the proof uses the signs only through this alternation (Section 4.4). Summing over all rows leaves $K[+; a, a] - K[\sigma_2; 2, 1]$, and $[\sigma_2; 2, 1] = [-\sigma_2; 1, 2] = [\mathsf{M}_2]$. This proves (6.5).

Step 2: the multiplicities. By their definition, the left side of (6.5) equals $\sum_{\sigma,r,b}(\mathrm{src}[\sigma; r, b] - \mathrm{out}[\sigma; r, b])\,[\sigma; r, b]$. Since $a \geq 2$, the symbols $[+; a, a]$ and $[\mathsf{M}_2]$ are distinct, so comparing coefficients gives equal multiplicities for every other symbol, and differences K for $[+; a, a]$ and $-K$ for $[\mathsf{M}_2]$. A consumed symbol is $[\sigma_r; r, b]$ with $r \geq 2$ and $b \geq r$ (either $b = b_r(x)$ for an interior state x, or $b = 4r - 1$); it and every symbol identified with it have both indices at least 2. Hence $\mathrm{src}[\mathsf{M}_2] = 0$ and $\mathrm{out}[\mathsf{M}_2] = K$. A produced symbol is $[\sigma_r; r, b_r(y)]$ or $[\sigma_r; r, r]$, with $r \geq 2$ and $b_r(y) \geq r - 1$, so an index 1 occurs only for $r = 2$ and $y = 0$, which gives $[\sigma_2; 2, 1] = [\mathsf{M}_2]$.

Step 3: the catalyst. By Step 2, the catalyst multiplicity equals the source multiplicity for every symbol except $[+; a, a]$, where it is $\mathrm{out}[+; a, a] = \mathrm{src}[+; a, a] - K$, and it equals the output multiplicity for every symbol except $[\mathsf{M}_2]$, where it is $\min(0, K) = 0$. Hence every symbol with positive catalyst multiplicity is consumed, so its indices are at least 2; summing over all symbols gives $\sum_{\sigma,r,b} c_{\sigma,r,b} = \bar{w} + \bar{z} - K$; and the consumed symbols, counted with multiplicity, are those of $\mathcal{C}$ and K copies of $[+; a, a]$, while the produced ones are those of $\mathcal{C}$ and K copies of $[\mathsf{M}_2]$. The tensor statement follows by (2.6) and $\mathsf{M}_2^{\otimes K} \cong \mathsf{M}_{2^K}$. □

For $2 \leq r < a$, the symbol $[\sigma_r; r, r+1] = [\sigma_{r+1}; r+1, r]$ of Step 1 belongs to state 2 of row r and to state 0 of row $r+1$; a comparison of the column ranges $[r-1, 4r-1]$ shows that different rows share no other symbol. The auxiliary matrix factors are not ledger symbols; Section 7 accounts for them separately.

6.4 The rounding loss

Rounding changes the frequencies with which each split produces its children, and so it changes the gain of the construction, which we measure by the rounded potential G below: NG approximates the logarithm Δ_N of the ratio of the one-step target side to the one-step source side of Section 7, by (7.11). The term $K \log 2$ of G comes from the output $\mathsf{M}_{2^{NK}}$; the term of a split at (r, x) compares the roughly $\exp(N w_{r,x} H(\hat{q}_{r,x}))$ sectors of its $N w_{r,x}$ uses with their auxiliary side $d_{r,x}^{N w_{r,x}}$; and $z_r \log 3$ counts the roughly $3^{N z_r}$ sectors of the triplings of row r.

For probability vectors $\hat{q}$ and q on the same finite set, with q positive, let $H(\hat{q}) = -\sum_y \hat{q}_y \log \hat{q}_y$, with $0 \log 0 = 0$, be the entropy, and let $D(\hat{q} \| q) = \sum_{y:\, \hat{q}_y > 0} \hat{q}_y \log(\hat{q}_y / q_y)$ be the relative entropy, which is nonnegative (Gibbs' inequality). For an active split at (r, x), let $\hat{q}_{r,x} = (\hat{q}_{r,x,y})_y$, with

$\hat{q}_{r,x,y} = n_{r,x,y}/w_{r,x}$, be its empirical distribution of children, and let $q_{r,x} = (q_{r,x,y})_y$. The *rounded potential* and the *rounding loss* are

$$G = K \log 2 + \sum_{r,x:\, w_{r,x}>0} w_{r,x}\big(H(\hat{q}_{r,x}) - \log d_{r,x}\big) + \bar{z} \log 3, \qquad \mathcal{L} = \sum_{r,x:\, w_{r,x}>0} w_{r,x} D(\hat{q}_{r,x} \| q_{r,x}). \tag{6.7}$$

Finally, let J_a be the number of transition coordinates, that is, the number of transition edges in all row graphs, loops included:

$$\begin{aligned} J_a &= \sum_{r=2}^{a} \sum_{x=1}^{3r-1} d_{r,x} = \sum_{r=2}^{a} \Big(\Big\lfloor \frac{9r^2}{4} \Big\rfloor + 3r - 1 \Big) \\ &= \frac{12a^3 + 42a^2 + 12a - 65 + (-1)^a}{16} = O(a^3). \end{aligned} \tag{6.8}$$

The second equality holds because $\sum_{x=1}^{3r-1} \min(x, 3r-x) = \lfloor 9r^2/4 \rfloor$. The third follows by summing over r, using $\lfloor 9r^2/4 \rfloor = 9r^2/4$ for even r and $\lfloor 9r^2/4 \rfloor = (9r^2-1)/4$ for odd r.

Proposition 6.3 (Uniform entropy loss)**.** *For every rounding scale $K \geq 1$, the rounded potential and the rounding loss of (6.7) satisfy*

$$G = K \log F(a,a) - \mathcal{L}, \qquad 0 \leq \mathcal{L} \leq 4J_a. \tag{6.9}$$

Proof. The identity comes from evaluating the potential $\log F$ on the ledger. The bound comes from comparing each empirical distribution with the transition probabilities edge by edge, using that no child probability is smaller than $1/(3d_{r,x})$.

Step 1: a lower bound for the child probabilities. Fix a row r and an interior state x, and write $h = h_{r,x}$ and $d = d_{r,x}$. By (5.2), $q_{r,x,y} = f_r(y)/(d\, f_r(x))$, where $f_r(y) = \frac{3(r-1)}{2} + y$ is increasing; so the smallest probability belongs to the smallest child $x - h$ and equals $f_r(x-h)/(d\, f_r(x))$. If $x \leq 3r/2$, then $h = x$, and the ratio $f_r(0)/f_r(x)$ decreases with x. If $x \geq 3r/2$, then $h = 3r - x$, and the ratio $f_r(2x-3r)/f_r(x) = 2 - (f_r(0) + 3r)/f_r(x)$ increases with x. Both expressions, read as functions of a real variable x, take the common value $f_r(0)/f_r(3r/2) = (r-1)/(2r-1)$ at $x = 3r/2$, whether or not $3r/2$ is an integer. Therefore every child y of x satisfies

$$q_{r,x,y} \geq \frac{f_r(0)}{d_{r,x} f_r(3r/2)} = \frac{r-1}{(2r-1)d_{r,x}} \geq \frac{1}{3d_{r,x}}, \tag{6.10}$$

where the last step uses $r \geq 2$. The bound is attained for $r = 2$ and $x = 3$, where $q_{2,3,0} = 1/12$.

Step 2: the loss of one split. Fix an active split at (r,x). In this step we omit the subscripts r, x: thus $\lambda = \lambda_{r,x}$, $w = w_{r,x}$ and $d = d_{r,x}$, and n_y, $\hat{q}_y$, q_y stand for $n_{r,x,y}$, $\hat{q}_{r,x,y}$, $q_{r,x,y}$. By (6.2), $n_y < K\lambda q_y + 1$ for every child y, and $K\lambda < w + d$; by (6.10), $1/q_y \leq 3d$. Hence every child with $n_y > 0$ satisfies

$$\frac{\hat{q}_y}{q_y} = \frac{n_y}{w q_y} < \frac{K\lambda}{w} + \frac{1}{w q_y} < 1 + \frac{d}{w} + \frac{3d}{w} = 1 + \frac{4d}{w}.$$

Averaging the logarithm of this inequality with the weights $\hat{q}_y$ gives $D(\hat{q}_{r,x} \| q_{r,x}) \leq \log(1 + 4d/w)$, and therefore

$$w\, D(\hat{q}_{r,x} \| q_{r,x}) \leq w \log\Big(1 + \frac{4d}{w}\Big) \leq 4d.$$

The argument divides only by the positive integer w, never by the occupation λ, so the bound holds at every scale K, even when some occupations are tiny or zero. Summing over the active splits, and using $D \geq 0$, gives $0 \leq \mathcal{L} \leq \sum_{r,x} 4d_{r,x} = 4J_a$.

Step 3: the identity. By (5.1), $F(r, b_r(y)) = s_r f_r(y)$ for $0 \le y \le 3r$. Hence the transition probabilities (5.2) are ratios of profile values,

$$q_{r,x,y} = \frac{F(r, b_r(y))}{d_{r,x} F(r, b_r(x))}, \quad \text{that is,} \quad \log F(r, b_r(y)) = \log F(r, b_r(x)) + \log d_{r,x} + \log q_{r,x,y}.$$

Multiply by $n_{r,x,y}$ and sum over the children. For an active split, $\sum_y n_{r,x,y} = w_{r,x}$, and the definitions of H and D give $-\sum_{y:\, n_{r,x,y}>0} \hat{q}_{r,x,y} \log q_{r,x,y} = H(\hat{q}_{r,x}) + D(\hat{q}_{r,x} \| q_{r,x})$. Hence

$$w_{r,x} \log F(r, b_r(x)) - \sum_y n_{r,x,y} \log F(r, b_r(y)) = w_{r,x}\big(H(\hat{q}_{r,x}) - \log d_{r,x}\big) + w_{r,x} D(\hat{q}_{r,x} \| q_{r,x}). \quad (6.11)$$

For an inactive split both sides are zero. For a tripling, $F(r, 4r-1) = 3F(r, r)$ (Step 1 of the proof of Theorem 3.2), so

$$z_r \big(\log F(r, 4r-1) - \log F(r, r)\big) = z_r \log 3. \quad (6.12)$$

Now assign to each symbol $[\sigma; r, b]$ the value $\log F(r, b)$. This is well defined, because F is symmetric, the value does not depend on σ, and $F(1, b) = b$, so that the boundary symbol $[\mathsf{M}_b] = [\sigma; 1, b]$ receives $\log b$. Extend the assignment linearly and apply it to (6.5). By (6.11) and (6.12), the left side becomes

$$\sum_{r,x:\, w_{r,x}>0} w_{r,x}\big(H(\hat{q}_{r,x}) - \log d_{r,x}\big) + \mathcal{L} + \bar{z} \log 3,$$

while the right side becomes $K \log F(a, a) - K \log 2$. Adding $K \log 2$ to both sides gives $G = K \log F(a, a) - \mathcal{L}$. □

Thus $w_{r,x} D(\hat{q}_{r,x} \| q_{r,x})$ is the price of producing children with the frequencies $\hat{q}_{r,x}$ instead of $q_{r,x}$; it vanishes if K clears the denominators of all scaled values. For a distribution $\hat{p}$ with entries in $\frac{1}{w_{r,x}}\mathbb{Z}$ that minimizes $D(\hat{p} \| q_{r,x})$, the bound of Böcherer and Geiger [5, Section IV.D, Eq. (42)] and (6.10) give $3d_{r,x}$ in place of $4d_{r,x}$, and so does Step 2 when $w_{r,x} \ge K\lambda_{r,x}$; the extra $d_{r,x}$ pays for the splits with $w_{r,x} < K\lambda_{r,x}$.

Remark 6.4 (Size of the loss in examples). The bound $4J_a$ is far from tight. A floating-point evaluation for the rounding rule of Theorem 8.4, with $2 \le a \le 12$ and $K \in \{1, 2, 7, a^3, a^3\lceil \log_2 a \rceil\}$, gives $\mathcal{L}/(4J_a) < 0.029$ in every case. The largest ratio occurs at $a = 2$ and $K = 1$, where $\mathcal{L} = \log 5$ and $4J_2 = 56$. For $a = 12$ and the scale $K = a^3\lceil \log_2 a \rceil = 6912$ of Theorem 8.1 (Section 10 uses a larger K), $\mathcal{L} \approx 3.96$, against $4J_{12} = 6716$.

6.5 Bounds on the rounded counts

Put $W = \sum_{r,x} w_{r,x} \log d_{r,x}$; at type scale N, the source auxiliary matrix factor of all splits has side $\exp(NW)$. Recall from (5.5) that $\sum_{r,x} \lambda_{r,x}$ and $\sum_{r,x} \lambda_{r,x} \log d_{r,x}$ are both at most Γ_a, and from Proposition 5.3 that $\lambda_{r,x} \log d_{r,x} \le 180$ for every split. These bounds and (6.2) give

$$\begin{aligned} W &\le K\Gamma_a + J_a \log(2a), \qquad \bar{w} \le K\Gamma_a + J_a, \\ \bar{z} &\le (2K+1)(a-1), \\ w_{r,x} \log d_{r,x} &\le 180K + 2a \log(2a). \end{aligned} \quad (6.13)$$

Indeed, $d_{r,x} \le \lfloor 3a/2 \rfloor + 1 \le 2a$. Summing $w_{r,x} < K\lambda_{r,x} + d_{r,x}$ over all splits gives $\bar{w} < K \sum_{r,x} \lambda_{r,x} + J_a$. Multiplying the inequality $w_{r,x} < K\lambda_{r,x} + d_{r,x}$ of a single split by $\log d_{r,x} \le \log(2a)$ gives $w_{r,x} \log d_{r,x} < K\lambda_{r,x} \log d_{r,x} + d_{r,x} \log(2a)$; with $\lambda_{r,x} \log d_{r,x} \le 180$ this proves the last bound, and summing over all splits proves the bound for W. By (5.4), $\mu_r \le 2$, so $z_r < 2K + 1$ in each of the $a - 1$ rows. The bound for a single split controls the largest sector count in Section 8; the bound for W alone would be weaker by a factor of order a.

7 The finite construction

This section turns the integer counts of Section 6 into a relative scheme and computes its exponent. Proposition 7.1 combines all splits and triplings into one catalytic degeneration, and Proposition 7.3 makes it exact and reuses the catalyst. Section 7.5 bounds every term of the resulting exponent (7.9) by quantities computed from the counts and proves $p \geq \tau_a > 9/4$.

7.1 Parameters and sector counts

Fix the target index $a \geq 2$ and a rounding scale $K \geq 1$, and let $n_{r,x,y}$, $w_{r,x}$ and z_r be the integer counts of Section 6. Only relations with a positive count are used. Let $\mathcal{I}_c$ be the set of active splits, that is, of pairs $i = (r, x)$ with $w_i = w_{r,x} > 0$, and let $\mathcal{I}_t$ be the set of active triplings, each indexed by its row r, so that $z_r > 0$. Put $\mathcal{J} = \mathcal{I}_c \sqcup \mathcal{I}_t$ and $I = |\mathcal{J}|$; we call an element of $\mathcal{J}$ a relation. A split $i = (r, x)$ has width $h_{r,x}$, auxiliary dimension $d_i = d_{r,x}$, child counts $n_{i,y} = n_{r,x,y}$, and empirical distribution of children $\hat{q}_i = \hat{q}_{r,x}$, with $\hat{q}_{i,y} = n_{i,y}/w_i$. Its child y is column $r - 1 + y$, which is the block of Lemma 4.3 with filtration index $(x + h_{r,x} - y)/2$.

Three further positive integers amortize finite overheads (Table 1): the type scale N multiplies all counts, the interpolation power k is the tensor power used in exactification, and the reuse count j is the number of uses of one copy of the catalyst. All results of Sections 8 and 10 take $N = 1$; a larger N matters only when K is held fixed, since it makes the type penalty and the copy overhead small compared with Δ_N.

At type scale N, a split $i = (r, x)$ is applied to $(C(r, r - 1 + x)B_X(d_i))^{\otimes Nw_i}$ with the type $(Nn_{i,y})_y$, and a tripling r is applied to $C(r, 4r - 1)^{\otimes Nz_r}$ with middle count m_r. The numbers of sectors retained by Lemmas 4.3 and 4.5 are the sector counts

$$M_i = \frac{(Nw_i)!}{\prod_y (Nn_{i,y})!} \quad (i \in \mathcal{I}_c), \qquad M_r = \binom{Nz_r}{m_r} 2^{Nz_r - m_r}, \quad m_r = \left\lfloor \frac{Nz_r + 1}{3} \right\rfloor \quad (r \in \mathcal{I}_t). \tag{7.1}$$

Zero child counts contribute $0! = 1$. The middle count m_r is the maximizing choice of Lemma 4.5, so $M_r \geq 3^{Nz_r}/(Nz_r + 1)$ by (4.3).

Proposition 7.1 below applies the separator to $L_i = 2M_i - 1$ copies of each relation i in each of three orientations. It turns L source copies into M^3 output copies, with degree at most D, the sum of the degree bounds $(M_i - 1)^2$ of all separators. The one-step source side A_1 collects the auxiliary dot products $B_X(d_i)$ of the splits; the one-step target side B_1 collects NK boundary factors M_2 and the separator auxiliaries:

$$M = \prod_{i \in \mathcal{J}} M_i, \qquad L = \prod_{i \in \mathcal{J}} L_i^3, \quad L_i = 2M_i - 1, \tag{7.2}$$

$$D = 3 \sum_{i \in \mathcal{J}} (M_i - 1)^2, \qquad A_1 = \prod_{i \in \mathcal{I}_c} d_i^{Nw_i}, \quad B_1 = 2^{NK} M. \tag{7.3}$$

Finally, $\mathcal{C}_N = \mathcal{C}^{\otimes N}$, where $\mathcal{C}$ is the catalyst of (6.4).

7.2 The catalytic degeneration

Proposition 7.1 (Catalytic degeneration)**.** *The maps described in the proof give a degeneration of degree at most D:*

$$L\, \mathcal{C}_N (S_{a,a}^{+})^{\otimes NK} \mathsf{M}_{A_1} \trianglerighteq M^3\, \mathcal{C}_N \mathsf{M}_{B_1}. \tag{7.4}$$

Proof. The idea is to apply Corollary 4.8 to every relation, to tensor the results with one degeneration parameter, and to read off the factors on the two sides from the ledger.

Step 1: one relation. Let $i = (r, x) \in \mathcal{I}_c$ be a split. Apply Corollary 4.8(a) with $\sigma = \sigma_r$, $b = r - 1 + x$, $h = h_{r,x}$, $n = Nw_i$, and the type that has $Nn_{i,y}$ positions of filtration index $(x + h_{r,x} - y)/2$ for every child y. The hypotheses hold: the state x is interior, so $h \geq 1$, and $b \geq x+1 \geq h+1$ because $r \geq 2$. The multinomial coefficient of the corollary is M_i, its auxiliary side is $(h+1)^n = d_i^{Nw_i}$, and the block of filtration index $(x+h-y)/2$ has column $b+h-(x+h-y) = r-1+y$. Hence

$$L_i^3 \, (S^{\sigma_r}_{r,r-1+x})^{\otimes Nw_i} \mathsf{M}_{d_i^{Nw_i}} \trianglerighteq M_i^3 \Big(\prod_y (S^{\sigma_r}_{r,r-1+y})^{\otimes Nn_{i,y}} \Big) \mathsf{M}_{M_i}.$$

For a tripling $r \in \mathcal{I}_t$ (so $r \geq 2$), part (b) with $n = Nz_r \geq 1$ and $m = m_r$ gives $L_r^3 \, (S^{\sigma_r}_{r,4r-1})^{\otimes Nz_r} \trianglerighteq M_r^3 \, (S^{\sigma_r}_{r,r})^{\otimes Nz_r} \mathsf{M}_{M_r}$. Each of these degenerations has degree at most $3(M_i - 1)^2$, and its maps before and after the separators are fixed linear maps of the full variable spaces that do not depend on t; Appendix B writes them out, including the terms of positive degree.

Step 2: all relations with one parameter. Tensor the degenerations of Step 1 over all $i \in \mathcal{J}$, with one common parameter t for all separators. Tensor products distribute over direct sums, so the left side has $\prod_i L_i^3 = L$ copies and the constant term has $\prod_i M_i^3 = M^3$ copies. By Lemma 2.1, the result is a polynomial in t of degree at most $\sum_i 3(M_i - 1)^2 = D$, and its constant term is the tensor product of the constant terms of Step 1.

Step 3: the convolution factors. In one source copy, every split $i = (r, x)$ contributes Nw_i factors with symbol $[\sigma_r; r, r-1+x]$, and every tripling contributes Nz_r factors $[\sigma_r; r, 4r-1]$; in one output copy, the splits contribute $Nn_{i,y}$ factors $[\sigma_r; r, r-1+y]$ and the triplings Nz_r factors $[\sigma_r; r, r]$. These are N times the source and output multiplicities of the ledger (6.5). By Proposition 6.2, the two multiplicities agree for every symbol except that $[+; a, a]$ has K more source than output copies and $[\mathsf{M}_2]$ has K output copies and no source copy. Since the catalyst multiplicities (6.4) are the minima of the two, the source factors form $\mathcal{C}_N (S^{+}_{a,a})^{\otimes NK}$ and the output factors form $\mathcal{C}_N \mathsf{M}_2^{\otimes NK} \cong \mathcal{C}_N \mathsf{M}_{2^{NK}}$. These identifications of symbols are the coordinate isomorphisms of (2.6), applied within each global leg (on the source side before the type restrictions, on the output side after the separators); they do not depend on t.

Step 4: the auxiliary factors. On the left, the auxiliary factors give $\bigotimes_{i \in \mathcal{I}_c} \mathsf{M}_{d_i^{Nw_i}} \cong \mathsf{M}_{A_1}$. On the right, the separator auxiliaries give $\bigotimes_{i \in \mathcal{J}} \mathsf{M}_{M_i} \cong \mathsf{M}_M$, and together with $\mathsf{M}_{2^{NK}}$ from Step 3 they give $\mathsf{M}_{2^{NK} M} = \mathsf{M}_{B_1}$, by (2.1). This proves (7.4). □

Example 7.2 (The ledger for $a = 2$)**.** For $a = 2$ there is one row, $r = 2$, with sign $\sigma_2 = +$ and injection state $x_2^{\mathrm{in}} = 1$. Its occupations are $\lambda_{2,1} = 2$, $\lambda_{2,2} = 14/5$, $\lambda_{2,4} = 11/5$ and $\mu_2 = 1$, and states 3 and 5 are never visited (Example 5.2). For $K = 15$ all scaled values $K\lambda_{2,x} q_{2,x,y}$ and $K\mu_2$ are integers, so the rounding leaves them unchanged. The split at state 1 (column 2, width 1, $d = 2$) has children $0, 2$ with counts $9, 21$; the split at state 2 (column 3, width 2, $d = 3$) has children $0, 2, 4$ with counts $6, 14, 22$; the split at state 4 (column 5, width 2, $d = 3$) has children $2, 4, 6$ with counts $7, 11, 15$; and $z_2 = 15$. Thus $\bar{w} = 30 + 42 + 33 = 105$ and $\bar{z} = 15$. Child y is column $1 + y$, and the tripling consumes column 7 and produces column 2. The ledger involves the symbols $[+; 2, b]$ for $b = 1, 2, 3, 5, 7$, where $[+; 2, 1] = [-; 1, 2] = [\mathsf{M}_2]$:

symbol	source multiplicity	output multiplicity	$c_{+,2,b}$
$[\mathsf{M}_2] = [+;2,1]$	0	$9+6=15$	0
$[+;2,2]$	30	15	15
$[+;2,3]$	42	$21+14+7=42$	42
$[+;2,5]$	33	$22+11=33$	33
$[+;2,7]$	15	15	15

The sources exceed the outputs by $K = 15$ copies of $[+;2,2] = [+;a,a]$, and the outputs exceed the sources by 15 copies of $[\mathsf{M}_2]$. The catalyst is

$$\mathcal{C} = (S^+_{2,2})^{\otimes 15}(S^+_{2,3})^{\otimes 42}(S^+_{2,5})^{\otimes 33}(S^+_{2,7})^{\otimes 15},$$

with $105 = \bar{w} + \bar{z} - K$ factors. In Proposition 7.1, $A_1 = 2^{30N}3^{75N}$, and M is the product of the three multinomial sector counts of the splits and the sector count of the tripling (7.1).

7.3 Exactification and repeated use of one catalyst

The degeneration (7.4) is approximate, and it needs the catalyst as an input. We make it exact by interpolation, use the resulting restriction j times while passing one copy of the catalyst from one use to the next, and supply that copy once from a rank decomposition, so that the catalyst is paid for once and not j times. Exactification comes first because restrictions compose freely, whereas iterating the degeneration would require a bound on the accumulated degree.

The catalyst is supplied by tensoring the evaluation decompositions (2.4) of its convolution factors. Each factor $S^\sigma_{r,b}$ has rank at most $(r+b-1)^3$ by (2.6), so

$$\kappa_N = \prod_{\sigma,r,b} (r+b-1)^{3Nc_{\sigma,r,b}} \geq \mathrm{R}(\mathcal{C}_N), \tag{7.5}$$

where the product runs over the symbols, as in (6.4). For positive integers k and j, put $E = kD+1$ (the number of interpolation points), $A = A_1^{kj}$, $B = B_1^{kj}$, and

$$r_* = \kappa_N^k E^j L^{kj} (2a-1)^{3NKkj}, \qquad m_* = M^{3kj}. \tag{7.6}$$

Proposition 7.3 (Relative scheme)**.** *For all positive integers k and j, the maps described in the proof give a restriction $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$. When $B > A$, it is a relative scheme (Definition 2.3).*

Proof. Step 1: exactification. The substitutions of Proposition 7.1 transform the left side of (7.4) into a polynomial in t of degree at most D whose constant term is the right side. Apply Lemma 2.2 with parameter k. Since tensor products distribute over direct sums and $\mathsf{M}_d\mathsf{M}_e \cong \mathsf{M}_{de}$ by (2.1), the result reads

$$EL^k\,\mathcal{C}_N^{\otimes k}(S^+_{a,a})^{\otimes NKk}\mathsf{M}_{A_1^k} \succeq M^{3k}\,\mathcal{C}_N^{\otimes k}\mathsf{M}_{B_1^k}.$$

Step 2: decomposing the source factors. A rank decomposition of a tensor T with R terms is the same as a restriction $R\,\mathsf{M}_1 \succeq T$, where M_1 is the $1\times 1\times 1$ tensor: the s-th copy of M_1 is sent to the s-th term. By (2.6), $\mathrm{R}(S^+_{a,a}) \leq (2a-1)^3$, and hence $(2a-1)^{3NKk}\mathsf{M}_1 \succeq (S^+_{a,a})^{\otimes NKk}$. Tensoring this with $EL^k\,\mathcal{C}_N^{\otimes k}\mathsf{M}_{A_1^k}$ and composing with Step 1 gives the one-step restriction

$$\alpha\,\mathcal{C}_N^{\otimes k}\mathsf{M}_{A_1^k} \succeq \beta\,\mathcal{C}_N^{\otimes k}\mathsf{M}_{B_1^k}, \qquad \alpha = EL^k(2a-1)^{3NKk}, \quad \beta = M^{3k}. \tag{7.7}$$

Step 3: reuse. We show by induction that, for every $j' \geq 1$,

$$\alpha^{j'} \mathcal{C}_N^{\otimes k} \mathsf{M}_{A_1^{kj'}} \succeq \beta^{j'} \mathcal{C}_N^{\otimes k} \mathsf{M}_{B_1^{kj'}}.$$

The case $j' = 1$ is (7.7). For the induction step, tensor the restriction for j' with the identity on $\alpha \mathsf{M}_{A_1^k}$:

$$\alpha^{j'+1} \mathcal{C}_N^{\otimes k} \mathsf{M}_{A_1^{k(j'+1)}} \succeq \beta^{j'} (\alpha\, \mathcal{C}_N^{\otimes k} \mathsf{M}_{A_1^k}) \mathsf{M}_{B_1^{kj'}}.$$

Each of the $\beta^{j'}$ direct summands on the right is the left side of (7.7) tensored with $\mathsf{M}_{B_1^{kj'}}$; in it, $\mathcal{C}_N^{\otimes k}$ occurs once as a tensor factor. Apply (7.7) inside each summand, with the identity on $\mathsf{M}_{B_1^{kj'}}$. The summand becomes $\beta\, \mathcal{C}_N^{\otimes k} \mathsf{M}_{B_1^k} \mathsf{M}_{B_1^{kj'}} \cong \beta\, \mathcal{C}_N^{\otimes k} \mathsf{M}_{B_1^{k(j'+1)}}$, which proves the claim for $j'+1$. Since the one-step restriction acts summand by summand, every summand keeps the single tensor factor $\mathcal{C}_N^{\otimes k}$; the catalyst is never raised to a higher tensor power.

Step 4: supplying and removing the catalyst. By (7.5), $\mathrm{R}(\mathcal{C}_N^{\otimes k}) \leq \kappa_N^k$, so $\kappa_N^k \mathsf{M}_1 \succeq \mathcal{C}_N^{\otimes k}$, and hence $\kappa_N^k \alpha^j \mathsf{M}_A \succeq \alpha^j \mathcal{C}_N^{\otimes k} \mathsf{M}_A$. On the right side, choose one monomial in every convolution factor of $\mathcal{C}_N^{\otimes k}$. All coefficients of a convolution tensor are 0 or 1, so the product monomial has coefficient one, and setting all other variables to zero restricts $\mathcal{C}_N^{\otimes k}$ to M_1. Hence $\beta^j \mathcal{C}_N^{\otimes k} \mathsf{M}_B \succeq \beta^j \mathsf{M}_B$. Composing these two restrictions with Step 3 for $j' = j$ gives $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ with $r_* = \kappa_N^k \alpha^j$ and $m_* = \beta^j$, which are the values in (7.6). □

7.4 The exponent of the relative scheme

Lemma 2.4 turns a relative scheme into a recursive algorithm. For the restriction of Proposition 7.3, put

$$\Delta_N = \log \frac{B_1}{A_1} = NK \log 2 + \log M - N \sum_{i \in \mathcal{I}_c} w_i \log d_i, \tag{7.8}$$

the logarithm of the ratio of the target side to the source side in one use of the catalytic degeneration. Since $B/A = (B_1/A_1)^{kj}$, the restriction is a relative scheme exactly when $\Delta_N > 0$. In that case, by (7.6),

$$\log \frac{r_*}{m_*} = kj \left(3NK \log(2a-1) + \log \frac{L}{M^3} + \frac{\log E}{k} + \frac{\log \kappa_N}{j} \right), \qquad \log \frac{B}{A} = kj\, \Delta_N,$$

so the pooled exponent $p = \log(r_*/m_*)/\log(B/A)$ of the scheme is

$$p = \frac{1}{\Delta_N} \left(3NK \log(2a-1) + \log \frac{L}{M^3} + \frac{\log E}{k} + \frac{\log \kappa_N}{j} \right). \tag{7.9}$$

The numerator consists of the main term $3NK \log(2a-1)$, which pays for the rank of the NK source copies of $S_{a,a}^+$, and of the copy overhead $\log(L/M^3)$, the interpolation overhead $(\log E)/k$ and the catalyst overhead $(\log \kappa_N)/j$. The source M_A is the matrix product to which the recursion is applied again, so no decomposition of it enters the formula; its side $A = A_1^{kj}$ comes from the auxiliary factor M_{A_1} of Proposition 7.1.

The condition $\Delta_N > 0$ is not automatic. For $a = 2$ and $N = 1$, every rounding allowed by Lemma 6.1 gives $B_1 < A_1$, that is, $\Delta_1 < 0$, for every $K \leq 7$, as an exhaustive computation in exact integer arithmetic shows (script `extraction_a2_small_scale.py`; see the supporting files in Section 11). For the counts of Example 7.2 ($K = 15$), a floating-point evaluation gives $\Delta_1 \approx 7.28$. A proof of $B > A$ therefore needs the lower bound on Δ_N in (7.11) below; Theorem 8.1 supplies such a proof for its parameters.

7.5 Type losses and finite cost bounds

The terms of (7.9) involve the sector counts M_i, which are large multinomial coefficients. This subsection bounds every term by quantities computed directly from the integer counts, starting with a comparison of multinomial coefficients and entropies.

Lemma 7.4 (Counting the words of one type)**.** *Let $\mathcal{Y}$ be a finite nonempty set, let n_y, for $y \in \mathcal{Y}$, be positive integers with sum n, and put $\theta = (n_y/n)_{y\in\mathcal{Y}}$. Then*

$$\frac{\exp(nH(\theta))}{(n+1)^{|\mathcal{Y}|-1}} \le \frac{n!}{\prod_{y\in\mathcal{Y}} n_y!} \le \exp(nH(\theta)).$$

Proof. Draw n independent samples from the distribution θ on $\mathcal{Y}$. For nonnegative integers g_y with sum n, the probability that each y is drawn exactly g_y times is

$$\mathbb{P}(g) = n! \prod_y \frac{\theta_y^{g_y}}{g_y!} = \frac{n!}{n^n} \prod_y \frac{n_y^{g_y}}{g_y!}.$$

Since $\prod_y \theta_y^{n_y} = \exp(-nH(\theta))$, the probability of $g = (n_y)_y$ equals the multinomial coefficient times $\exp(-nH(\theta))$, and $\mathbb{P} \le 1$ gives the upper bound. The factor $n_y^{g_y}/g_y!$ changes by the ratio $n_y/(g_y+1)$ when g_y increases by one, so it is largest at $g_y = n_y$, and $g = (n_y)_y$ maximizes $\mathbb{P}$. At most $(n+1)^{|\mathcal{Y}|-1}$ vectors g occur (all entries but one lie in $\{0,\dots,n\}$), and their probabilities add up to one; so the largest is at least $(n+1)^{-(|\mathcal{Y}|-1)}$, which is the lower bound. □

For a split i, let $\mathcal{Y}_i^+ = \{y : n_{i,y} > 0\}$ be the set of its children with positive count; its auxiliary dimension is d_i even when some children have count zero, because Lemma 4.3 uses the whole factor $B_X(d_i)$. The entropy charge of a relation is $H_i = w_i H(\hat{q}_i)$ for a split and $H_r = z_r \log 3$ for a tripling. Define the type penalty $\mathcal{T}_N$ and the largest scaled entropy charge $\bar{H}_N$ by

$$\begin{aligned} \mathcal{T}_N &= \sum_{i\in\mathcal{I}_c} (|\mathcal{Y}_i^+| - 1)\log(Nw_i + 1) + \sum_{r\in\mathcal{I}_t} \log(Nz_r + 1), \\ \bar{H}_N &= N \max_{i\in\mathcal{J}} H_i. \end{aligned} \tag{7.10}$$

The type penalty bounds the distance of Δ_N from its ideal value NG, where G is the rounded potential and $\mathcal{L}$ the rounding loss of (6.7):

$$NG - \mathcal{T}_N \le \Delta_N \le NG, \qquad G = K\log F(a,a) - \mathcal{L}, \qquad 0 \le \mathcal{L} \le 4J_a. \tag{7.11}$$

Here J_a is the number of transition coordinates, and the second and third relations are (6.9).

To prove the first relation, recall from (6.7) that $G = K\log 2 + \sum_{i\in\mathcal{I}_c} w_i(H(\hat{q}_i) - \log d_i) + \bar{z}\log 3$. Since $\bar{z}\log 3 = \sum_{r\in\mathcal{I}_t} H_r$, comparison with (7.8) gives

$$\Delta_N - NG = \sum_{i\in\mathcal{J}} (\log M_i - NH_i).$$

For a split, apply Lemma 7.4 to the positive counts $Nn_{i,y}$, $y \in \mathcal{Y}_i^+$. Their sum is Nw_i and their distribution is $\hat{q}_i$; zero counts change neither M_i nor $H(\hat{q}_i)$. This gives $-(|\mathcal{Y}_i^+| - 1)\log(Nw_i + 1) \le \log M_i - NH_i \le 0$. For a tripling, $M_r \le 3^{Nz_r}$, because the numbers $\binom{n}{m}2^{n-m}$, $0 \le m \le n$, add up to 3^n; together with $M_r \ge 3^{Nz_r}/(Nz_r+1)$ this gives $-\log(Nz_r+1) \le \log M_r - NH_r \le 0$. Summing over $\mathcal{J}$ proves (7.11). We call the nonnegative difference $NG - \Delta_N = \sum_{i\in\mathcal{J}}(NH_i - \log M_i)$ the *type loss*; the type penalty bounds it.

In Example 7.2 with $N = 1$, every empirical distribution equals the transition probabilities, so $\mathcal{L} = 0$ and $G = 15\log(10/3) \approx 18.06$; the type penalty is $\mathcal{T}_1 = \log 31 + 2\log 43 + 2\log 34 + \log 16 \approx 20.78$, and $\Delta_1 \approx 7.28$ lies between $G - \mathcal{T}_1 \approx -2.72$ and G.

The numerator terms of (7.9) satisfy

$$\begin{aligned} 0 \le \log(L/M^3) &< 3I\log 2, \\ \log E &\le \log(3kI+1) + 2\bar{H}_N, \\ \log \kappa_N &\le 3N(\bar{w}+\bar{z}-K)\log(5a). \end{aligned} \tag{7.12}$$

For the first bound, $L_i/M_i = 2 - 1/M_i$ lies in $[1,2)$ for every relation, and $L/M^3 = \prod_i (L_i/M_i)^3$. For the second, Lemma 7.4 and $M_r \le 3^{Nz_r}$ give $M_i \le \exp(NH_i) \le \exp(\bar{H}_N)$ for every relation, and therefore $E = 1 + 3k\sum_i (M_i - 1)^2 \le 1 + 3kI\exp(2\bar{H}_N) \le (3kI+1)\exp(2\bar{H}_N)$. For the third, the catalyst has $\sum_{\sigma,r,b} c_{\sigma,r,b} = \bar{w}+\bar{z}-K$ factors by (6.6). Every symbol with positive catalyst multiplicity is consumed by some relation (Proposition 6.2), so it has a representative $[\sigma_r; r, b]$ with $2 \le r \le a$ and $r \le b \le 4r-1$; hence the base of its rank bound $(r+b-1)^3$ satisfies $r+b-1 \le 5a-2 < 5a$. These bounds use only the integer counts.

Without any overhead, and with Δ_N equal to $NK\log F(a,a)$, the formula (7.9) would give the profile exponent

$$\tau_a = \frac{3\log(2a-1)}{\log F(a,a)} \longrightarrow \frac{9}{4}. \tag{7.13}$$

The overheads can only increase the exponent: since $\Delta_N \le NG \le NK\log F(a,a)$ by (7.11) and all three overheads in the numerator of (7.9) are nonnegative, every scheme of this section with $B > A$ satisfies $p \ge \tau_a$. The strict bound $F(a,a) < (2a-1)^{4/3}$ of Theorem 3.2 gives $\tau_a > 9/4$.

8 Explicit rate of convergence and rational seeds

This section fixes the parameters of the construction of Section 7 as functions of the target index a and adds up the bounds of Section 7.5, which gives the rate of Theorem 8.1. It then fixes every remaining choice, identifies the field of coefficients (Theorem 8.4, Remark 8.5), and passes to a rational seed (Corollary 8.7); together these results prove Theorems 1.1 and 1.2.

8.1 An explicit rate of convergence

We bound the terms of the pooled exponent (7.9) by (7.11) and (7.12). Without the rounding loss $\mathcal{L}$, the type penalty $\mathcal{T}_N$ and the three overheads, p would equal the profile exponent τ_a of (7.13); we choose the parameters so that each of these quantities is at most an absolute constant times K. The rounding loss is at most $4J_a$ and the type penalty at $N = 1$ is $O(J_a \log K)$, with $J_a = O(a^3)$, so the rounding scale $K = a^3\ell_a$ makes both $O(K)$ and the type scale $N = 1$ suffices. The copy overhead is $O(I) = O(a^2)$. The interpolation overhead is at most about twice the largest entropy charge of one relation, which is $O(K)$ already for $k = 1$; a larger interpolation power would lower it but enlarge $A = A_1^{kj}$. The catalyst has $O(aK)$ factors, each of rank at most $(5a)^3$, and the reuse count $j = a\ell_a$ brings its overhead down to $O(K)$. Hence the numerator of p is $K(3\log a + O(1))$ and $\Delta_1 \ge K(\frac{4}{3}\log a - O(1))$, so that $p - 9/4 = O(1/\log a)$, like $\tau_a - 9/4$. At small a, larger k, j and K reduce the overheads further (Section 10, which keeps $N = 1$).

Theorem 8.1 (An explicit parameter family)**.** *Let $a \ge 2^{64}$ be an integer, and put*

$$\ell_a = \lceil \log_2 a \rceil, \qquad K = a^3\ell_a, \qquad N = k = 1, \qquad j = a\ell_a.$$

Take the occupations $\lambda_{r,x}, \mu_r$ of the row walk of Proposition 5.3, in which the split at state x has width $h_{r,x} = \min(x, 3r - x)$; round them at scale K as in Section 6, by one of the roundings that Lemma 6.1 allows; and let $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ be the relative scheme of Proposition 7.3 built from these rounded counts. Then $B > A$, and the pooled exponent p and the source side A of this scheme satisfy

$$\frac{9}{4} < \tau_a \le p < \frac{9}{4} + \frac{900}{\log a}, \qquad \log A \le 181 a^5 \ell_a^2. \tag{8.1}$$

In particular, $p \to 9/4$ as $a \to \infty$, and $\log A = O(a^5 (\log a)^2)$.

Proof. We bound Δ_1 from below and each term of the numerator from above, all divided by K. Throughout, $N = k = 1$.

Elementary facts. Since $a \ge 2^{64}$, we have $\log a \ge 64 \log 2 > 32$ and $\ell_a \ge 64$; since $\log_2 a \le \ell_a \le a$, also $\log K \le 4 \log a$ and $(\log a)/\ell_a \le \log 2$. We use the numerical bounds $\log 2 < 7/10$, $\log 4 < 2$, $\log 5 < 2$ and $\log 181 < 6$. The number of transition coordinates satisfies $J_a \le 2a^3$ by (6.8). The number of relations satisfies $I \le 3(a+2)(a-1)/2 \le 2a^2$, because row r has $3r - 1$ interior states and one tripling. Finally, by (6.2), $w_{r,x} < K\lambda_{r,x} + d_{r,x}$ and $z_r < K\mu_r + 1$, where $\lambda_{r,x} < 180$ and $\mu_r \le 2$ by Proposition 5.3, and $d_{r,x} \le \lfloor 3a/2 \rfloor + 1 \le 2a$. Hence $w_i < 180K + 2a$ for every split i and $z_r < 2K + 1$ for every tripling r.

Step 1: a lower bound for Δ_1. We show that $\Delta_1/K > \frac{4}{3} \log a - 7 > 0$. The type penalty (7.10) at $N = 1$ satisfies

$$\mathcal{T}_1 \le J_a \log(180K + 2a + 1) + (a - 1) \log(2K + 2). \tag{8.2}$$

Indeed, the coefficient of $\log(w_i + 1)$ in $\mathcal{T}_1$ is the number of children of split i with positive count, minus one; it is less than d_i, so these coefficients sum to at most J_a. There are at most $a - 1$ triplings, and each contributes $\log(z_r + 1) \le \log(2K + 2)$. Since $180K + 2a + 1 \le 181K$ and $2K + 2 \le 4K$, the two logarithms in (8.2) are less than $6 + 4 \log a$ and $2 + 4 \log a$, respectively. Combining this with the bound $\mathcal{L} \le 4J_a$ on the rounding loss (Proposition 6.3), with $J_a \le 2a^3$, and with $K = a^3 \ell_a$, we obtain

$$\frac{4J_a + \mathcal{T}_1}{K} \le \frac{20 + 8 \log a + \frac{a-1}{a^3}(2 + 4 \log a)}{\ell_a} \le \frac{\frac{41}{2} + 9 \log a}{\ell_a} < \frac{41}{128} + \frac{63}{10}.$$

The middle inequality uses $(a - 1)/a^3 \le 1/a^2 \le 1/4$; the last uses $\ell_a \ge 64$ and $(\log a)/\ell_a \le \log 2 < 7/10$.

The inequality $\log(1 + \xi) \ge \xi - \xi^2/2$ for $\xi \ge 0$, applied with $\xi = 1/(3h)$, together with $\sum_{h=1}^{a-1} 1/h \ge \log a$ and $\sum_{h \ge 1} 1/h^2 < 2$, gives

$$\log s_a \ge \frac{1}{3} \sum_{h=1}^{a-1} \frac{1}{h} - \frac{1}{18} \sum_{h=1}^{a-1} \frac{1}{h^2} \ge \frac{1}{3} \log a - \frac{1}{9}.$$

Since $F(a, a) = s_a(3a - 1)/2 \ge a s_a$, it follows that $\log F(a, a) \ge \frac{4}{3} \log a - \frac{1}{9}$. By (7.11) with $N = 1$ and by Proposition 6.3, $\Delta_1 \ge K \log F(a, a) - \mathcal{L} - \mathcal{T}_1$. Therefore

$$\frac{\Delta_1}{K} \ge \log F(a, a) - \frac{4J_a + \mathcal{T}_1}{K} > \frac{4}{3} \log a - 7 > 0, \tag{8.3}$$

because $\frac{1}{9} + \frac{41}{128} + \frac{63}{10} < 7$ and $\log a > 32$. As $\log(B/A) = kj\Delta_1$, this proves $B > A$.

Step 2: copy overhead. By (7.12) and $I \le 2a^2$,

$$\frac{\log(L/M^3)}{K} < \frac{3I \log 2}{K} \le \frac{6}{a\ell_a} \le 1.$$

Step 3: interpolation overhead. By (7.12), $\log E \le \log(3I+1) + 2\bar{H}_1$, where $\bar{H}_1 = \max_i H_i$ is the largest entropy charge. For a split, $H_i \le w_i \log d_i$, because its empirical child distribution has at most d_i outcomes, and $w_i \log d_i \le 180K + 2a\log(2a)$ by (6.13). For a tripling, $H_r = z_r \log 3 < (2K+1)\log 3$, which is smaller. Hence $\bar{H}_1 \le 180K + 2a\log(2a)$, and

$$\frac{\log E}{K} \le 360 + \frac{\log(6a^2+1) + 4a\log(2a)}{K} \le 360 + \frac{2a+4a^2}{a^3\ell_a} \le 360 + \frac{5}{a\ell_a} \le 361,$$

where we used $\log(6a^2+1) \le 2a$ and $\log(2a) \le a$, both valid for $a \ge 2$.

Step 4: catalyst overhead. By (6.6), the catalyst has $\bar{w} + \bar{z} - K$ factors. The count bounds (6.13), together with $\Gamma_a \le 180a$ and $J_a \le 2a^3$, give

$$\bar{w} + \bar{z} - K \le 180aK + 2a^3 + (2K+1)(a-1) - K \le 182aK + 2a^3 + a < 184aK.$$

Hence (7.12) gives $\log \kappa_1 \le 3 \cdot 184aK \log(5a)$, and with $j = a\ell_a$,

$$\frac{\log \kappa_1}{jK} \le \frac{552\log(5a)}{\ell_a} \le 552\log 2\left(1 + \frac{\log 5}{\log a}\right) < 552 \cdot \frac{7}{10}\left(1 + \frac{2}{8}\right) = 483.$$

The middle inequality uses $\ell_a \ge (\log a)/\log 2$; the last uses $\log 2 < 7/10$, $\log 5 < 2$ and $\log a > 8$.

Step 5: combination. Since $3\log(2a-1) < 3\log a + 3$, Steps 2–4 bound the numerator of p by $K(3\log a + 3 + 1 + 361 + 483) < K(3\log a + 850)$. Together with Step 1, this gives

$$p < \frac{3\log a + 850}{\frac{4}{3}\log a - 7} = \frac{9}{4} + \frac{850 + \frac{63}{4}}{\frac{4}{3}\log a - 7} < \frac{9}{4} + \frac{900}{\log a},$$

because $\frac{4}{3}\log a - 7 \ge \log a$ when $\log a \ge 21$. The fixed-batch bound Φ of (10.1) obeys the same bound, since the extra term $(\log 2)/(kj) < 1 \le K$ of its numerator still leaves it below $K(3\log a + 850)$. The lower bound $p \ge \tau_a > 9/4$ holds for every scheme of Section 7 with $B > A$ (see (7.13)), hence here by Step 1.

Step 6: source side. With $N = 1$, $\log A_1 = W$ is the sum of the logarithms of the auxiliary dimensions, weighted by the split counts. By (6.13), $W \le K\Gamma_a + J_a \log(2a) \le 180a^4\ell_a + 2a^3\log(2a)$, so

$$\log A = kj \log A_1 \le a\ell_a(180a^4\ell_a + 2a^3\log(2a)) = a^5\ell_a^2\left(180 + \frac{2\log(2a)}{a\ell_a}\right) \le 181a^5\ell_a^2,$$

since $\log(2a) \le a$ and $\ell_a \ge 64$. The limit statements follow because $\ell_a = O(\log a)$. □

Corollary 8.2 (An index for a given allowance)**.** *Let ε be a positive rational, and put*

$$\ell = \max\left\{64, \left\lceil \frac{1800}{\varepsilon} \right\rceil\right\}, \qquad a = 2^\ell.$$

The relative scheme of Theorem 8.1 for this a has pooled exponent $p < 9/4 + \varepsilon$, and its source side satisfies $\log A \le 181 \cdot 2^{5\ell}\ell^2$. The index a and the parameters K, N, k, j are computed from ε by integer and rational arithmetic.

Proof. Here $a \ge 2^{64}$ and $\ell_a = \ell$. Since $\log a = \ell \log 2 > \ell/2$, we get $900/\log a < 1800/\ell \le \varepsilon$, and (8.1) gives both claims. □

Remark 8.3 (Calibration). The family of Theorem 8.1 is chosen to keep the proof short, and its bound is weak for moderate a: at $a = 2^{64}$, (8.1) gives only $p < 22.54$; the bound $9/4 + 900/\log a$ is below 3 exactly when $\log a > 1200$; and for $\varepsilon = 1/10$, Corollary 8.2 takes $a = 2^{18000}$. Section 10 obtains much better bounds for small a (Table 2).

8.2 An effective algorithm over a cyclotomic field

Theorem 8.1 allows every rounding of Lemma 6.1. To obtain one algorithm for each allowance, we fix every choice by a rule and record which roots of unity the maps use.

Theorem 8.4 (Effective construction)**.** *There is a terminating procedure that, given a positive rational ε, specifies a finite arithmetic algorithm that multiplies arbitrary complex $n \times n$ matrices in $O_\varepsilon(n^{9/4+\varepsilon})$ operations. All coefficients of the algorithm lie in a cyclotomic field $\mathbb{Q}(\zeta_\Lambda)$ whose order Λ the procedure computes. The procedure uses only finite integer and rational computations, coordinate permutations, tensor products, and the maps of Sections 2 and 4–7.*

Proof. We list the rules in the order in which they are applied, and then collect the roots of unity.

Step 1: parameters. Compute ℓ and a from ε by Corollary 8.2, and K, N, k, j by Theorem 8.1.

Step 2: occupations and rounded counts. Solve the rational linear systems of Proposition 5.3 by exact Gaussian elimination. Round each row separately by Lemma 6.1, using breadth-first (shortest) augmenting paths. List the arcs in the following order, which is also the order in which the search scans the arcs at each vertex (the reverse residual arc of an arc is scanned at that arc's position): the transition edges by increasing state x and then by increasing child y; then the tripling edge $3r \to 1$; then the source and sink arcs by increasing state. From the rounded counts $n_{r,x,y}$ and z_r, form the ledger and the catalyst multiplicities $c_{\sigma,r,b}$ of Proposition 6.2. The examples in Section 10 use the same rule.

Step 3: maps. Order tensor positions and sector words lexicographically. Use the determinant bases (4.2), the interval sectors of Lemma 4.5, the Fourier substitutions of Proposition 4.6, the rank decompositions (2.4), and the root-of-unity averages of Lemma 2.2. Separate the sectors before applying their individual coordinate maps, and compose the maps in the order of the proofs of Propositions 7.1 and 7.3; Appendix B writes out the composed relation maps. This gives the relative scheme of Proposition 7.3.

Step 4: the field of the scheme. The maps use roots of unity of the following orders only: $E = kD + 1$, in the exactification of Lemma 2.2; $2a - 1$, in the rank decomposition (2.4) of $C(a, a)$, through which Proposition 7.3 replaces each source copy of $S^+_{a,a}$ by $(2a-1)^3$ rank-one tensors; $2M_i - 1$ for $i \in \mathcal{J}$, in the separator of relation i; and $r + b - 1$ for every catalyst factor $S^\sigma_{r,b}$ with $c_{\sigma,r,b} > 0$, in the rank decompositions that supply the catalyst (7.5). Every other map has rational coefficients: the determinant bases, the interval sectors, the coordinate permutations, the scalars $1/E$ and $1/(2M_i - 1)$, and the restriction of the output catalyst to a coefficient equal to one; catalyst reuse introduces no further roots of unity. Hence all coefficients of the scheme lie in $\mathbb{Q}(\zeta_\Lambda)$, where

$$\Lambda = \operatorname{lcm}\big(E,\ 2a-1,\ \{2M_i - 1 : i \in \mathcal{J}\},\ \{r + b - 1 : c_{\sigma,r,b} > 0\}\big). \tag{8.4}$$

We represent $\mathbb{Q}(\zeta_\Lambda)$ by the power basis modulo the Λ-th cyclotomic polynomial.

Step 5: the algorithm. By Theorem 8.1 and Corollary 8.2, $2 < p < 9/4 + \varepsilon$, so Lemma 2.4 turns the scheme into an algorithm with $O(n^p) \subseteq O_\varepsilon(n^{9/4+\varepsilon})$ operations. Its zero padding and its recursion introduce no further roots of unity, so all its coefficients lie in $\mathbb{Q}(\zeta_\Lambda)$. The procedure thus specifies finite coefficient arrays and the order of their composition. □

Remark 8.5 (The field for every index). Steps 2–4 of this proof do not use the special form $a = 2^\ell$ of Step 1. Hence, for every integer $a \geq 2^{64}$, the relative scheme of Theorem 8.1, built with the rounding rule of Step 2 and the orders of Step 3, is produced from a by a terminating procedure, and its coefficients lie in $\mathbb{Q}(\zeta_\Lambda)$, with Λ given by (8.4). Together with Theorem 8.1, this proves Theorem 1.2.

8.3 Rational coefficients and a square seed

Lemma 8.6 (Restriction of coefficients to the rationals)**.** *Let T be a tensor with rational coefficients, and suppose that a decomposition of T into R rank-one tensors is given over a number field $\mathbb{K}$ of degree $g = [\mathbb{K} : \mathbb{Q}]$. Then the construction below gives a decomposition of T into at most $g^2 R$ rank-one tensors with rational coefficients.*

Proof. Choose a basis $\theta_1 = 1, \theta_2, \dots, \theta_g$ of $\mathbb{K}$ over $\mathbb{Q}$ (for $\mathbb{Q}(\zeta_\Lambda)$, the power basis, in which all field operations are effective), and let $\varphi\colon \mathbb{K} \to \mathbb{Q}$ be the $\mathbb{Q}$-linear map that extracts the coefficient of θ_1. In one summand of the decomposition, write the first two linear forms as $\sum_s \theta_s \alpha_s$ and $\sum_{s'} \theta_{s'} \beta_{s'}$, where the forms α_s and $\beta_{s'}$ have rational coefficients, and call the third form γ. Apply φ to every coefficient of the tensor identity. The summand becomes

$$\sum_{s,s'=1}^{g} \alpha_s \otimes \beta_{s'} \otimes \varphi(\theta_s \theta_{s'} \gamma),$$

where φ acts on the coefficients of γ. This is a sum of at most g^2 rank-one tensors with rational coefficients; terms with $\alpha_s = 0$ or $\beta_{s'} = 0$ are omitted, so a summand with rational coefficients is left unchanged. The tensor T itself is unchanged, because its coefficients are rational and φ is the identity on $\mathbb{Q}$. □

Corollary 8.7 (Rational seeds)**.** *For every positive rational ε, a terminating procedure produces an integer $d_0 \geq 2$ and a decomposition of M_{d_0} into $R_{d_0} < d_0^{9/4+\varepsilon}$ rank-one tensors with rational coefficients. Consequently, a fixed rational bilinear recursion attains an exponent strictly below $9/4 + \varepsilon$.*

Proof. Apply Theorem 8.4 with allowance $\varepsilon/2$. For each side n, unroll the pooled recursion of Lemma 2.4 down to its classical leaves, with the zero padding of that lemma. By the last assertion of Lemma 2.4, every leaf multiplication multiplies a linear form in the entries of the left input by a linear form in the entries of the right input. The unrolled algorithm is therefore a decomposition of M_n into rank-one tensors, one for each leaf multiplication. Their number is at most the arithmetic cost, hence $O_\varepsilon(n^{9/4+\varepsilon/2})$, and their coefficients lie in $\mathbb{Q}(\zeta_\Lambda)$, a field that does not depend on n. Apply Lemma 8.6, and let R_n be the number of rank-one tensors after descent. Since the degree $g = [\mathbb{Q}(\zeta_\Lambda) : \mathbb{Q}]$ does not depend on n, we have $R_n \leq g^2 \cdot O_\varepsilon(n^{9/4+\varepsilon/2}) = O_\varepsilon(n^{9/4+\varepsilon/2})$.

Write $9/4 + \varepsilon = \alpha_\varepsilon/\beta_\varepsilon$ with positive integers $\alpha_\varepsilon, \beta_\varepsilon$. For $n = 2, 3, \dots$, compute R_n and stop at the first n with $R_n^{\beta_\varepsilon} < n^{\alpha_\varepsilon}$; this n is d_0. The loop terminates because the exponent of the bound on R_n is smaller than $9/4 + \varepsilon$ by $\varepsilon/2$. Ordinary block recursion on the seed has exponent $\log_{d_0} R_{d_0} < 9/4 + \varepsilon$; its fixed linear maps cost $O(n^2)$ operations per node. □

Remark 8.8 (Size of the seed). Suppose that $\varepsilon \leq 3/4$, the only case of interest, since classical multiplication has exponent 3. Let A and B be the sides of the scheme that the proof of Corollary 8.7 uses, for the allowance $\varepsilon/2$. Every side $n \leq A$ lies below the leaf threshold $\lceil A/(1 - A/B) \rceil + 1$ of Lemma 2.4, so for such n the unrolled algorithm is the classical one, with $n^3 \geq n^{9/4+\varepsilon}$ rational products, which descent leaves unchanged; hence $d_0 > A$. Moreover $A = A_1^{kj} = A_1^j \geq 2^j$ with $j = a\ell_a \geq 2^{70}$, so $d_0 > 2^{2^{70}}$; here $A_1 \geq 2$ because, by (6.1), the splits that enter state 0 of a row have total count K, so some split is active. Since the seed has rational coefficients, its identity holds over every field of characteristic zero. Theorem 9.2 gives an explicit upper bound for the side of a seed obtained from the rational variant of the construction.

9 Explicit bounds for a rational seed

Corollary 8.7 finds its seed side by testing $n = 2, 3, \dots$ in turn, and it bounds neither the side nor the coefficients of the seed. This section makes both explicit for a variant of the construction with rational maps (Sections 9.1–9.3), then bounds the bit cost and the precision of recursion with any fixed rational seed (Section 9.4) and combines the results into one algorithm for all sizes (Section 9.5).

9.1 A rational variant of the construction

The construction uses roots of unity in the rank decompositions (2.4) of convolution tensors, in the separators and in exactification (Step 4 of the proof of Theorem 8.4). In the first and the last they extract one coefficient of a polynomial of bounded degree, which interpolation at the integer nodes $1, 2, \dots$ extracts with rational weights. For the separators we use the polynomial separator of Lü [18, Proposition 2.1] (Remark 4.7), which has the same $2M - 1$ source copies as Proposition 4.6. For positive integers n and $1 \le \ell \le n$, let

$$\varpi_{n,\ell}(\xi) = \prod_{\substack{1 \le \ell' \le n \\ \ell' \ne \ell}} \frac{\xi - \ell'}{\ell - \ell'} = \sum_{\gamma=0}^{n-1} \varpi_{n,\ell,\gamma}\, \xi^{\gamma}$$

be the ℓ-th Lagrange basis polynomial of the nodes $1, \dots, n$, and write $[\xi^{\gamma}]$ for the coefficient of ξ^{γ}. Thus $f(\xi) = \sum_{\ell=1}^{n} f(\ell)\, \varpi_{n,\ell}(\xi)$ for every polynomial f of degree less than n with coefficients in a vector space over $\mathbb{Q}$.

Lemma 9.1 (Rational interpolation)**.**

(a) For positive integers r, b and $n = r + b - 1$,

$$C(r,b) = \sum_{\ell=1}^{n} \Big(\sum_{\alpha=0}^{r-1} \ell^{\alpha} x_{\alpha} \Big) \Big(\sum_{\beta=0}^{b-1} \ell^{\beta} y_{\beta} \Big) \Big(\sum_{\gamma=0}^{n-1} \varpi_{n,\ell,\gamma} z_{\gamma} \Big). \tag{9.1}$$

(b) (The separator of Remark 4.7 for $K_0 = M$.) *Let $M \ge 1$, and let $T = \sum_{g=0}^{M-1} T_g$ be as in Proposition 4.6, with the coordinates and target variables of its proof. Index the source copies of $(2M-1)\,T$ by $\ell = 1, \dots, 2M-1$, and in copy ℓ substitute, with $0^0 = 1$,*

$$x_{\alpha} \longmapsto \sum_{g'=0}^{M-1} \varpi_{M,g'+1}(\ell)\, X_{\alpha,g'}, \qquad y_{g,\beta} \longmapsto \sum_{e=0}^{M-1} (\ell - g - 1)^{e}\, t^{-e}\, Y_{g,\beta,e},$$

$$z_{g,\gamma} \longmapsto \sum_{e'=0}^{M-1} \big([\xi^{e'}]\, \varpi_{2M-1,\ell}(g + 1 + \xi) \big)\, t^{e'}\, Z_{g,\gamma,e'}.$$

These maps have rational coefficients and give a degeneration

$$(2M-1)\,T \trianglerighteq \bigoplus_{g=0}^{M-1} T_g\, B_X(M)$$

of degree at most $M-1$, whose target variables and constant term are those of Proposition 4.6. Permuting the three legs gives the same statement with any chosen shared leg.

(c) In Lemma 2.2, the restriction $E\,T^{\otimes k} \succeq \widetilde{T}_0^{\otimes k}$ is also given by the maps $\Psi(\ell)^{\otimes k}$, $1 \le \ell \le E$, and the scalars $(-1)^{\ell-1}\binom{E}{\ell}$ in place of $1/E$.

Proof. The idea is that each identity recovers a polynomial of degree less than the number of nodes, or one of its coefficients, from its values at the nodes, which interpolation does exactly.

(a) The product $P(\xi) = (\sum_\alpha x_\alpha \xi^\alpha)(\sum_\beta y_\beta \xi^\beta)$ has degree less than n, and its coefficient of ξ^γ is $\sum_{\alpha+\beta=\gamma} x_\alpha y_\beta$. Interpolation gives this coefficient as $\sum_\ell P(\ell)\varpi_{n,\ell,\gamma}$. Multiplying by z_γ and summing over γ gives (9.1), by (2.3).

(b) We write the proof of Lü [18] in our coordinates: copy ℓ evaluates the shared leg at the node ℓ through the Lagrange basis of the nodes $1,\dots,M$, and expands the other two legs around the node $g+1$ of their sector, with t scaling the distance from that node. After summing over the copies, the term $x_\alpha y_{g,\beta} z_{g,\gamma}$ becomes

$$\sum_{g',e,e'} t^{e'-e}\Big([\xi^{e'}] \sum_{\ell=1}^{2M-1} f(\ell)\,\varpi_{2M-1,\ell}(g+1+\xi)\Big) X_{\alpha,g'} Y_{g,\beta,e} Z_{g,\gamma,e'},$$

where $f(\xi) = \varpi_{M,g'+1}(\xi)\,(\xi-g-1)^e$. Since f has degree at most $2M-2$, interpolation at the nodes $1,\dots,2M-1$ gives $\sum_\ell f(\ell)\varpi_{2M-1,\ell}(\xi) = f(\xi)$. Substituting $g+1+\xi$ for ξ and using $f(g+1+\xi) = \varpi_{M,g'+1}(g+1+\xi)\,\xi^e$, we find that the coefficient in parentheses is $[\xi^{e'-e}]\,\varpi_{M,g'+1}(g+1+\xi)$, which vanishes unless $0 \le e'-e \le M-1$, because $\varpi_{M,g'+1}$ has degree $M-1$. Hence the image of $(2M-1)\,T$ is a polynomial in t of degree at most $M-1$. Its constant term consists of the terms with $e'=e$, whose coefficient $\varpi_{M,g'+1}(g+1)$ is one if $g'=g$ and zero otherwise; this is the constant term of Step 3 of the proof of Proposition 4.6.

(c) For a polynomial f of degree less than E, with coefficients in any vector space, the E-th finite difference $\sum_{\ell=0}^{E}(-1)^\ell\binom{E}{\ell}f(\ell)$ vanishes, so $f(0) = \sum_{\ell=1}^{E}(-1)^{\ell-1}\binom{E}{\ell}f(\ell)$. In the proof of Lemma 2.2, apply this identity to $f = \widetilde{T}(t)^{\otimes k}$, which has degree at most $kD = E-1$, in place of the average over the E-th roots of unity. The nodes are nonzero, so the Laurent entries of Ψ can be evaluated there, and the weights go on one leg of each copy. □

We now build the relative scheme of Proposition 7.3 with $N=k=1$, as in Theorem 8.1, but use part (a) of Lemma 9.1 for every rank decomposition of a convolution tensor (in Step 2 of the proof of Proposition 7.3 and for the catalyst (7.5)), part (b) for every separator, and part (c) for exactification. Part (b) has the source, the target variables and the constant term of Proposition 4.6, and the proofs of Corollary 4.8 and Proposition 7.1 use the separator only through these and its degree bound; in particular, the reindexing (B.6) is defined on all target variables, as Lemma 2.1 requires. Appendix B writes these maps out for Proposition 4.6; with part (b), terms of positive degree may also have $g'=g$, and their exponents are $1,\dots,M-1$ instead of δ^2, but the same fixed coordinate maps apply to them. Since $M-1 \le (M-1)^2$, these proofs give (7.4) for the new maps with the same L, M and D as in (7.2) and (7.3); D is now only an upper bound for the degree, which we keep so that the counts coincide. The proof of Proposition 7.3, with $E = D+1$ and with the decompositions of part (a), which have the same numbers of terms as (2.4), then gives $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ with the counts (7.6), and every other map of the construction has rational coefficients (Step 4 of the proof of Theorem 8.4). Thus the *rational variant* has rational coefficients and, for the same rounded counts, the same r_*, m_*, A and B as the scheme of Theorem 8.1; only its maps differ. Since p is a function of these numbers, Theorem 8.1 shows that the rational variant is a relative scheme whose pooled exponent p and source side A satisfy (8.1).

9.2 An explicit seed

We now unroll the rational variant at a side given by a formula. Let ε be a positive rational, put $\varepsilon' = \min(\varepsilon, 1/2)$, and let $a = 2^{\ell_a}$, with $\ell_a = \lceil 3600/\varepsilon' \rceil \geq 7200$, be the index that Corollary 8.2 assigns to the allowance $\varepsilon'/2$. The cap $\varepsilon' \leq 1/2$ makes the pooled exponent less than $5/2$, which Section 9.3 uses. Build the rational variant $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ with the parameters of Theorem 8.1 for this a, and with any rounding allowed by Lemma 6.1. Let $n_* = \lceil AB/(B-A) \rceil + 1$ be its leaf threshold (Lemma 2.4), and put

$$\Xi = r_*(1 + r_* m_*)\, n_*^3, \qquad d_0 = (\Xi + 1)^{\lceil 2/\varepsilon' \rceil}. \tag{9.2}$$

Theorem 9.2 (An explicit seed)**.** *Unrolling the pooled recursion of Lemma 2.4 for this scheme at side d_0 gives a rational seed of side d_0 with R products, where*

$$R \leq \Xi d_0^p < d_0^{9/4+\varepsilon'} \leq d_0^{9/4+\varepsilon}.$$

With the rounding rule fixed in Section 9.3 and block recursion (Lemma 9.4), this gives a second proof of Theorem 1.1, with the seed side given by (9.2) instead of a search.

Proof. The idea is that the unrolled seed has one rank-one term for each scalar product at the last depth of the pooled recursion, and Step 3 of the proof of Lemma 2.4 bounds their number. As in the proof of Corollary 8.2, $\log a = \ell_a \log 2 > \ell_a/2$ gives $900/\log a < 1800/\ell_a \leq \varepsilon'/2$, so (8.1), which holds for the rational variant (Section 9.1), gives $p < 9/4 + \varepsilon'/2$.

Run the pooled recursion at side d_0, put $\rho = A/B$, and let $\ell_{\max}$ be its last depth. Step 3 of the proof of Lemma 2.4 gives, also when $\ell_{\max} = 0$,

$$\nu_{\ell_{\max}} \leq \Big(1 + \frac{r_* m_*}{r_* - m_*}\Big)\Big(\frac{r_*}{m_*}\Big)^{\ell_{\max}}, \qquad \rho^{\ell_{\max}} d_0 > \rho.$$

Since $r_*/m_* = \rho^{-p}$, the second inequality gives $(r_*/m_*)^{\ell_{\max}} < (r_*/m_*)\, d_0^p$. With $r_* - m_* \geq 1$ and $m_* \geq 1$, it follows that $\nu_{\ell_{\max}} < r_*(1 + r_* m_*)\, d_0^p$. Every product at the last depth has side at most n_*, and the classical algorithm computes it with at most n_*^3 scalar products, each of a linear form in the left input by a linear form in the right input (Lemma 2.4). As in the proof of Corollary 8.7, these products give a decomposition of M_{d_0} into $R \leq \nu_{\ell_{\max}} n_*^3 \leq \Xi d_0^p$ rank-one tensors, with rational coefficients because the scheme has rational coefficients. Finally, $d_0^{\varepsilon'/2} \geq \Xi + 1 > \Xi$ and $p < 9/4 + \varepsilon'/2$ give $R < d_0^{9/4+\varepsilon'}$. □

9.3 The cost of computing the seed

To make the seed of Theorem 9.2 a function of ε, we round by the rule of Step 2 of the proof of Theorem 8.4. We count *steps*: one step is an arithmetic operation or a comparison on two rational numbers, each stored as a reduced pair of binary integers, or one operation on an integer index or loop counter. Let b_ε be the total bit length of the numerator and denominator of ε. It is not bounded by any function of the value of ε, and it enters the bounds through the bit lengths of the numbers computed from ε (Step 5 of the proof below). With the parameters of Section 9.2, put

$$\mathfrak{T} = \bar{w} + \bar{z} + K + I + 1, \qquad \mathfrak{F} = 100(j+1)\mathfrak{T},$$
$$\mathfrak{W} = 2 + b_\varepsilon + a + K + j + \bar{w} + \bar{z} + I + E + \kappa_1 + r_* + m_* + A + B + \sum_{i \in \mathcal{J}} (M_i + L_i),$$
$$\mathfrak{H} = \mathfrak{W}^{\mathfrak{F}}, \qquad \mathfrak{Q} = \mathfrak{H} + d_0 + 2, \qquad \mathfrak{S} = \mathfrak{Q}^{50}.$$

These numbers are computed from ε by the formulas above, the occupation systems of Proposition 5.3 and the rounding rule.

Theorem 9.3 (Preprocessing cost)**.** *The dense implementation described in the proof computes the seed of Theorem 9.2 from ε in at most $\mathfrak{S}$ steps, and outputs its three rational coefficient arrays, which have at most $3\Xi d_0^5$ entries. Every reduced numerator and denominator that it computes has bit length at most $2^{2\mathfrak{S}}$. With schoolbook integer arithmetic, the whole computation, including writing the arrays, takes $O(2^{8\mathfrak{S}})$ bit operations, with an absolute implied constant.*

Proof. The idea is to fix one simple implementation and to bound every count, dimension and exponent that it uses by $\mathfrak{W}$, every array of the assembled scheme by $\mathfrak{H}$, and the arrays of the unrolled recursion by $\mathfrak{Q}^{14}$; the bit lengths then follow because one rational operation at most doubles them, plus one bit.

Step 1: the implementation. Store rational numbers as reduced pairs of binary integers. Use exact Gaussian elimination for the occupations and for basis changes, the rounding rule of Theorem 8.4 for the counts, repeated multiplication for factorials and binomial coefficients, and binary powering for integer powers. Compute the interpolation coefficients $\varpi_{n,\ell,\gamma}$ by polynomial multiplication, the separator coefficients of Lemma 9.1(b) from them by evaluation and by the identity $[\xi^{e'}]\varpi_{2M_i-1,\ell}(g+1+\xi) = \sum_{s\ge e'}\binom{s}{e'}(g+1)^{s-e'}\varpi_{2M_i-1,\ell,s}$, and the primitive map entries by evaluating Laurent monomials at the integer nodes. Form tensor products, direct sums, restrictions and compositions as dense arrays, expanding after every operation and exploiting no sparsity.

Step 2: the primitive data. There are $a-1$ row systems, each of order at most $3a$. By the proof of Lemma 6.1, the rounding of a row makes at most β_{tot} augmentations. Since β_{tot} is at most the sum of the fractional parts η_e of that proof, it is less than the number of nonintegral edges of the row, so each row needs at most J_a+1 augmentations, each a breadth-first search over fewer than J_a+3a+3 arcs. Every factorial or binomial argument (a split or tripling count, E, or an interpolation size M_i, L_i or $r+b-1$) is at most $\mathfrak{W}$, determinant-basis matrices have size at most $10a^2$, and primitive Laurent exponents are less than $\max_i M_i$ in absolute value. The separator of relation i has $(2M_i-1)M_i(2M_i+1) < 4M_i^3$ coefficients, computed in fewer than $\mathfrak{W}^5$ steps. Hence generating these data, and computing the numbers $\mathfrak{T},\dots,\mathfrak{S}$, takes fewer than $\mathfrak{W}^{20}$ steps.

Step 3: dense assembly. One catalytic use has $3(\bar w+\bar z)$ convolution positions and $3\bar w$ dot-product positions, at most $6\mathfrak{T}$ primitive factors in total. Each leg of a primitive factor, including the combined spaces on which the determinant bases act, has dimension at most $10a^2 \le \mathfrak{W}^3$. The source-copy labels of the separators, the sector and auxiliary labels of their outputs, and the exactification label add at most $3I+6I+1 \le 10\mathfrak{T}$ further factors, each of dimension at most $\mathfrak{W}$. Counting all of them at once, every leg of a local intermediate array has dimension at most $\mathfrak{W}^{18\mathfrak{T}+10\mathfrak{T}} = \mathfrak{W}^{28\mathfrak{T}}$. The rank decompositions of the source and of the catalyst have at most $3\mathfrak{T}$ convolution positions and $3\mathfrak{T}$ rank labels, so their legs, with the matrix and copy labels, have dimension at most $\mathfrak{W}^{12\mathfrak{T}+3}$. During reuse the catalyst stays a single factor (Step 3 of the proof of Proposition 7.3). The identity factors on matrix tensors add at most four powers of $\mathfrak{W}$, since $A^2B^2 \le \mathfrak{W}^4$, and the copy counts add at most two more, since by (7.7) they are at most $(r_*/\kappa_1)\,m_* \le r_*m_* \le \mathfrak{W}^2$. Hence every intermediate leg has dimension at most $\mathfrak{W}^{28\mathfrak{T}+6} \le \mathfrak{W}^{34\mathfrak{T}}$.

Before its direct-sum labels are expanded, one catalytic use takes at most $40\mathfrak{T}$ dense assembly stages: two per primitive factor for basis changes and coordinate selection (at most $12\mathfrak{T}$), at most $3I$ oriented separators, at most $3I$ reindexings, at most $16\mathfrak{T}$ tensor products or direct sums, which combine the primitive factors and the labels, and three stages for exactification and the ledger. Expanding the labels multiplies the number of stages by at most the number of summands, which is at most $\mathfrak{W}^{34\mathfrak{T}}$. The j reuse steps and the source and catalyst decompositions therefore take at most $40(j+1)\mathfrak{T}\,\mathfrak{W}^{34\mathfrak{T}} \le \mathfrak{W}^{40(j+1)\mathfrak{T}}$ stages. Both the dimensions and the number of stages are below $\mathfrak{H} = \mathfrak{W}^{100(j+1)\mathfrak{T}}$, so the assembly, including generating and copying entries, takes fewer than $\mathfrak{H}^{20}$ steps.

Step 4: unrolling. The assembled scheme is a restriction given by its maps, not yet a rank decomposition of M_{d_0}, so we unroll the pooled recursion at side d_0. We have $n_* \le AB + 2 \le \mathfrak{W}^3$ and $\Xi \le \mathfrak{W}^{13} \le \mathfrak{H}$. Since $\nu_{\ell+1} = r_* \lceil \nu_\ell / m_* \rceil \ge (r_*/m_*)\nu_\ell > \nu_\ell$, every depth has at most $\nu_{\ell_{\max}} < \Xi d_0^p \le \Xi d_0^3 < \mathfrak{Q}^4$ products (proof of Theorem 9.2), each of side at most d_0. Above the leaf threshold the side strictly decreases, since $A\lceil n/B \rceil < (A/B)n + A < n$ for $n > AB/(B-A)$, so there are at most d_0 depths. Padding adds at most m_* zero products per depth and at most B to each side. Hence each depth has at most $(\nu_\ell + m_*)(n_\ell + B)^2 \le 2\mathfrak{Q}^6$ coordinates per input matrix, and all coordinate spaces at one depth, including the intermediate source products, have dimension less than $\mathfrak{Q}^{14}$. Propagate the linear forms of the two inputs downward, and the output forms upward, by dense matrix multiplication, as linear forms in the d_0^2 entries of each original input; each depth costs at most a fixed multiple of $\mathfrak{Q}^{42}$ steps. Each classical product at the last depth then selects two downward forms and a column of the upward map, which gives one rank-one term. The at most d_0 depths, the assembly of Step 3, and writing the $3Rd_0^2 \le 3\Xi d_0^5$ output entries take fewer than $\mathfrak{Q}^{50} = \mathfrak{S}$ steps.

Step 5: bit lengths. Write $\mathfrak{b}(\theta)$ for the larger of the bit lengths of the reduced numerator and denominator of a rational number θ. A sum, difference, product or defined quotient of two rational numbers θ and θ' has $\mathfrak{b}$ at most $2\max\{\mathfrak{b}(\theta), \mathfrak{b}(\theta')\} + 1$. All initial integer literals, including ε, can be chosen with at most $\mathfrak{Q}$ bits, since $b_\varepsilon \le \mathfrak{W}$; factorials, powers and interpolation coefficients are computed by the counted operations. After at most $\mathfrak{S}$ steps, every rational number therefore has

$$\mathfrak{b} \le 2^{\mathfrak{S}}(\mathfrak{Q} + 1) - 1 \le 2^{2\mathfrak{S}}.$$

Schoolbook multiplication, division and Euclidean gcd on integers of this length cost $O(2^{6\mathfrak{S}})$ bit operations per step. Including loop control, indexing and output, the total is $O(2^{8\mathfrak{S}})$. □

9.4 Recursion with a fixed rational seed

Here we fix one rational seed, for instance that of Corollary 8.7 or of Theorem 9.2, and bound two other costs of block recursion with it: the bit operations needed for integer inputs (Proposition 9.5) and a precision that suffices for a prescribed forward error (Proposition 9.6). Let the seed have side $d_0 \ge 2$ and R products, and put $p_{\text{seed}} = \log_{d_0} R$; this is not the pooled exponent p. Write the seed as

$$\operatorname{vec}(XY) = \mathsf{W}((\mathsf{U} \operatorname{vec} X) \odot (\mathsf{V} \operatorname{vec} Y)), \qquad \mathsf{U}, \mathsf{V} \in \mathbb{Q}^{R \times d_0^2}, \quad \mathsf{W} \in \mathbb{Q}^{d_0^2 \times R},$$

for $d_0 \times d_0$ matrices X and Y, where $\odot$ is the entrywise product. *Block recursion* multiplies $n \times n$ matrices by padding them with zeros to side d_0^ℓ, where $\ell = \lceil \log_{d_0} n \rceil$, applying this identity to the $d_0 \times d_0$ arrays of blocks, computing the R block products recursively, and cropping the result to its leading $n \times n$ block. Constants in O_{seed} depend only on the seed.

Lemma 9.4 (Block recursion with a rational seed)**.** *We have $R > d_0^2$, and block recursion multiplies $n \times n$ matrices with at most $c_{\text{seed}} n^{p_{\text{seed}}}$ arithmetic operations, where $c_{\text{seed}} = (1 + 6Rd_0^2)R$.*

Proof. The rows of U are the left forms of the seed, linear forms in the entries of X. The flattening of M_{d_0} along leg 1 has rank d_0^2 (Step 1 of the proof of Lemma 2.4), and its image lies in the span of the left forms, so $R \ge d_0^2$. If equality held, the left forms would be a basis of the dual space, so some nonzero X would make $\mathsf{U} \operatorname{vec} X$ the first unit vector. Then $\operatorname{vec}(XY)$ would be a multiple of the first column of W for every Y, so the map $Y \mapsto XY$ would have rank at most one, whereas its rank is $d_0 \operatorname{rank}(X) \ge d_0 \ge 2$. Hence $R > d_0^2$.

At a node of side N, a multiple of d_0, forming the R left and the R right block combinations costs at most $2R(2d_0^2 - 1)(N/d_0)^2$ operations, and forming the d_0^2 output blocks costs at most

$d_0^2(2R-1)(N/d_0)^2$; together this is less than $6RN^2$. Let $\theta_{\ell'}$ be the cost at side $d_0^{\ell'}$. Then $\theta_0 = 1$ and $\theta_{\ell'} \le R\theta_{\ell'-1} + 6Rd_0^{2\ell'}$, so, using $R - d_0^2 \ge 1$,

$$\theta_\ell \le R^\ell + 6Rd_0^{2\ell} \sum_{\ell'=0}^{\ell-1} \Big(\frac{R}{d_0^2}\Big)^{\ell'} \le (1 + 6Rd_0^2)R^\ell.$$

Since $d_0^\ell < d_0 n$, we have $R^\ell = (d_0^\ell)^{p_{\text{seed}}} < (d_0 n)^{p_{\text{seed}}} = Rn^{p_{\text{seed}}}$. □

For the integer bound, choose a positive integer c_{den} that clears all denominators of U, V and W, and put

$$c_{\text{coef}} = \max\{1, \max|c_{\text{den}}\mathsf{U}|, \max|c_{\text{den}}\mathsf{V}|, \max|c_{\text{den}}\mathsf{W}|\},$$
$$c_{\text{in}} = \lceil \log_2(d_0^2 c_{\text{coef}}) \rceil, \qquad c_{\text{out}} = \lceil \log_2(R c_{\text{coef}}) \rceil,$$

where $\max|\cdot|$ is the largest absolute entry.

Proposition 9.5 (Integer bit complexity). *Let $n \ge 2$, $\ell = \lceil \log_{d_0} n \rceil$ and $b_{\text{in}} \ge 1$. Block recursion with the integer matrices $c_{\text{den}}\mathsf{U}$, $c_{\text{den}}\mathsf{V}$, $c_{\text{den}}\mathsf{W}$, followed by one exact division of every entry by $c_{\text{den}}^{3\ell}$, multiplies two $n \times n$ integer matrices whose entries have absolute value below $2^{b_{\text{in}}}$ exactly, with intermediate integers of bit length at most*

$$2b_{\text{in}} + \ell(2c_{\text{in}} + c_{\text{out}}) + 2,$$

including a sign bit, and with $O_{\text{seed}}(n^{p_{\text{seed}}}(b_{\text{in}} + \log n)^2)$ bit operations, using schoolbook integer arithmetic. The cost of constructing the seed is not included.

Proof. Use the integer matrices $c_{\text{den}}\mathsf{U}$, $c_{\text{den}}\mathsf{V}$, $c_{\text{den}}\mathsf{W}$ at every level, without intermediate divisions. One level then computes c_{den}^3 times the product, so by induction on the depth the root computes $c_{\text{den}}^{3\ell}XY$; divide every entry of the result by $c_{\text{den}}^{3\ell}$ at the end. These divisions are exact.

The absolute row sums of $c_{\text{den}}\mathsf{U}$ and $c_{\text{den}}\mathsf{V}$ are at most $d_0^2 c_{\text{coef}}$, so each input-transform level multiplies the largest absolute entry by at most $d_0^2 c_{\text{coef}}$, and a leaf product has absolute value below $2^{2b_{\text{in}}}(d_0^2 c_{\text{coef}})^{2\ell}$. Each output-transform level multiplies the bound by at most Rc_{coef}. Hence every intermediate value, including partial sums and products by coefficients, has absolute value below $2^{2b_{\text{in}}}(d_0^2 c_{\text{coef}})^{2\ell}(Rc_{\text{coef}})^\ell \le 2^{2b_{\text{in}}+\ell(2c_{\text{in}}+c_{\text{out}})}$, so the stated bit length suffices, with one bit to spare. Evaluating the one-level integer identity on all-one inputs gives $c_{\text{den}}^3 d_0 \le Rd_0^4 c_{\text{coef}}^3$, so the divisor $c_{\text{den}}^{3\ell}$ also has this bit length. By Lemma 9.4, the recursion uses $O_{\text{seed}}(n^{p_{\text{seed}}})$ integer operations, and forming the divisor and dividing take $O(n^2 + \ell)$ more. Since $\ell = O(\log n)$, each operation costs $O_{\text{seed}}((b_{\text{in}} + \log n)^2)$ bit operations. □

For entries with a bounded number of bits, the cost is $O_{\text{seed}}(n^{p_{\text{seed}}} \log^2 n)$. For the seed of Theorem 9.2 or Corollary 8.7 for a rational $\varepsilon > 0$, we have $p_{\text{seed}} < 9/4 + \varepsilon$, so such matrices are multiplied with $O_\varepsilon(n^{9/4+\varepsilon})$ bit operations. Rational inputs can be handled by first clearing one common denominator for each matrix, with b_{in} bounding the resulting integer entries; if the n^2 entries have unrelated b_{in}-bit denominators, this common denominator may need up to $n^2 b_{\text{in}}$ bits.

For the precision bound, write $\|Z\|_{\max}$ for the largest absolute entry of a matrix Z and $\|Z\|_\infty$ for its largest absolute row sum, and put $\Upsilon = \|\mathsf{U}\|_\infty\|\mathsf{V}\|_\infty\|\mathsf{W}\|_\infty$.

Proposition 9.6 (A sufficient precision). *Let X and Y be $n \times n$ matrices with $n \ge 2$, and put*

$$\ell = \lceil \log_{d_0} n \rceil, \qquad \chi = \ell(2d_0^2 + R + 3) + 1, \qquad \gamma_\chi = \frac{\chi u}{1 - \chi u}.$$

Compute XY by block recursion, evaluating every entry $\sum_k c_k w_k$ of each linear combination by recursive summation $(\cdots((c_1w_1 + c_2w_2) + c_3w_3) + \cdots)$ in a fixed order of k, and the leaves as scalar products. Suppose that each arithmetic operation and each stored nonzero coefficient has relative error at most u, that zeros are stored exactly, and that no operation underflows or overflows. If $\chi u < 1$, the computed product $\widehat{XY}$ satisfies

$$\|\widehat{XY} - XY\|_{\max} \le \gamma_\chi \Upsilon^\ell \|X\|_{\max}\|Y\|_{\max}. \tag{9.3}$$

Here X and Y are the stored inputs; errors in storing the original inputs are not counted.

Proof. The idea is to expand each computed entry into bilinear monomials and to count the rounding factors along one path of the recursion. Expand a computed output entry into signed bilinear monomials in the entries of the padded inputs, each carrying the factors $1 + \delta$ of the operations that produced it. At each input-transform level, a monomial picks up one factor from storing a coefficient, one from multiplying by it, and at most $d_0^2 - 1$ from additions, so its two input paths contribute at most $2\ell(d_0^2 + 1)$ factors. Each output-transform level contributes at most $R + 1$ factors, and the leaf product one. The total is at most χ, also when a stored coefficient is reused, and a product of at most χ factors $1 + \delta$ with $|\delta| \le u$ differs from one by at most γ_χ. Applying the three row-sum bounds at each level shows that the absolute values of the exact monomials sum to at most $\Upsilon^\ell \|X\|_{\max}\|Y\|_{\max}$. The triangle inequality proves (9.3). □

Applying the seed to all-one inputs shows $\Upsilon \ge d_0 \ge 2$. Hence, for $0 < \vartheta \le 1$, the condition

$$u \le \frac{\vartheta}{2\chi\Upsilon^\ell}$$

gives $\chi u \le 1/4$ and an error of at most $\vartheta\|X\|_{\max}\|Y\|_{\max}$. In binary floating point with t_{sig} significand bits, including the leading bit, the unit roundoff is $2^{-t_{\text{sig}}}$ for round-to-nearest and at most $2^{1-t_{\text{sig}}}$ for directed rounding. Hence the condition holds with

$$\left\lceil \log_2 \frac{2\chi}{\vartheta} \right\rceil + \ell\lceil \log_2 \Upsilon \rceil + 2$$

significand bits, that is, with $O_{\text{seed}}(\log n + \log(1/\vartheta))$ bits. Demmel et al. [10], following Bini and Lotti [4], give a forward error analysis of this kind for a large class of recursive matrix multiplication algorithms; Proposition 9.6 states the bound with explicit constants for one rational seed and dot products evaluated from left to right.

9.5 One algorithm for all sizes

The algorithms of Section 2.4 depend on a fixed allowance ε. Letting the allowance shrink with the input size, and counting the preprocessing, gives one algorithm for all sizes.

Corollary 9.7 (One algorithm for all sizes)**.** *There is one algorithm that, for every n, multiplies two $n \times n$ matrices over a field of characteristic zero in $n^{9/4+o(1)}$ steps, where a step is one step of a fixed multitape Turing machine during the preprocessing, or one addition, subtraction, multiplication or division in the field on matrix entries or seed coefficients.*

Proof. The idea is to try the allowances $1/v$, $v = 1, 2, \ldots$, within a preprocessing budget of n machine steps each, and to use the smallest allowance that succeeds. At side n, for each integer $1 \le v \le \lfloor \log_2 n \rfloor$, run a fixed Turing-machine implementation of the procedure of Theorem 9.3 for the allowance $1/v$ for at most n steps; within the same budget compute c_{seed} of Lemma 9.4 for the

resulting seed and test $c^v_{\mathrm{seed}} \le n$. Interrupt any attempt that exceeds its budget. Use the seed of the largest successful v, or classical multiplication if there is none. Including binary counters for the budgets, this preprocessing takes $O(n(\log n)^2)$ steps. The coefficients of the selected seed have total bit length at most n, so converting them to field elements by Horner's rule in base 2 and one division each takes $O(n)$ field operations.

For every fixed v, the procedure and the test finish within the budget once n is large enough, and then $c^v_{\mathrm{seed}} \le n$. Hence the selected $v = v(n)$ tends to infinity. Once some v succeeds, Theorem 9.2 gives $p_{\mathrm{seed}} < 9/4 + 1/v$, and Lemma 9.4 bounds the multiplication by $c_{\mathrm{seed}} n^{p_{\mathrm{seed}}} \le n^{1/v} n^{9/4+1/v} = n^{9/4+2/v(n)}$ operations, which proves the claim. □

Remark 9.8 (The rate of the $o(1)$ term). For a suitable implementation, the index $v(n)$ selected in the proof of Corollary 9.7 grows like $\log\log n$, which makes the term $o(1)$ explicit. As there, the preprocessing is counted in steps of a multitape Turing machine; such a machine can carry out the at most $\mathfrak{S}$ steps of Theorem 9.3, locating each operand by a scan of its memory, in at most a constant times the square of their bit operations. Step 3 of the proof of that theorem counts at most $40(j+1)\mathfrak{T}$ assembly stages before the direct-sum labels are expanded. Suppose that the implementation performs each of these stages at once on the expanded arrays, forming every entry as a sum of at most $\mathfrak{H}$ products of at most four previously formed numbers. Since every leg has dimension at most $\mathfrak{W}^{34\mathfrak{T}}$, each stage then takes at most $\mathfrak{H}^2$ steps in the sense of that theorem, so its bounds still hold. Then, for all sufficiently large n,

$$\frac{\log_2\log_2 n}{273600} \le v(n) < \frac{\log_2\log_2 n}{3600}, \tag{9.4}$$

and the algorithm uses $O(n^{9/4+547200/\log_2\log_2 n})$ steps. The upper bound in (9.4) holds for every implementation, because it comes from the size of the seed alone. Hence the guarantee $p_{\mathrm{seed}} < 9/4 + 1/v(n)$ of Theorem 9.2 for the selected seed is never better than $9/4 + 3600/\log_2\log_2 n$, and for this argument the double-logarithmic rate is optimal up to the constant. With the bit-length bound $2^{2\mathfrak{S}}$ of Theorem 9.3 in place of (9.5) below, the same argument gives only $v(n) \ge \log_2\log_2\log_2 n/172800$ and the exponent $9/4 + 345600/\log_2\log_2\log_2 n$.

Sketch. The idea is that the seed for the allowance $1/v$ has a side that is doubly exponential in v, which gives the upper bound, and that a bit-length bound governed by the depth of the computation rather than by its length makes computing the seed doubly exponential in v as well, which gives the lower bound. The attempt for v uses $\varepsilon' = \min(1/v, 1/2)$ and $\ell_a = \lceil 3600/\varepsilon' \rceil \ge 3600v$.

Upper bound. By Lemma 9.4 the seed has $R > d_0^2$ products, so a successful attempt writes more than $3d_0^4$ coefficients within n steps, and $\log_2 n \ge 4\log_2 d_0 - O(1)$. As in Remark 8.8, $A = A_1^j \ge 2^j = 2^{a\ell_a}$, and $d_0 > \Xi \ge n_* > A$ by (9.2). Hence $\log_2\log_2 n > \ell_a \ge 3600v$.

Sizes. Now let $v \ge 2$, so that $\varepsilon' = 1/v$, $\ell_a = 3600v$, $a = 2^{3600v}$ and $b_\varepsilon \le 2 + \log_2 v$; the attempt for $v = 1$ builds the same seed as for $v = 2$. The rational variant has the counts of Section 7. Since $p < 3$ and $m_* = M^{3j} \le B^3$ by (7.3) and (7.6), we have $r_* = m_*(B/A)^p < m_* B^3 \le B^6$; every other summand of $\mathfrak{W}$ is also at most B^6, so $\mathfrak{W} \le 15B^6$. By (8.1), (7.11) and $F(a,a) < (2a-1)^{4/3}$, $\log B = \log A + j\Delta_1 \le 181a^5\ell_a^2 + jK\log F(a,a) < 182a^5\ell_a^2$, so $\log\mathfrak{W} < 1100a^5\ell_a^2$. By (6.13), as in Step 4 of the proof of Theorem 8.1, $\mathfrak{T} < 185aK$, so $\mathfrak{F} < 2\cdot 10^4 a^5\ell_a^2$; and $\Xi \le \mathfrak{W}^{13}$ (Step 4 of the proof of Theorem 9.3) gives $\log d_0 \le 2v\log(\mathfrak{W}^{13}+1) < 9a^5\ell_a^3$. As $a = 2^{\ell_a}$ with $\ell_a \ge 7200$, the numbers $\mathfrak{F}$, $\log\mathfrak{W}$ and $\log d_0$ are less than a^6. Since $\log d_0 < \mathfrak{F}\log\mathfrak{W}$, also $\log\mathfrak{S} = 50\log\mathfrak{Q} < 50(\mathfrak{F}\log\mathfrak{W} + 2) < 2\cdot 10^9 a^{10}\ell_a^4 < a^{11}$.

Depth of the recursion. By (8.3), which holds for the rational variant because it has the same counts, $\log(B/A) = j\Delta_1 > \log 2$. Hence $\rho = A/B < 1/2$, and $\rho^{\ell_{\max}} d_0 > \rho$ (Step 3 of the proof of Lemma 2.4) gives $\ell_{\max} \le \lceil \log_2 d_0 \rceil$; Step 4 of the proof of Theorem 9.3 used only $\ell_{\max} \le d_0$.

Bit lengths. If each factor has $\mathfrak{b} \le \mathfrak{b}_0$, then a sum of at most $\mathfrak{H}$ products of at most four factors, and each of its partial sums in any order, has $\mathfrak{b}+1 \le 4\mathfrak{H}(\mathfrak{b}_0+1)$, because the product of the denominators of the terms is a common denominator with at most $4\mathfrak{H}\mathfrak{b}_0$ bits. The data of Step 2 of the proof of Theorem 9.3, including the occupations, the inverse basis changes and the intermediate values of Gaussian elimination, which are ratios of minors, have $\mathfrak{b} \le \mathfrak{W}^3$; the separator coefficients of Lemma 9.1(b), for instance, are ratios of integers of absolute value at most $(2M_i)^{2M_i}$. Each of the at most $40(j+1)\mathfrak{T} \le \mathfrak{F}$ assembly stages forms every entry as a sum of at most $\mathfrak{H}$ products of at most four numbers formed earlier, and each depth of Step 4 forms sums of at most $\mathfrak{Q}^{14}$ products of a coefficient of the scheme and an entry of the previous depth. Hence every rational number of the computation satisfies

$$\mathfrak{b} < (4\mathfrak{H})^{\mathfrak{F}}(4\mathfrak{Q}^{14})^{\lceil \log_2 d_0 \rceil}(\mathfrak{W}^3+1) \le \mathfrak{Q}^{2\mathfrak{F}+15\lceil \log_2 d_0 \rceil+3} \le 2^{a^{18}}, \tag{9.5}$$

in place of $\mathfrak{b} \le 2^{2\mathfrak{S}}$.

Lower bound. With schoolbook arithmetic, each of the at most $\mathfrak{S} < 2^{2a^{11}}$ steps costs $O(2^{3a^{18}})$ bit operations, so the attempt for v, including its test, takes $O(2^{a^{19}}) = O(2^{2^{68400v}})$ Turing-machine steps, with an absolute implied constant. Also $R < d_0^{11/4}$ gives $c_{\mathrm{seed}} < 7d_0^8$, so $c_{\mathrm{seed}}^v < e^{a^7}$. Suppose $\log_2 \log_2 n \ge 273600$, and let $v = \lfloor \log_2 \log_2 n/136800 \rfloor \ge 2$. Then $a^7 < 2^{68400v} \le (\log_2 n)^{1/2}$, so for large n the attempt for v succeeds, and $v(n) \ge v \ge \log_2 \log_2 n/273600$; the multiplication then takes at most $n^{9/4+2/v(n)}$ operations, as in the proof of Corollary 9.7. With $2^{2\mathfrak{S}}$ in place of (9.5), the attempt takes $O(2^{16\mathfrak{S}}) \subseteq O(2^{2^{a^{12}}})$ steps, and $v = \lfloor \log_2 \log_2 \log_2 n/86400 \rfloor$ gives the weaker bound. □

10 Finite examples

Theorem 8.1 applies only to target indices $a \ge 2^{64}$, and its bound is weak for moderate a (Remark 8.3). For a small fixed index we can instead compute the rounded counts and bound the exponent directly. This section does so for $a = 4$, 8 and 12; no theorem of the paper depends on it. The examples are computer-assisted. The bounds $\overline{\Phi}$ and digit counts in Table 2, and the other bounds quoted from the report, are certified by exact rational interval arithmetic (Section 10.3) and recorded in the report `results/multiway-finite-costs.json`. The column Φ and the other derived values (τ_a, the terms of $\overline{\Phi}$, differences and percentages) are floating-point evaluations. The supplementary script `finite_table_checks.py` reproduces every number of this section from the report.

10.1 An upper expression for the exponent

Let $\Delta_N > 0$, so that Proposition 7.3 gives a relative scheme with pooled exponent p as in (7.9). The examples bound the slightly larger *fixed-batch bound*

$$\Phi = \frac{1}{\Delta_N}\left(3NK\log(2a-1) + \log\frac{L}{M^3} + \frac{\log E}{k} + \frac{\log \kappa_N}{j} + \frac{\log 2}{kj}\right) = p + \frac{\log 2}{kj\Delta_N}, \tag{10.1}$$

which also bounds the exponent of the recursion without pooling. Indeed, if the r_* products created by one batch of m_* products of side n are batched only among themselves, they have side $A\lceil n/B \rceil$ and fill $\beta_0 = \lceil r_*/m_* \rceil$ batches, the last one completed by zero products, so the cost of one batch satisfies $\mathrm{cost}(n) \le \beta_0 \,\mathrm{cost}(A\lceil n/B \rceil) + O(n^2)$, where $O(n^2)$ accounts for the linear maps of the scheme. The estimate in the proof of Lemma 2.4, with β_0 in place of r_*/m_*, gives the exponent $(\log \beta_0)/\log(B/A)$. Since $r_*/m_* \ge (B/A)^2 > 1$ by Step 1 of that proof, $\beta_0 \le 2r_*/m_*$, and this exponent is at most Φ.

The sector counts (7.1) enter Φ through Δ_N, the number E of interpolation points and the copy overhead $\log(L/M^3)$. Rather than computing the large integers M_i, we bound $\log M_i$. Lemma 7.4 gives such bounds with an error of at most $\log(Nw_i+1)$ for each used child beyond the first; the following factorial bounds are usually sharper, and we use the intersection of both.

Lemma 10.1 (Elementary factorial bounds)**.** *Put $c_0 = (1+\log 2)/2$. For every integer $\nu \geq 1$,*

$$(\nu+\tfrac12)\log\nu - \nu + c_0 \leq \log(\nu!) \leq (\nu+\tfrac12)\log\nu - \nu + 1. \tag{10.2}$$

Proof. Since log is concave, its chord over $[m, m+1]$ lies below its graph, so $\frac12(\log m + \log(m+1)) \leq \int_m^{m+1} \log\xi\, d\xi$. Summing over $1 \leq m < \nu$ gives the trapezoid inequality

$$\log(\nu!) - \tfrac12\log\nu \leq \int_1^\nu \log\xi\, d\xi = \nu\log\nu - \nu + 1,$$

which is the upper bound. By Jensen's inequality, $\int_{m-1/2}^{m+1/2}\log\xi\, d\xi \leq \log m$; summing over $1 \leq m \leq \nu$ gives

$$\log(\nu!) \geq \int_{1/2}^{\nu+1/2}\log\xi\, d\xi = (\nu+\tfrac12)\log(\nu+\tfrac12) - \nu + \tfrac12\log 2.$$

Finally, $(\nu+\frac12)\log(\nu+\frac12) - (\nu+\frac12)\log\nu = (\nu+\frac12)\log(1+\frac{1}{2\nu}) \geq \frac12$ by $\log(1+\delta) \geq \delta/(1+\delta)$, and $\frac12 + \frac12\log 2 = c_0$. □

Now let ν_y, $y \in \mathcal{Y}$, be positive integers with sum ν. Applying the lower bound of (10.2) to $\nu!$ and the upper bound to each $\nu_y!$, and then the other way round, we obtain, since the linear terms cancel,

$$c_0 - |\mathcal{Y}| \leq \log\frac{\nu!}{\prod_y \nu_y!} - \nu H((\nu_y/\nu)_y) - \frac12\log\frac{\nu}{\prod_y \nu_y} \leq 1 - |\mathcal{Y}|c_0, \tag{10.3}$$

where $\nu H((\nu_y/\nu)_y) = \nu\log\nu - \sum_y \nu_y\log\nu_y$.

We apply (10.3) to each relation separately. For a split $i \in \mathcal{I}_c$, take $\mathcal{Y} = \mathcal{Y}_i^+ = \{y : n_{i,y} > 0\}$ and $\nu_y = Nn_{i,y}$; children with $n_{i,y} = 0$ contribute $0! = 1$. Then $\nu = Nw_i$ and $\nu H((\nu_y/\nu)_y) = NH_i$, where $H_i = w_i H(\hat q_i)$ is the entropy charge. We intersect the resulting interval for $\log M_i$ with the interval $[NH_i - (|\mathcal{Y}_i^+|-1)\log(Nw_i+1),\ NH_i]$ of Lemma 7.4 and with $[0,\infty)$; if $|\mathcal{Y}_i^+| = 1$, then $M_i = 1$ and we use $\log M_i = 0$. For the tripling of row $r \in \mathcal{I}_t$, with $M_r = \binom{Nz_r}{m_r}2^{Nz_r - m_r}$, we bound the logarithm of the binomial coefficient in the same way, using the two counts m_r and $Nz_r - m_r$ (if $m_r = 0$, the coefficient is 1), add $(Nz_r - m_r)\log 2$, and intersect with $[Nz_r\log 3 - \log(Nz_r+1),\ Nz_r\log 3]$. The lower end of this interval is (4.3); the upper end holds because the numbers $M_{\mathrm{t}}(Nz_r, m)$, $0 \leq m \leq Nz_r$, add up to 3^{Nz_r}.

For each relation $i \in \mathcal{J}$, write $[\underline\eta_i, \overline\eta_i]$ for the interval obtained in this way, so that $\underline\eta_i \leq \log M_i \leq \overline\eta_i$. Put

$$\underline\Delta_N = NK\log 2 - N\sum_{i\in\mathcal{I}_c} w_i\log d_i + \sum_{i\in\mathcal{J}}\underline\eta_i \tag{10.4}$$

and define the *sufficient upper expression*

$$\overline\Phi = \frac{3NK\log(2a-1) + 3I\log 2 + \frac1k\bigl(\log(3kI+1) + 2\max_{i\in\mathcal{J}}\overline\eta_i\bigr) + \frac1j\log\kappa_N + \frac{\log 2}{kj}}{\underline\Delta_N}, \tag{10.5}$$

where $\log\kappa_N = 3N\sum_{\sigma,r,b} c_{\sigma,r,b}\log(r+b-1)$ is computed from the catalyst multiplicities, as in (7.5).

Lemma 10.2 (A sufficient upper expression)**.** *We have $\underline\Delta_N \leq \Delta_N$. If $\underline\Delta_N > 0$, then $B > A$ and $p < \Phi \leq \overline\Phi$.*

Proof. By (7.8) and $\log M_i \geq \underline{\eta}_i$, $\underline{\Delta}_N \leq \Delta_N$. If $\underline{\Delta}_N > 0$, then $B/A = (B_1/A_1)^{kj} > 1$, and $p < \Phi$ by (10.1). We compare the numerators of Φ and $\overline{\Phi}$ term by term. The copy overhead is below $3I \log 2$ by (7.12). Since every $M_i \geq 1$, we have $E = 1 + 3k \sum_{i \in \mathcal{J}} (M_i - 1)^2 \leq (3kI + 1) \max_i M_i^2$, so $\log E \leq \log(3kI + 1) + 2 \max_i \overline{\eta}_i$. The other terms agree. Hence the numerator of $\overline{\Phi}$ is at least that of Φ. Both numerators are positive, so replacing Δ_N by the smaller positive number $\underline{\Delta}_N$ can only increase the quotient. □

10.2 Three examples

Table 2 reports $\overline{\Phi}$ for $a = 4, 8, 12$, with the parameters

$$K = a^3 \ell_a^2, \qquad N = 1, \qquad k = 32, \qquad j = 128\, a \ell_a,$$

where $\ell_a = \lceil \log_2 a \rceil$; thus $K = 256$, 4 608, 27 648 and $j = 1\,024$, 3 072, 6 144. For each a, the occupations of Proposition 5.3 were computed exactly and rounded at scale K by the deterministic rule of Theorem 8.4; other roundings allowed by Lemma 6.1 give other counts. The report lists the counts $n_{r,x,y}$ and z_r and the catalyst multiplicities $c_{\sigma,r,b}$ of Proposition 6.2.

		terms of $\overline{\Phi}$, divided by $\underline{\Delta}_N$						
a	τ_a	main	copy	interp.	catalyst	bound $\overline{\Phi}$	Φ (computed)	digits of A
4	2.5923	3.3267	0.0926	0.1397	0.0699	3.628926097	3.5754691	84 456 434
8	2.5116	2.7169	0.0119	0.0822	0.0376	2.848599165	2.8406393	11 877 489 885
12	2.4790	2.5786	0.0036	0.0674	0.0262	2.675820359	2.6728401	234 905 651 862

Table 2: Examples with $N = 1$, $k = 32$, $K = a^3 \ell_a^2$ and $j = 128\, a \ell_a$. Column $\overline{\Phi}$: upper end of a rational enclosure of (10.5), rounded upwards, hence a certified upper bound for Φ and p. The four columns before it: the main term $3NK \log(2a - 1)$, the copy bound $3I \log 2$, the interpolation bound and the catalyst overhead, each divided by $\underline{\Delta}_N$; the last term $(\log 2)/(kj\underline{\Delta}_N)$ is below 10^{-7}. Column Φ: floating-point value of (10.1). Last column: number of decimal digits of the source side A.

In all three examples $\underline{\Delta}_N > 0$, so by Lemma 10.2 each entry $\overline{\Phi}$ bounds Φ and hence p, while $p \geq \tau_a$ (Section 7). For example, the relative scheme with $a = 12$ has $2.479 < p < 2.675820359$. The column Φ is computed from the same counts, with $\log M_i$ from the log-gamma function and with E and L/M^3 from the actual sector counts. The relaxations in $\overline{\Phi}$ raise the bound by about 1.5% at $a = 4$ and 0.1% at $a = 12$, and pooling lowers Φ by $(\log 2)/(kj\Delta_N) < 10^{-7}$.

The parameters of Theorem 8.1, $K = a^3 \ell_a$, $k = 1$ and $j = a\ell_a$, suffice for large a. For $a \leq 12$, however, the overheads per unit of K are large compared with $\log F(a, a) < 3.8$, and the table amortizes them further: $k = 32$ divides the interpolation overhead by 32, the factor 128 in j divides the catalyst overhead by 128, and the extra factor ℓ_a in K reduces the shares of the copy overhead, the rounding loss and the type penalty, which grow more slowly than K. With the same K and N, the report gives the bounds $\overline{\Phi} \leq 16.826$, 10.175 and 8.097 for $a = 4, 8, 12$ when $(k, j) = (1, a\ell_a)$, and 12.501, 7.629 and 6.009 when $(k, j) = (32, a\ell_a)$. With k and j as in Theorem 8.1 (but the larger K of the table), the catalyst overhead is therefore the largest cost at these indices. Each increase enlarges the source side $A = A_1^{kj}$, whose logarithm is linear in k and j and roughly linear in K.

The main term would equal the profile exponent τ_a if $\underline{\Delta}_N$ were $NK \log F(a, a)$. By Proposition 6.3, the rounded potential satisfies $NG = NK \log 2 - N \sum_{i \in \mathcal{I}_c} w_i \log d_i + N \sum_{i \in \mathcal{J}} H_i =$

$NK \log F(a,a) - N\mathcal{L}$, and therefore

$$\underline{\Delta}_N = NK \log F(a,a) - N\mathcal{L} - \sum_{i \in \mathcal{J}} (NH_i - \underline{\eta}_i).$$

The rounding loss $N\mathcal{L}$ is below one in all three examples. The last sum bounds the type loss $\sum_i (NH_i - \log M_i)$ of Section 7.5 and is at most the type penalty $\mathcal{T}_N$; it removes about 22%, 7.6% and 3.9% of $NK \log F(a,a)$ for $a = 4, 8, 12$. So the type loss is the largest finite cost, and interpolation is the largest overhead in the numerator. At $a = 12$, all finite costs together raise the bound by about 0.197 above τ_{12}, which is less than the remaining gap $\tau_{12} - 9/4 \approx 0.229$.

10.3 Rational enclosures of logarithms

The report builds $\overline{\Phi}$, the $\underline{\eta}_i$ and $\overline{\eta}_i$, and $\log A$ from logarithms of positive integers by rational operations, maxima and minima. Write such an integer as $\nu = 2^e\theta$ with an integer $e \geq 0$ and $1 \leq \theta < 2$, and put $\varrho = (\theta - 1)/(\theta + 1)$. For $1 \leq \theta \leq 2$ we have $0 \leq \varrho \leq 1/3$, and the series $\log\theta = 2\operatorname{artanh}\varrho = 2(\varrho + \varrho^3/3 + \cdots)$ has nonnegative terms, those after $2\varrho^{79}/79$ adding up to at most $2\varrho^{81}/(81(1-\varrho^2))$. Therefore

$$2\Big(\varrho + \frac{\varrho^3}{3} + \cdots + \frac{\varrho^{79}}{79}\Big) \leq \log\theta \leq 2\Big(\varrho + \frac{\varrho^3}{3} + \cdots + \frac{\varrho^{79}}{79}\Big) + \frac{2\varrho^{81}}{81(1-\varrho^2)}, \tag{10.6}$$

and the two sides differ by less than 10^{-40}. The report evaluates the lower and upper bounds at multiples $\theta^- \leq \theta \leq \theta^+$ of 2^{-80} in $[1,2]$ with $\theta^+ - \theta^- \leq 2^{-80}$, which encloses $\log\theta$ because log is increasing; it encloses $\log 2$ in the same way and puts $\log\nu = e\log 2 + \log\theta$. Logarithms of rationals are differences of such enclosures. All further operations are interval arithmetic in exact rationals, and all roundings (of each logarithm to a multiple of 2^{-160}, and of printed values) are outward. Finally, A has $\lfloor \log_{10} A \rfloor + 1$ decimal digits, and the report encloses $\log_{10} A = (\log A)/(\log 10)$ in an interval whose endpoints have the same integer part.

11 Limits, formalization and open questions

This section collects reusable components and possible improvements (Section 11.1), proves lower bounds for the source side within the construction (Section 11.2), describes a Lean formalization of a rational variant (Section 11.3), lists the supporting files (Section 11.4) and states open questions (Section 11.5).

11.1 Reusable components and possible improvements

Several components can be used separately in other tensor arguments. The type restriction of Lemma 4.3 turns a filtration that multiplication preserves, such as the determinant filtration, into restrictions of fixed type. The occupation estimates of Proposition 5.3 bound both the total occupation of a row and the sum of the logarithms of its auxiliary dimensions, weighted by the occupations, uniformly in the length of the row. The rounding of Lemma 6.1 keeps integer balances exact, and its entropy estimate (Proposition 6.3) holds even when some occupations are arbitrarily small or zero. The separator of Proposition 4.6 applies to any sectors that share one leg.

The exponent formula (7.9) separates the overheads, so possible improvements can be judged one term at a time. A separator with fewer source copies would reduce the copy overhead, and one with a smaller degree the interpolation overhead: with the separator of Lü [18] (Remark 4.7), (7.3) could

use $D = 3\sum_i (M_i - 1)$, which replaces $2\bar{H}_N$ by $\bar{H}_N$ in the bound on $\log E$ in (7.12); Appendix C discusses two refinements. A cheaper decomposition of the catalyst than the rank bound κ_N would reduce the catalyst overhead, and other transition rules might reduce the occupation and type costs. Section 10.2 shows which terms matter at small indices. Using all six leg orientations instead of three squares the dimensions at a given interpolation power k but gives no better exponent: the sixfold construction with power k coincides with the threefold one with power $2k$ (Section A.4). Section 11.2 shows that within the construction the source side must be doubly exponential in $1/\varepsilon$, even with a free catalyst; we know of no lower bound that excludes the improvements of the overheads listed above.

11.2 Limits of the construction

The source sides in Sections 8–10 are astronomically large. This subsection proves that within the construction they must be large: a pooled exponent below 3 needs $A \geq 12^{18}$, and a pooled exponent at most $9/4 + \varepsilon$, with $\varepsilon \leq 1/4$, needs $\log\log A > 0.6/\varepsilon$, so the order $\log\log A = O(1/\varepsilon)$ of Corollary 8.2 is optimal up to the constant.

A *scheme of the construction* is a relative scheme $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ of Proposition 7.3 (so $B > A$), built for some target index $a \geq 2$ and positive integers K, N, k, j from the counts of any rounding that Lemma 6.1 allows; p denotes its pooled exponent. The rational variant of Section 9.1 has the same r_*, m_*, A and B as the scheme built from the same counts, so the results below apply to it as well. By (7.3) and $A = A_1^{kj}$, we have $\log A = NkjW$, with $W = \sum_{r,x} w_{r,x} \log d_{r,x}$ as in Section 6.5. We write $(2a-3)!! = \prod_{r=2}^{a-1}(2r-1)$, which is 1 for $a = 2$.

Proposition 11.1 (Lower bounds for the source side)**.** *Let $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ be a scheme of the construction, with pooled exponent p.*

(a) If $p < 3$, then $A \geq 12^{18} > 2^{64}$.

(b) If $p \leq 9/4 + \varepsilon$, where $0 < \varepsilon \leq 1/4$, then

$$\frac{3a-1}{2} \geq \Big(\frac{6}{5}\Big)^{9/(4\varepsilon)}, \qquad j > \frac{9(a-2)}{8\varepsilon}, \qquad \log A > \frac{\log 3}{8\varepsilon}\Big(\frac{6}{5}\Big)^{9/(2\varepsilon)} > \frac{1}{2}\Big(\frac{6}{5}\Big)^{9/(2\varepsilon)}.$$

Proof. The row identities force splits and triplings, which enlarge W. A small exponent forces a large gain Δ_N, which needs triplings, and a catalyst overhead that is small compared with the gain, which needs a large reuse count j.

Step 1: lower bounds for W. Fix a row r and a positive function ψ on the states $0, \dots, 3r$ such that $d_{r,x}\psi(x) \geq \psi(y)$ for every interior state x and every child $y \in \mathcal{Y}_{r,x}$. Applying the linear map $\mathbf{e}_x \mapsto \log\psi(x)$ to (6.1), as in Step 3 of the proof of Proposition 6.3, gives $\sum_{x,y} n_{r,x,y} \log(\psi(y)/\psi(x)) = K\log(\psi(0)/\psi(x_r^{\mathrm{in}})) + z_r \log(\psi(3r)/\psi(1))$. Every term on the left is at most $n_{r,x,y}\log d_{r,x}$, so

$$\sum_x w_{r,x} \log d_{r,x} \geq K \log\frac{\psi(0)}{\psi(x_r^{\mathrm{in}})} + z_r \log\frac{\psi(3r)}{\psi(1)}. \tag{11.1}$$

The children of x are $x-h, x-h+2, \dots, x+h$ with $h = h_{r,x} \leq x$; so a child $y \geq 1$ satisfies $y \leq x+h \leq (h+1)x$, the child 0 occurs only if $h = x$, and the child 1 only if $h = x-1$. The function with $\psi(0) = 2$ and $\psi(x) = x$ for $x \geq 1$ satisfies the condition, because $(x+1)x \geq 2$; by (11.1) it gives $K\log 2 + z_a \log(3a)$ in the top row, where $x_a^{\mathrm{in}} = 1$, and $z_r \log(3r)$ in a lower row, where $x_r^{\mathrm{in}} = 2$. In a lower row, the function with $\psi(0) = 3$, $\psi(1) = 3/2$ and $\psi(x) = x/2$ for $x \geq 2$ also satisfies the condition. At $x = 1$, whose children are 0 and 2, this is direct. For $x \geq 2$, the children $y \geq 2$, $y = 1$

and $y = 0$ satisfy it because $x + h \le (h+1)x$, $x^2 \ge 3$ and $x(x+1) \ge 6$, respectively. This potential gives $K \log 3 + z_r \log r$. Summing over the rows,

$$W \ge K \log(2 \cdot 3^{a-2}) + \bar{z} \log 2, \qquad W \ge K \log 2 + \bar{z} \log 6. \tag{11.2}$$

For $a = 2$, the states $1, \dots, 5$ have the auxiliary dimensions $2, 3, 4, 3, 2$ and the children $\{0, 2\}$, $\{0, 2, 4\}$, $\{0, 2, 4, 6\}$, $\{2, 4, 6\}$, $\{4, 6\}$ (Example 5.2), and $\psi = (2, 1, 2, 6, 6, 9, 18)$ on the states $0, \dots, 6$ satisfies the condition; hence

$$W \ge K \log 2 + \bar{z} \log 18 \qquad (a = 2). \tag{11.3}$$

Finally, state 0 is not interior and $3r \notin \{0, x_r^{\mathrm{in}}\}$, so the coefficients of $\mathbf{e}_0$ and $\mathbf{e}_{3r}$ in (6.1) show that K splits of row r have the child 0 and z_r splits have the child $3r$. Hence row r has at least $K + z_r$ splits.

Step 2: the catalyst. By Step 3 of the proof of Proposition 6.2, every catalyst multiplicity equals the source multiplicity, except that of $[+; a, a]$, which is smaller by K. The base $r + b - 1$ in (7.5) does not depend on the representative of a symbol. A split at (r, x) consumes a symbol with base $2r - 2 + x \ge 2r - 1$, a tripling of row r consumes one with base $5r - 2$, and $[+; a, a]$ has base $2a - 1$. By Step 1, and since $(2r - 1)(5r - 2) \ge 24$ for $r \ge 2$,

$$\frac{\log \kappa_N}{3N} \ge \sum_{r=2}^{a} \big((K + z_r) \log(2r-1) + z_r \log(5r-2)\big) - K \log(2a-1) \ge K \log\big((2a-3)!!\big) + \bar{z} \log 24. \tag{11.4}$$

Step 3: the gain. In (6.7), $H(\hat{q}_{r,x}) \le \log d_{r,x}$, because $\hat{q}_{r,x}$ is a distribution on $d_{r,x}$ children. With (7.11), (6.9) and $B > A$, this gives

$$0 < \Delta_N \le NG \le N \min\big(K \log 2 + \bar{z} \log 3,\ K \log F(a, a)\big). \tag{11.5}$$

The three overheads in (7.9) are nonnegative (Section 7.5), so

$$p\Delta_N \ge 3NK \log(2a - 1) + \frac{\log \kappa_N}{j}. \tag{11.6}$$

Step 4: part (a). Let $p < 3$, and put $\eta_a = F(a, a)/(2a - 1)$. By (11.6), both $3NK \log(2a - 1)$ and $(\log \kappa_N)/j + 3NK \log(2a - 1)$ are less than $3\Delta_N$. By (11.5) and (11.4),

$$\bar{z} \log 3 > K \log \frac{2a - 1}{2}, \qquad j \log \eta_a > \log\big((2a - 3)!!\big) + \frac{\bar{z}}{K} \log 24. \tag{11.7}$$

In particular $\bar{z} \ge 1$. Since $\log A = Nkj W \ge jW$, it suffices to show $jW \ge 18 \log 12$.

If $a = 2$, then $\eta_2 = 10/9$ and $(2a-3)!! = 1$. Since $(3/2)^3 > 3$, the first inequality gives $\bar{z}/K > 1/3$, and the second gives $jK \log(10/9) > \bar{z} \log 24 \ge \log 24$. By (11.3),

$$jW \ge jK\Big(\log 2 + \frac{\bar{z}}{K} \log 18\Big) > \frac{\log 24}{\log(10/9)}\Big(\log 2 + \frac{1}{3} \log 18\Big) = \frac{2}{3} \cdot \frac{\log 24}{\log(10/9)} \log 12 > 18 \log 12,$$

the last because $24 \cdot 9^{27} > 10^{27}$.

If $a = 3$, then $\eta_3 = 56/45$ and $(2a-3)!! = 3$. Since $5^6 > 2^6 \cdot 3^5$, the first inequality gives $\bar{z}/K > 5/6$, and the second gives $j \log(56/45) > \log 3 + \frac{5}{6} \log 24 > 17 \log(56/45)$, because $3^6 \cdot 24^5 \cdot 45^{102} > 56^{102}$. Hence $j \ge 18$, and (11.2) with $K, \bar{z} \ge 1$ gives $W \ge \log 12$.

If $a \ge 4$, then $(2a-1)/2 > 3$, so $\bar{z} > K$, and the second inequality gives $j \log \eta_a > \log(24\,(2a-3)!!)$. We claim $24\,(2a - 3)!! > \eta_a^{19}$. For $a = 4$, $(2a - 3)!! = 15$, $\eta_4 = 110/81$ and $360 \cdot 81^{19} > 110^{19}$.

By (3.4), $\eta_{a+1}/\eta_a = (3a+1)(6a^2+a-2)/(3a(6a^2+a-1)) < 13/12$ for $a \ge 4$, and $(13/12)^{19} < 7 \le 2a-1 = (2a-1)!!/(2a-3)!!$, which gives the induction step. Hence $j > 19$, and (11.2) gives $W \ge \log 18$, so $jW \ge 20 \log 18 > 18 \log 12$.

Step 5: part (b). Let $p \le 9/4 + \varepsilon$ with $0 < \varepsilon \le 1/4$. By (11.6) and (11.5), $p \ge 3\log(2a-1)/\log F(a,a) = \tau_a$. Step 4 of the proof of Theorem 3.2 gives $s_a^3 \le (3a-1)/2$, that is, $\log F(a,a) \le \frac{4}{3}\log\frac{3a-1}{2}$. Hence $\frac{9}{4} + \varepsilon \ge \frac{9}{4}\log(2a-1)/\log\frac{3a-1}{2}$, that is,

$$\varepsilon \log \frac{3a-1}{2} \ge \frac{9}{4}\log\frac{4a-2}{3a-1} \ge \frac{9}{4}\log\frac{6}{5},$$

because $(4a-2)/(3a-1)$ increases with a. This is the first inequality; with $\varepsilon \le 1/4$ and $6^9 > 5^{10}$ it gives $(3a-1)/2 > 5$, so $a \ge 4$.

Next, (11.6), (11.5) and $\log F(a,a) < \frac{4}{3}\log(2a-1)$ (Theorem 3.2) give

$$\frac{\log \kappa_N}{j} \le \Big(\frac{9}{4} + \varepsilon\Big) NK \log F(a,a) - 3NK\log(2a-1) < \varepsilon NK \log F(a,a) < \frac{4\varepsilon}{3} NK \log(2a-1),$$

and with (11.4), $j > 9\log((2a-3)!!)/(4\varepsilon\log(2a-1))$. Pairing the factors $2r-1$ and $2(a+1-r)-1$ gives $((2a-3)!!)^2 = \prod_{r=2}^{a-1}(2r-1)(2a+1-2r)$. Each factor is a concave function of r whose values at $r=2$ and $r=a-1$ equal $3(2a-3) > 2a-1$, so $((2a-3)!!)^2 > (2a-1)^{a-2}$. This gives the second inequality.

Finally, (11.2) gives $\log A \ge jW > j(a-2)\log 3 > 9(a-2)^2 \log 3/(8\varepsilon)$. Since $a \ge 4$, we have $6(a-2) \ge 3a-1$, so $(a-2)^2 \ge ((3a-1)/2)^2/9 \ge (6/5)^{9/(2\varepsilon)}/9$ by the first inequality. This gives the third inequality, and $\log 3/(8\varepsilon) \ge (\log 3)/2 > 1/2$. □

Corollary 11.2 (Exponent against source side)**.**

(a) Every scheme of the construction with $p < 3$ satisfies $p > \dfrac{9}{4} + \dfrac{9\log(6/5)}{2\log(2\log A)}$.

(b) For rational $\varepsilon \in (0, 1/4]$, the least source side $A_{\min}(\varepsilon)$ of a scheme of the construction with $p \le 9/4 + \varepsilon$ satisfies $0.6/\varepsilon < \log\log A_{\min}(\varepsilon) < 6300/\varepsilon$. Thus $A_{\min}(\varepsilon) = \exp(\exp(\Theta(1/\varepsilon)))$.

(c) Let $0 < \varepsilon \le 1/2$. If the pooled recursion of a scheme of the construction with $p \le 9/4 + \varepsilon/2$, unrolled at a side d_0 as in Corollary 8.7 or Theorem 9.2, gives fewer than $d_0^{9/4+\varepsilon}$ products, then $\log d_0 > \frac{1}{2}(6/5)^{9/\varepsilon}$. This applies to the seeds of both results.

Proof. (a) If $p \le 5/2$, put $\varepsilon = p - 9/4$. Then $0 < \varepsilon \le 1/4$, since $p \ge \tau_a > 9/4$ (Section 7.5), and Proposition 11.1(b) gives $2\log A > (6/5)^{9/(2\varepsilon)}$, which is the claim. If $5/2 < p < 3$, Proposition 11.1(a) gives $2\log A \ge 36 \log 12 > 27 > (6/5)^{18}$, so the right side is less than $9/4 + 1/4 < p$.

(b) Corollary 8.2 gives a scheme with $p < 9/4 + \varepsilon$, so $A_{\min}(\varepsilon)$ exists. By Proposition 11.1(b) and $\varepsilon \le 1/4$, $\log\log A_{\min}(\varepsilon) > (9/(2\varepsilon))\log(6/5) - \log 2 \ge (18\log(6/5) - \log 2)/(4\varepsilon) > 0.6/\varepsilon$. Conversely, the scheme of Corollary 8.2 has $\log A \le 181 \cdot 2^{5\ell}\ell^2$, where $\ell = \lceil 1800/\varepsilon \rceil \ge 7200$, so $\log\log A_{\min}(\varepsilon) \le \log 181 + 5\ell\log 2 + 2\log\ell < 3.48\,\ell \le 3.48 \cdot 1801/\varepsilon < 6300/\varepsilon$.

(c) Every side $n \le A$ lies below the leaf threshold of Lemma 2.4, where the unrolled algorithm is classical, with $n^3 \ge n^{9/4+\varepsilon}$ products (Remark 8.8); hence $d_0 > A$. Apply Proposition 11.1(b) with $\varepsilon/2 \le 1/4$ in place of ε. Both results unroll schemes with $p < 9/4 + \varepsilon/2$; in Theorem 9.2, $\varepsilon' = \varepsilon$. □

The double exponential does not come from the catalyst. The first inequality of Proposition 11.1(b) uses only $p \ge \tau_a$, which holds whenever the three overheads are nonnegative, so it holds even if the catalyst costs nothing. Together with $a \ge 4$ (Step 5) and $\log A \ge W > (a-2)\log 3 \ge$

$((3a-1)/6)\log 3$ it gives $\log A > \frac{1}{3}(6/5)^{9/(4\varepsilon)}$; the bound on the reuse count j only doubles the exponent $9/(4\varepsilon)$. A cheaper decomposition of the catalyst therefore cannot make $\log\log A$ smaller than of order $1/\varepsilon$.

Remark 11.3 (What the construction certifies). Choose, for each input side n, any scheme of the construction with $p < 3$ and $A \le n$. By Corollary 11.2(a), $n^p > n^{9/4}\exp(\frac{9}{2}\log\frac{6}{5}\cdot\log n/\log(2\log n))$, and the last factor grows faster than every power of $\log n$. So, even when the scheme depends on n, the bounds n^p that Lemma 2.4 certifies in this way are not of the form $O(n^{9/4}(\log n)^{O(1)})$. This concerns the certified bounds, not the operation counts of the algorithms; Remark 9.8 treats the algorithm of Corollary 9.7.

Remark 11.4 (Scope of the lower bounds). The proofs use the separator only through its output and $L \ge M^3$, the counts only through the row identities (6.1), and the widths only through $1 \le h_{r,x} \le \min(x, 3r-x)$, except in (11.3). For all such widths, $\psi = (2,1,2,4,6,12,16)$ satisfies the condition of Step 1 in row two and gives $\log 16$ in place of $\log 18$ (the case $a=2$ of Step 4 then uses $24\cdot 9^{30} > 10^{30}$ and $2^{34} > 3^{18}$), and the identity of Proposition 6.3, with $\mathcal{L} \ge 0$, holds for the transition probabilities $f_r(y)/((h+1)f_r(x))$ of a split of any such width h. Hence Proposition 11.1 and Corollary 11.2(a),(b) also hold for the schemes that the maps in the proofs of Propositions 7.1 and 7.3 build, with the separator of Proposition 4.6 or of Lemma 9.1(b), from any nonnegative integer solution of (6.1) with any such widths, possibly several at one state. They say nothing about separators with other outputs, other relations among the tensors $S^{\sigma}_{r,b}$, or decompositions of the source copies and the catalyst other than those of Proposition 7.3 (but see Remark 11.5).

Remark 11.5 (Arbitrary decompositions of the source and the catalyst). Suppose that, in Proposition 7.3, the source factors $(S^{+}_{a,a})^{\otimes NKk}$ of the j uses are not decomposed one use at a time, and that $(S^{+}_{a,a})^{\otimes NKkj} \otimes \mathcal{C}_N^{\otimes k}$ is supplied by one arbitrary rank decomposition with R terms, possibly joint across all factors, so that $r_* = RE^jL^{kj}$; the counts, the separator copies and the interpolation points are kept. On each leg of $C(r,b)$, the rows of the variables are nonzero and have disjoint supports, as for leg 3 in Section 2.2; so every flattening rank of $S^{\sigma}_{r,b}$ equals its leg dimension $rb(r+b-1)$, and flattening ranks multiply under tensor products. Hence R is at least the leg dimension of this tensor, and Step 2 of the proof of Proposition 11.1, with leg dimensions in place of rank bounds, gives $R \ge \big(a^2(2a-1)\big)^{NKkj}\big(((a-1)!)^{2K}((2a-3)!!)^K 1344^{\bar z}\big)^{Nk}$, where $1344 = 12\cdot 112$ is the product of the leg dimensions of $S^{\sigma}_{2,2}$ and $S^{\sigma}_{2,7}$. With $E \ge 1$, $L \ge M^3$ and (11.5), $p<3$ then implies that both $2^{3jK}3^{3j\bar z}$ and $F(a,a)^{3jK}$ exceed

$$\big(a^2(2a-1)\big)^{jK}((a-1)!)^{2K}((2a-3)!!)^K 1344^{\bar z}.$$

If $A \le 2^{24}$, then $jW \le 24\log 2$, and (11.2) gives $a \le 16$, $jK \le 24$ and $j\bar z \le 8$. An exhaustive search over these tuples in exact integer arithmetic (Section 11.4), with $\log A \ge jW$, (11.2) and, for $a=2$, the bound $W \ge K\log 2 + \bar z \log 16$ of Remark 11.4, finds that $(a,K,\bar z,j) = (2,2,1,4)$ is the only tuple that satisfies both conditions and whose lower bound for A is at most 2^{24}, and that this bound equals 2^{24}. Hence every such scheme with $p<3$ has $A \ge 2^{24}$, also in the generality of Remark 11.4. This bound is much weaker than Proposition 11.1(a).

11.3 A Lean formalization

The directory `formalization/` of the supplementary material contains a formalization in Lean 4 with Mathlib [9, 32] of a simpler rational variant of our construction.[1] Its final theorems, `algorithms_with_exponent_slack` and `omega_le_nine_fourths` in `RationalAlgorithmTrans`

[1] Checked with Lean 4.35.0, Mathlib at commit `ccbef15e`, and the repository of Filmus [12] at commit `8e38460c`.

`portBridge.lean`, state that, for every field of characteristic zero and every real $\varepsilon > 0$, division-free straight-line programs over that field multiply two $n \times n$ matrices with $O_\varepsilon(n^{9/4+\varepsilon})$ additions, subtractions and multiplications for all $n \geq 1$, and that therefore $\omega \leq 9/4$ over every such field. Programs, their cost and ω are those of the arithmetic model in the Lean code of OpenAI [24], as adapted by Navot [20] and distributed with Filmus [12]; from that library the proof uses only the model, generic program constructions and basic tensor definitions, not its proof of the bound. The formal proof follows Sections 4, 5 and 7 with rational maps as in Section 9.1: it includes Lemmas 4.1, 4.3 and 4.5, the occupations of Proposition 5.3 as the unique nonnegative rational solution of their linear system, with the row identity (5.3) (but not (5.4) or (5.5)), the three parts of Lemma 9.1, with the weights of part (c) written as values of Lagrange basis polynomials, the restriction of Proposition 7.3, and the pooled recursion of Lemma 2.4 for $p > 2$. It differs from the paper in three ways: it clears the denominators of the occupations instead of rounding them as in Section 6; it uses only the formula (3.4) and bounds on $\log F(a, a)$, not Theorem 3.2; and it reaches $9/4$ through limits, so it gives no explicit constants, rates or seed sides. Apart from Lemma 9.1, nothing in Sections 8–10 is formalized, in particular neither Theorem 1.2 nor the seeds of Theorem 1.1; nor are Proposition 11.1 and Corollary 11.2. Lean's kernel checks all proofs, which use only Lean's standard axioms (propositional extensionality, choice and quotient soundness). Other Lean developments prove $\omega \leq 9/4$ over $\mathbb{C}$ [24] and over every field [12, 20, 35], and the corresponding operation count over every associative unital ring [21].

11.4 Supporting files

The supplementary material contains scripts, most of them in exact arithmetic, in the directory `experiments/`, and their saved reports, in `results/`. The scripts check the determinant maps and the symmetrized convolution tensors on small cases (`multiway_determinant.py`, `cyclic_profile_extraction.py`), compute the occupations and the rounded counts (`multiway_profile_flows.py`, `multiway_rounded_flow.py`) and the exact binary certificate of Appendix A (`general_profile_extraction.py`), and evaluate the bounds of Section 10 in exact rational interval arithmetic (`multiway_finite_costs.py`, which writes the report `results/multiway-finite-costs.json`). The script `finite_table_checks.py` reproduces the numbers of Section 10 and the values of τ_a below from that report. The script `extraction_a2_small_scale.py` enumerates, for $a = 2$ and $N = 1$, all roundings allowed by Lemma 6.1, checks the sign of Δ_1 stated in Section 7.4 in exact integer arithmetic, and evaluates the numbers of Example 7.2. The scripts `flows_row_maxima.py` and `rounding_observed_loss.py` compute Example 5.2 exactly and the floating-point values in Remarks 5.4 and 6.4. The script `rational_interpolation_audit.py` checks, on small cases, the three parts of Lemma 9.1 and the pooled-recursion counts used in Theorem 9.2. The script `rational_seed_arithmetic_checks.py` writes `results/rational-seed-arithmetic-checks.json`; it checks the denominator clearing, the intermediate bit lengths and the forward-error bound of Section 9.4 for Strassen's seed with rescaled rational coefficients. Neither of these two scripts constructs the seed of Theorem 9.2. The script `source_side_bounds.py` (report `results/source-side-bounds.json`) checks the potentials of Step 1 of the proof of Proposition 11.1 on the rows $r \leq 60$, also for all widths, and every inequality between fixed numbers in the proofs of Proposition 11.1 and Corollary 11.2 and in open question 3; it evaluates those proved for all a on finite ranges, checks flattening ranks, and runs the search of Remark 11.5. The directory `formalization/` contains the Lean files of Section 11.3. No theorem, proposition or corollary of this paper depends on these files; Remark 11.5 rests on the exact search of `source_side_bounds.py`.

11.5 Open questions

1. *New relations.* Are there tensor relations (restrictions or degenerations) among the symmetrized convolution tensors $S^{\sigma}_{r,b}$, other than splits and triplings, that add inequalities to the profile axioms? By Theorem 3.2 and Remark 3.3, such relations are needed for a bound below $9/4$ of the form $3\log(2a-1)/\log P(a,a)$.

2. *Dependence on the allowance.* Within the construction, the least source side at allowance ε is $\exp(\exp(\Theta(1/\varepsilon)))$ (Corollary 11.2(b)), and a cheaper catalyst cannot remove the double exponential (Section 11.2). Is there a different finite construction with exponent below $9/4+\varepsilon$ and with $\log A$ bounded by a polynomial in $1/\varepsilon$?

3. *Moderate sizes.* A floating-point evaluation gives $\tau_a > 2.371339$ for $a \le 187$. Since $p \ge \tau_a$, a scheme of the construction has exponent below 2.371339, the bound of Alman et al. [2], only if $a \ge 188$; the later bounds near 2.371 in Section 1.5 are lower by less than $3\cdot 10^{-4}$. Every scheme with $p < 3$ has $A \ge 12^{18}$, and every scheme with $p < 2.371339$ has $\log A > 950$ (Proposition 11.1(b) with $\varepsilon = 0.121339$); since such a scheme also has $a \ge 188$ by the evaluation above, the bound $\log A > 9(a-2)^2\log 3/(8\varepsilon)$ from the proof of part (b) gives $\log A > 3\cdot 10^5$. With arbitrary decompositions of the source and the catalyst we know only $A \ge 2^{24}$ (Remark 11.5). All these bounds are far below the sizes of Sections 8–10. How small can the source side of a scheme with exponent below 2.371339 be made, in this construction or in a variant of it?

4. *Smaller seeds.* Theorems 9.2 and 9.3 bound the seed side, the bit lengths of the coefficients and the preprocessing cost by explicit but enormous functions of ε, and Remark 9.8 sketches how, for a suitable implementation, they give an explicit rate for the algorithm of Corollary 9.7. Every seed obtained by unrolling a scheme of the construction as in Corollary 11.2(c) has a side doubly exponential in $1/\varepsilon$. How far can the other bounds, and the constants of Remark 9.8, be reduced, in this construction or in a variant of it?

A The exact binary certificate and its denominator scale

This appendix explains why the construction rounds its counts instead of clearing denominators. A *certificate* is a set of positive rational weights on finitely many profile relations whose weighted sum telescopes; scaled to integers, the weights count how often each relation is used in a tensor construction. The certificate below uses only binary splits (splits of width one) and triplings, and it weights the two children of each split by their tangent probabilities. It reproves the diagonal bound $P(a,a) \ge F(a,a)$ of Theorem 3.2 (Proposition A.1), but the least scale K_a that clears its denominators grows at least exponentially in a (Theorem A.2), and the source of the corresponding tensor relation at least doubly exponentially. The rounding of Section 6 avoids these exponential denominators: it keeps every row identity exact at every scale $K \ge 1$, at the cost of a rounding loss of at most $4J_a = O(a^3)$ (Proposition 6.3). The main construction does not use this appendix.

Quantities of the certificate that correspond to quantities of Sections 5 and 6 carry the superscript ex; the tangent probabilities $q^{\pm}_{r,x}$ and the counts $w^{\pm}_{r,x}$ take the place of $q_{r,x,y}$ and $n_{r,x,y}$, and the tripling weights μ_r are those of (5.4).

A.1 Binary splits with tangent probabilities

Fix a target index $a \geq 2$. As in Section 5, state x of row r ($2 \leq r \leq a$, $0 \leq x \leq 3r$) corresponds to column $b = r - 1 + x$. By (5.1), $F(r, r-1+x) = s_r f_r(x)$, where $f_r(x) = \frac{3(r-1)}{2} + x$ has slope one in x. Hence $2F(r,b) = F(r,b-1) + F(r,b+1)$ for $r \leq b \leq 4r-2$: on these columns the concavity inequality (3.2) holds for F with equality.

A binary split at an interior state $1 \leq x \leq 3r-1$ is the relation of Lemma 4.3 with width one: it consumes $C(r,b)B_X(2)$, where $b = r-1+x$, and produces the children $C(r, b \pm 1)$ at the states $x \pm 1$. Its *tangent probabilities* are the shares of the two children in the concavity equality for F:

$$q^{\pm}_{r,x} = \frac{F(r,b\pm1)}{2F(r,b)} = \frac{f_r(x\pm1)}{2f_r(x)} = \frac{3r-3+2x\pm2}{2(3r-3+2x)}. \tag{A.1}$$

They are positive rationals, since $f_r(x-1) \geq f_r(0) = 3(r-1)/2 > 0$, and $q^+_{r,x} + q^-_{r,x} = 1$ because $f_r(x+1) + f_r(x-1) = 2f_r(x)$. They are the probabilities that (5.2) gives when the width $h_{r,x}$ is replaced by one, that is, with auxiliary dimension two and children $x \pm 1$.

The certificate applies to an arbitrary profile P. Put $\eta_{r,b} = \log(P(r,b)/F(r,b))$. These logarithmic ratios are symmetric, $\eta_{r,b} = \eta_{b,r}$, because P and F are, and normalization (3.1) together with $F(1,b) = b$ gives $\eta_{1,b} = 0$; in particular $\eta_{1,2} = 0$. For $1 \leq x \leq 3r-1$ and $b = r-1+x$, the concavity inequality takes the form

$$\eta_{r,b} - q^-_{r,x}\eta_{r,b-1} - q^+_{r,x}\eta_{r,b+1} \geq 0. \tag{A.2}$$

Indeed $b \geq 2$, so (3.2) applies, and dividing it by $2F(r,b)$ gives $e^{\eta_{r,b}} \geq q^-_{r,x}e^{\eta_{r,b-1}} + q^+_{r,x}e^{\eta_{r,b+1}}$; the weighted arithmetic–geometric mean inequality bounds the right side below by $\exp(q^-_{r,x}\eta_{r,b-1} + q^+_{r,x}\eta_{r,b+1})$, and taking logarithms proves (A.2). The instance of the tripling inequality (3.3) whose right side is $3P(r,r)$ reads $P(r,4r-1) \geq 3P(r,r)$. Since $f_r(3r) = 3f_r(1)$, we have $F(r,4r-1) = 3F(r,r)$, and therefore

$$\eta_{r,4r-1} - \eta_{r,r} \geq 0. \tag{A.3}$$

The certificate multiplies each inequality (A.2) by a weight $\lambda^{\mathrm{ex}}_{r,x}$ and each inequality (A.3) by a weight μ_r. In the final row $r = a$, put

$$\lambda^{\mathrm{ex}}_{a,x} = \frac{4f_a(x)(3a-x)}{3(a-1)(3a-1)} \quad (1 \leq x \leq 3a-1), \qquad \mu_a = \frac{1}{a-1}. \tag{A.4}$$

In each lower row $2 \leq r \leq a-1$, put

$$\lambda^{\mathrm{ex}}_{r,1} = \frac{2(3r-1)}{3(r-1)}, \quad \lambda^{\mathrm{ex}}_{r,x} = \frac{8f_r(x)(3r-x)}{3(r-1)(3r+1)} \ (2 \leq x \leq 3r-1), \quad \mu_r = \frac{2(3r-1)}{(r-1)(3r+1)}. \tag{A.5}$$

These weights are the occupations of the row walk of Section 5 with every width replaced by one: the proof of Proposition A.1 shows that they satisfy the row identity (5.3) of this walk, which determines them because the walk reaches state 0 from every state (as in the proof of Proposition 5.3). We call the numbers $\lambda^{\mathrm{ex}}_{r,x}q^+_{r,x}$, $\lambda^{\mathrm{ex}}_{r,x}q^-_{r,x}$ and $\mu_r/3$ the *type weights*; Section A.2 explains the division by three.

Proposition A.1 (Rational telescoping)**.** *Multiply* (A.2) *at each state* $1 \leq x \leq 3r-1$ *of each row* $2 \leq r \leq a$ *by* $\lambda^{\mathrm{ex}}_{r,x}$*, and multiply* (A.3) *for each row by* μ_r*. As a linear form in the variables* $\eta_{r,b} = \eta_{b,r}$*, the sum of the resulting left sides is exactly* $\eta_{a,a} - \eta_{1,2}$*. In particular* $\eta_{a,a} - \eta_{1,2} \geq 0$ *for every profile* P*. All weights* $\lambda^{\mathrm{ex}}_{r,x}, \mu_r$ *and all type weights are positive rationals.*

Proof. The idea is that within each row the weights cancel the coefficient of every column except column $r-1$ and the column of the injection state; the terms left over by adjacent rows then cancel by symmetry.

Fix a row r. Write $f = f_r$ and $\eta_x = \eta_{r,r-1+x}$ for $0 \le x \le 3r$; put $g(x) = \lambda^{\mathrm{ex}}_{r,x}/f(x)$ for $1 \le x \le 3r-1$, and $g(x) = 0$ for all other integers x. The weighted sum for this row is

$$R_r = \sum_{x=1}^{3r-1} \lambda^{\mathrm{ex}}_{r,x}(\eta_x - q^-_{r,x}\eta_{x-1} - q^+_{r,x}\eta_{x+1}) + \mu_r(\eta_{3r} - \eta_1).$$

By (A.1), $\lambda^{\mathrm{ex}}_{r,x} q^{\pm}_{r,x} = g(x)f(x \pm 1)/2$. The variable η_x occurs in the split at x, as its source, in the splits at $x-1$ and $x+1$, as a child (for those of these states that are interior), and, if $x = 1$ or $x = 3r$, in the tripling. Since g vanishes outside $1 \le x \le 3r-1$, the coefficient of η_x in R_r is

$$\beta_x = f(x)\Big(g(x) - \frac{g(x-1)+g(x+1)}{2}\Big) - \mu_r[x=1] + \mu_r[x=3r] \qquad (0 \le x \le 3r), \tag{A.6}$$

where $[\cdot]$ is 1 if the condition holds and 0 otherwise. In particular $\beta_x = 0$ whenever $x \ne 1, 3r$ and g is affine on $\{x-1, x, x+1\}$. We use the values $f(0) = 3(r-1)/2$, $f(1) = (3r-1)/2$, $f(2) = (3r+1)/2$ and $f(3r) = 3(3r-1)/2$.

The final row $r = a$. Here $g(x) = 4(3a-x)/(3(a-1)(3a-1))$ for $1 \le x \le 3a$, including $x = 3a$, where both sides vanish; so g is affine on $\{1, \dots, 3a\}$, and $\beta_x = 0$ for $2 \le x \le 3a-1$. The affine formula takes the value $4a/((a-1)(3a-1))$ at $x = 0$, so $g(1) - g(2)/2 = 2a/((a-1)(3a-1))$; moreover $g(1) = 4/(3(a-1))$ and $g(3a-1)f(3a)/2 = 1/(a-1)$. Therefore $\beta_0 = -g(1)f(0)/2 = -1$, $\beta_1 = f(1)(g(1) - g(2)/2) - \mu_a = a/(a-1) - 1/(a-1) = 1$ and $\beta_{3a} = \mu_a - g(3a-1)f(3a)/2 = 0$, so that $R_a = \eta_1 - \eta_0 = \eta_{a,a} - \eta_{a,a-1}$.

A lower row $r < a$. Here $g(1) = 4/(3(r-1))$, while $g(x) = 8(3r-x)/(3(r-1)(3r+1))$ for $2 \le x \le 3r$ is affine, so $\beta_x = 0$ for $3 \le x \le 3r-1$. The affine formula takes the value $g(1)+4/(3r+1)$ at $x = 1$, so (A.6) gives $\beta_2 = \frac{f(2)}{2} \cdot \frac{4}{3r+1} = 1$. Next, $g(1) - g(2)/2 = 4/((r-1)(3r+1))$ and $g(3r-1)f(3r)/2 = 2(3r-1)/((r-1)(3r+1)) = \mu_r$. Therefore $\beta_0 = -g(1)f(0)/2 = -1$, $\beta_1 = 2(3r-1)/((r-1)(3r+1)) - \mu_r = 0$ and $\beta_{3r} = \mu_r - \mu_r = 0$, so that $R_r = \eta_2 - \eta_0 = \eta_{r,r+1} - \eta_{r,r-1}$.

In both cases $R_r = \eta_{x^{\mathrm{in}}_r} - \eta_0$. With η_x replaced by the basis vector $\mathbf{e}_x$, this is the row identity (5.3) for the walk of width one.

Summing the rows. By symmetry, $\eta_{r,r-1} = \eta_{r-1,r}$, so the term $-\eta_{r,r-1}$ left by row r cancels the term $+\eta_{r-1,r}$ left by row $r-1$, and the sum of R_r over $2 \le r \le a$ is $\eta_{a,a} - \eta_{2,1} = \eta_{a,a} - \eta_{1,2}$. Only coefficient identities and the symmetry $\eta_{r,b} = \eta_{b,r}$ were used, so this is an identity of linear forms; since each weighted inequality is nonnegative, $\eta_{a,a} - \eta_{1,2} \ge 0$. Finally, $\lambda^{\mathrm{ex}}_{r,x} > 0$ because $f_r(x) > 0$ and $3r - x > 0$, the weights μ_r are positive, and $q^{\pm}_{r,x} > 0$ by (A.1); hence all weights and type weights are positive rationals. □

Since $\eta_{1,2} = 0$, Proposition A.1 gives $P(a,a) \ge F(a,a)$ for every profile P, the diagonal bound of Theorem 3.2. Since the weights depend only on F, they can be realized by tensor maps with fixed types once the type weights are scaled to integers.

A.2 Integer types and their entropy identity

A tensor realization of the certificate uses every relation at its tangent distribution. For a split at state x of row r, this is the type that sends the fractions $q^+_{r,x}$ and $q^-_{r,x}$ of the factors to the children at states $x+1$ and $x-1$. For a tripling, it is the uniform distribution on the three label classes (left, middle and right) of Lemma 4.5: on a tensor power $C(r, 4r-1)^{\otimes n}$ with $3 \mid n$, take the sectors of Lemma 4.5 with $n/3$ middle labels and keep only those whose outer labels are $n/3$ left and $n/3$ right labels; this is again a coordinate restriction, with $n!/((n/3)!)^3$ sectors. Because the certificate prescribes this uniform type for every tripling, the count of every tripling must be divisible by three; this is why $\mu_r/3$, rather than μ_r, is a type weight.

Let K_a be the least positive integer for which all the numbers $K_a\lambda^{\mathrm{ex}}_{r,x}q^{\pm}_{r,x}$ and $K_a\mu_r/3$ are integers, that is, the least common multiple of the denominators of all type weights in lowest terms. The counts and their totals

$$w^{\mathrm{ex}}_{r,x} = K_a\lambda^{\mathrm{ex}}_{r,x}, \quad w^{\pm}_{r,x} = K_a\lambda^{\mathrm{ex}}_{r,x}q^{\pm}_{r,x}, \quad z^{\mathrm{ex}}_r = K_a\mu_r, \quad \bar{w}^{\mathrm{ex}} = \textstyle\sum_{r,x} w^{\mathrm{ex}}_{r,x}, \quad \bar{z}^{\mathrm{ex}} = \textstyle\sum_r z^{\mathrm{ex}}_r \tag{A.7}$$

are integers, with $w^{\mathrm{ex}}_{r,x} = w^{+}_{r,x} + w^{-}_{r,x}$ and $3 \mid z^{\mathrm{ex}}_r$. Since every weight is positive, the certificate uses all $\sum_{r=2}^{a}(3r-1) = (3a^2+a-4)/2$ splits and all $a-1$ triplings, $3(a+2)(a-1)/2$ relations in total.

We record tensor factors by formal symbols in a free abelian group, as in Section 6.3, but with one symbol $[r,b] = [b,r]$ for each unordered pair of positive integers, standing for the sixfold product $Q_{r,b}$ of (2.7); the identification is the isomorphism $Q_{r,b} \cong Q_{b,r}$ of (2.8), and a coefficient n on $[r,b]$ stands for $Q_{r,b}^{\otimes n}$. Since Proposition A.1 is an identity of linear forms in variables subject only to $\eta_{r,b} = \eta_{b,r}$, multiplying it by K_a and replacing each $\eta_{r,b}$ by $[r,b]$ gives the integer identity

$$\begin{aligned}\sum_{r=2}^{a}\sum_{x=1}^{3r-1}\bigl(w^{\mathrm{ex}}_{r,x}[r,r-1+x] - w^{+}_{r,x}[r,r+x] - w^{-}_{r,x}[r,r-2+x]\bigr)\\ + \sum_{r=2}^{a} z^{\mathrm{ex}}_r\bigl([r,4r-1]-[r,r]\bigr) = K_a[a,a] - K_a[1,2].\end{aligned} \tag{A.8}$$

As in the signed ledger (6.5), the split at state x of row r consumes $w^{\mathrm{ex}}_{r,x}$ factors of its column and produces $w^{\pm}_{r,x}$ factors of its children at $x \pm 1$, and the tripling of row r consumes z^{ex}_r factors of column $4r-1$ and produces as many of column r.

For example, let $a = 2$: there is one row, with states $0,\dots,6$ and columns $1,\dots,7$. The weights are $\lambda^{\mathrm{ex}}_{2,x} = 10/3$, $56/15$, $18/5$, $44/15$, $26/15$ for $x = 1,\dots,5$ and $\mu_2 = 1$, so $K_2 = 15$; the counts $(w^{+}_{2,x}, w^{-}_{2,x})$ are $(35,15)$, $(36,20)$, $(33,21)$, $(26,18)$ and $(15,11)$, $z^{\mathrm{ex}}_2 = 15$, $\bar{w}^{\mathrm{ex}} = 230$ and $\bar{z}^{\mathrm{ex}} = 15$. In (A.8), column 2 is consumed 50 times (by the split at state 1) and produced $20+15 = 35$ times (by the split at state 2 and the tripling), each of the columns $3,\dots,7$ is consumed and produced equally often, and column 1 is produced 15 times; since $[2,1] = [1,2]$, the left side reduces to $15[2,2] - 15[1,2]$.

As in (6.4), the factors common to the consumed and the produced side form a catalyst: for $r \le b$, let $c^{\mathrm{ex}}_{r,b}$ be the smaller of the total coefficients of $[r,b]$ in the positive and in the negative terms on the left of (A.8), and put

$$\mathcal{C}^{\mathrm{ex}} = \bigotimes_{r\le b} Q_{r,b}^{\otimes c^{\mathrm{ex}}_{r,b}}, \qquad \mathcal{C}^{\mathrm{ex}}_N = (\mathcal{C}^{\mathrm{ex}})^{\otimes N}. \tag{A.9}$$

Both kinds of terms have total coefficient $\bar{w}^{\mathrm{ex}} + \bar{z}^{\mathrm{ex}}$, because every relation produces as many factors as it consumes, and by (A.8) they agree symbol by symbol except at $[a,a]$, which is consumed K_a more times than it is produced, and at $[1,2]$, which is produced K_a times and never consumed. Hence $\sum_{r\le b} c^{\mathrm{ex}}_{r,b} = \bar{w}^{\mathrm{ex}} + \bar{z}^{\mathrm{ex}} - K_a$.

The certificate describes a tensor relation, built with the maps of Section 4 at type scale N as in Section 7, but with every relation used in all six leg orientations $\pi \in S_3$, so that no row signs are needed; each split applies Lemma 4.3 with width one and type $(Nw^{+}_{r,x}, Nw^{-}_{r,x})$, and each tripling the three-class restriction above. By (2.2), the six orientations of the auxiliary $B_X(2)^{\otimes Nw^{\mathrm{ex}}_{r,x}}$ of a split combine into $\mathsf{M}_{4^{Nw^{\mathrm{ex}}_{r,x}}}$, so the source auxiliary is $\mathsf{M}_{2^{2N\bar{w}^{\mathrm{ex}}}}$. By (A.8), the relation consumes the convolution factors $\mathcal{C}^{\mathrm{ex}}_N\, Q_{a,a}^{\otimes NK_a}$ and produces $\mathcal{C}^{\mathrm{ex}}_N\, Q_{1,2}^{\otimes NK_a} \cong \mathcal{C}^{\mathrm{ex}}_N\, \mathsf{M}_{4^{NK_a}}$, retaining the catalyst as in Section 7.

Because every relation is used at its tangent distribution, the certificate loses nothing in the potential $K_a \log F(a,a)$. In multiplicative form, this is the identity

$$2^{K_a-\bar{w}^{\mathrm{ex}}}\,3^{\bar{z}^{\mathrm{ex}}}\prod_{r,x}\frac{(w^{\mathrm{ex}}_{r,x})^{w^{\mathrm{ex}}_{r,x}}}{(w^{+}_{r,x})^{w^{+}_{r,x}}(w^{-}_{r,x})^{w^{-}_{r,x}}}=F(a,a)^{K_a}, \tag{A.10}$$

where the product runs over all splits. To prove it, fix a split at state x of row r and put $b=r-1+x$. By (A.1) and $w^{\pm}_{r,x}=w^{\mathrm{ex}}_{r,x}q^{\pm}_{r,x}$, we have $F(r,b\pm1)=2F(r,b)\,w^{\pm}_{r,x}/w^{\mathrm{ex}}_{r,x}$, and hence

$$(2F(r,b))^{w^{\mathrm{ex}}_{r,x}}=\frac{(w^{\mathrm{ex}}_{r,x})^{w^{\mathrm{ex}}_{r,x}}}{(w^{+}_{r,x})^{w^{+}_{r,x}}(w^{-}_{r,x})^{w^{-}_{r,x}}}\,F(r,b+1)^{w^{+}_{r,x}}F(r,b-1)^{w^{-}_{r,x}}.$$

Each tripling gives $F(r,4r-1)^{z^{\mathrm{ex}}_r}=3^{z^{\mathrm{ex}}_r}F(r,r)^{z^{\mathrm{ex}}_r}$. Multiply these identities over all relations. The map $[r,b]\mapsto\log F(r,b)$ is well defined because F is symmetric; applied to (A.8), it shows that the profile values cancel between the two sides except for $F(a,a)^{K_a}$ on the consumed side and $F(1,2)^{K_a}=2^{K_a}$ on the produced side, which gives (A.10). In logarithmic form, (A.10) is the analogue, for the walk of width one, of (6.9) with $\mathcal{L}=0$: every empirical distribution $(w^{+}_{r,x},w^{-}_{r,x})/w^{\mathrm{ex}}_{r,x}$ equals its tangent distribution $(q^{+}_{r,x},q^{-}_{r,x})$.

A.3 The minimal denominator scale

Theorem A.2 (Minimal denominator scale)**.** *The scale K_a of Section A.2 has the closed form*

$$K_a=\operatorname{lcm}\left(\frac{3(a-1)(3a-1)}{\gcd(3a-1,4)},\quad\left\{\frac{3(r-1)(3r+1)}{\gcd(r-1,2)^2}:2\le r\le a-1\right\}\right). \tag{A.11}$$

In particular,

$$3\operatorname{lcm}(1,\dots,a-1)\mid 2K_a,\qquad K_a\mid 3\operatorname{lcm}(1,\dots,3a-1), \tag{A.12}$$

$$K_a\ge\frac{3}{2}\operatorname{lcm}(1,\dots,a-1)\ge\frac{3\cdot 2^{a-2}}{a}. \tag{A.13}$$

For example, $K_2=15$, $K_3=84$, $K_4=3465$, $K_6=1113840$ and $K_{10}=5956750800$.

Proof. The idea is to compute, row by row, the least common multiple of the reduced denominators of the type weights, by putting them over one common denominator and finding the greatest common divisor of that denominator and all numerators.

Step 1: from numerators to a row denominator. If the type weights of row r are the numbers ν/δ with ν in a finite set $\mathcal{N}$ of positive integers, their reduced denominators $\delta/\gcd(\delta,\nu)$ have the least common multiple $\rho_r=\delta/\gcd(\delta,\mathcal{N})$, where $\gcd(\delta,\mathcal{N})$ is the greatest common divisor of δ and all elements of $\mathcal{N}$. Indeed, for every prime $\mathfrak{p}$, with v the exponent of $\mathfrak{p}$, both numbers have exponent $v(\delta)-\min(v(\delta),\min_{\nu\in\mathcal{N}}v(\nu))$. Hence $K_a=\operatorname{lcm}_{2\le r\le a}\rho_r$, and it remains to compute $\gcd(\delta,\mathcal{N})$ in each row.

Step 2: the final row. The type weights of row a have the common denominator $\delta=3(a-1)(3a-1)$. The numerator of $\mu_a/3=1/(3(a-1))$ is $3a-1$. By (A.4) and (A.1), $\lambda^{\mathrm{ex}}_{a,x}q^{\pm}_{a,x}=2f_a(x\pm1)(3a-x)/\delta=(3a-3+2x\pm2)(3a-x)/\delta$ for $1\le x\le 3a-1$. Since the numerator $3a-1$ divides δ, the number $\gcd(\delta,\mathcal{N})$ equals the greatest common divisor of $3a-1$ and the other numerators, which may be reduced modulo $3a-1$. Modulo $3a-1$ we have $3a-3+2x\pm2\equiv 2x-2\pm2$ and $3a-x\equiv-(x-1)$, so the products are congruent to $-2x(x-1)$ for the plus sign and to $-2(x-1)(x-2)$ for the minus sign. Each is -2 times a product of two consecutive integers and is therefore divisible by four, and the plus

sign at $x = 2$ gives -4. Hence $\gcd(\delta, \mathcal{N}) = \gcd(3a-1, 4)$ and $\rho_a = 3(a-1)(3a-1)/\gcd(3a-1, 4)$, the first entry of (A.11).

Step 3: a lower row. Let $2 \le r \le a-1$ and use the common denominator $\delta = 3(r-1)(3r+1)$. By (A.5) and (A.1), the state $x = 1$ contributes $\lambda^{\mathrm{ex}}_{r,1} q^{+}_{r,1} = (3r+1)/(3(r-1))$ and $\lambda^{\mathrm{ex}}_{r,1} q^{-}_{r,1} = 1$, with numerators $(3r+1)^2$ and $3(r-1)(3r+1)$. The tripling contributes $\mu_r/3$, with numerator $2(3r-1)$, and the states $2 \le x \le 3r-1$ contribute the numerators $2(3r-3+2x \pm 2)(3r-x)$. An odd prime dividing both $(3r+1)^2$ and $2(3r-1)$ would divide $3r+1$ and $3r-1$, hence their difference 2, which is impossible; so $\gcd(\delta, \mathcal{N})$ is a power of two. If r is even, $(3r+1)^2$ is odd and $\gcd(\delta, \mathcal{N}) = 1$. If r is odd, then $r-1$, $3r+1$, $3r-1$ and $3r-3+2x \pm 2$ are even, so δ and every numerator are divisible by four. Moreover $(3r+1)^2$ has exactly two factors of two when $r \equiv 3 \pmod 4$, and $2(3r-1)$ has exactly two when $r \equiv 1 \pmod 4$; so $\gcd(\delta, \mathcal{N}) = 4$. In both cases $\gcd(\delta, \mathcal{N}) = \gcd(r-1, 2)^2$, and $\rho_r = 3(r-1)(3r+1)/\gcd(r-1, 2)^2$. These are the remaining entries of (A.11).

Step 4: divisibility. Since $3a-1 = 3(a-1)+2$ and $3r+1 = 3(r-1)+4$, we have $\gcd(a-1, 3a-1) = \gcd(a-1, 2)$ and $\gcd(r-1, 3r+1) = \gcd(r-1, 4)$. By Steps 2 and 3,

$$\rho_a = 3\,\mathrm{lcm}(a-1, 3a-1)\,\frac{\gcd(a-1,2)}{\gcd(3a-1,4)}, \qquad \rho_r = 3\,\mathrm{lcm}(r-1, 3r+1)\,\frac{\gcd(r-1,4)}{\gcd(r-1,2)^2} \quad (r < a).$$

Each ratio of greatest common divisors is 1 or $1/2$: it is 1 when a (respectively r) is even, and otherwise it is $2/\gcd(3a-1, 4)$ (respectively $\gcd(r-1, 4)/4$), where the greatest common divisor is 2 or 4. The integers $a-1$, $3a-1$, $r-1$ and $3r+1$ lie between one and $3a-1$, so every ρ_r divides $3\,\mathrm{lcm}(1, \dots, 3a-1)$, and so does K_a. In every row, $2\rho_r$ is a multiple of the corresponding $3\,\mathrm{lcm}(\cdot, \cdot)$, hence of $3(r-1)$. As r runs from 2 to a, the number $r-1$ runs through $1, \dots, a-1$, so $3\,\mathrm{lcm}(1, \dots, a-1)$ divides $2K_a$. This proves (A.12).

Step 5: growth. The first divisibility gives $2K_a \ge 3\,\mathrm{lcm}(1, \dots, a-1)$. Every binomial coefficient $\binom{n}{\alpha}$ divides $\mathrm{lcm}(1, \dots, n)$: by Legendre's formula, the exponent of a prime $\mathfrak{p}$ in $\binom{n}{\alpha}$ is the sum over $e \ge 1$ of $\lfloor n/\mathfrak{p}^e \rfloor - \lfloor \alpha/\mathfrak{p}^e \rfloor - \lfloor (n-\alpha)/\mathfrak{p}^e \rfloor \in \{0, 1\}$, whose terms vanish for $\mathfrak{p}^e > n$; so it is at most the exponent of $\mathfrak{p}$ in $\mathrm{lcm}(1, \dots, n)$, the largest e with $\mathfrak{p}^e \le n$. The $n+1$ binomial coefficients $\binom{n}{\alpha}$ sum to 2^n, so the largest of them is at least $2^n/(n+1)$. With $n = a-1$ this gives $\mathrm{lcm}(1, \dots, a-1) \ge 2^{a-1}/a$, and (A.13) follows. □

The scale K_a also bounds the size of the tensor relation of Section A.2 from below. By (A.8), the symbol $[a, a]$ has net coefficient K_a, and it is consumed only by the split at state 1 of row a, because a consumed symbol $[r, r-1+x]$ or $[r, 4r-1]$ equals $[a, a]$ only when $r = a$ and $x = 1$. Hence $\bar{w}^{\mathrm{ex}} \ge w^{\mathrm{ex}}_{a,1} \ge K_a$, and the source auxiliary $\mathsf{M}_{2^{2N\bar{w}^{\mathrm{ex}}}}$ has side at least 2^{2NK_a}, which by (A.13) grows at least doubly exponentially in a.

A.4 Comparison with the main construction

The splits of larger width reduce the auxiliary cost. If the rounded counts $w_{r,x}$ are replaced by the scaled occupations $K\lambda_{r,x}$, then $\log A_1$ becomes $NK \sum_{r,x} \lambda_{r,x} \log d_{r,x}$, and Proposition 5.3 bounds this sum by $\Gamma_a = 105 + 180(a-2)$. With splits of width one, every auxiliary dimension is 2, and the weights (A.4) and (A.5) add up to $5a + O(1)$ in the final row and to $10r + O(1)$ in each lower row r, so the corresponding sum $\log 2 \sum_{r,x} \lambda^{\mathrm{ex}}_{r,x}$ is $(5 \log 2 + o(1))a^2$. The auxiliary cost of order a and the polynomial rounding scale give the bound $\log A = O(a^5 (\log a)^2)$ of Theorem 8.1.

The *sixfold construction* uses every relation in all six leg orientations, that is, $Q_{r,b} = S^{+}_{r,b} S^{-}_{r,b}$ in place of $S^{\sigma}_{r,b}$, as the certificate above does; since $Q_{r,b} \cong Q_{b,r}$, it needs no row signs. For the same integer counts it has the same ledger and catalyst multiplicities as the signed ledger of Section 6.3: a symbol of row r involves a pair $\{r, b\}$ with $r - 1 \le b \le 4r - 1$, so two rows $r < r'$ share a

symbol only if $r' = r+1$, and then only $[r, r+1] = [r+1, r]$, which the signed ledger identifies as well, $[\sigma_r; r, r+1] = [\sigma_{r+1}; r+1, r]$. Every factor is squared: A_1, because the six orientations of $B_X(d_i)^{\otimes Nw_i}$ give $\mathsf{M}_{d_i}^{2Nw_i}$ by (2.2), and likewise B_1, L, M^3 and κ_N; the source factor $Q_{a,a}^{\otimes NK}$ has rank at most $(2a-1)^{6NK}$; and the degree becomes $2D$, since each relation has six separators. Hence the sixfold construction with interpolation power k has $E = 2kD+1$ and, for every reuse count j, gives through Proposition 7.3 the same r_*, m_*, A and B as the threefold construction of Section 7 with power $2k$. With the other parameters fixed, k enters (7.9) and (10.1) only through $(\log(kD+1))/k$ and $(\log 2)/(kj)$, which do not increase with k because $\log(1+u)/u$ decreases for $u > 0$. So the threefold construction gives smaller dimensions at a given interpolation power, but with power k its pooled exponent is never smaller than that of the sixfold construction with power $\lceil k/2 \rceil$.

Finally, the source side A is a property of the specified relative scheme, not a lower bound for schemes derived from it with the same exponent: tensoring $r_*\mathsf{M}_A \succeq m_*\mathsf{M}_B$ with M_2 gives $r_*\mathsf{M}_{2A} \succeq m_*\mathsf{M}_{2B}$ by (2.1), with the same exponent and source side $2A$, and restricting the factor M_2 to a single coefficient recovers a scheme with source side A.

B The composed relation maps

This appendix supplies the full-space formulas behind Corollary 4.8 and Proposition 7.1; it is not a second proof of either. Both proofs use Lemma 2.1, which requires the maps applied after the separator to be fixed linear maps on all target variables, including those of terms of positive degree, whose first-leg label g' differs from the sector label g of the other two legs. For one relation we write out the separated polynomial with all of its terms, (B.5), and the reindexing on the full target spaces, (B.6); Section B.4 checks the later regroupings.

B.1 The data of one relation

We use the integer counts of Section 6 at rounding scale K, the type scale N, and the set $\mathcal{J} = \mathcal{I}_c \sqcup \mathcal{I}_t$ of active relations of Section 7. The row of a relation $i \in \mathcal{J}$ is denoted r_i.

Splits. A split $i = (r, x) \in \mathcal{I}_c$ has row $r_i = r$, source column $b_i = r - 1 + x$, width $h_i = h_{r,x}$ and auxiliary dimension $d_i = h_i + 1$, and it is applied $w_i = w_{r,x}$ times. A child y of the split has column $r - 1 + y = b_i + h_i - 2j$ with $j = (x + h_i - y)/2$, so it is the diagonal block of Lemma 4.3 with filtration index j. At type scale N the split uses the type $(n_0^{(i)}, \dots, n_{h_i}^{(i)})$ given by

$$n_j^{(i)} = N\, n_{r,x,y} \quad \text{with } y = x + h_i - 2j \qquad (0 \le j \le h_i), \tag{B.1}$$

so that $\sum_j n_j^{(i)} = Nw_i$ and the sector count (7.1) is $M_i = (Nw_i)!/\prod_j n_j^{(i)}!$. These counts are indexed by filtration index, whereas the child counts $n_{i,y} = n_{r,x,y}$ of Section 7 are indexed by child state: for instance, $n_0^{(i)} = N n_{i,x+h_i}$ refers to the longest child, of column $b_i + h_i$, whereas $n_{i,0}$ refers to the child at state 0.

Triplings. A tripling $i \in \mathcal{I}_t$ is applied z_{r_i} times. Its middle count is $m_{r_i} = \lfloor (Nz_{r_i} + 1)/3 \rfloor$, and its sector count M_i is the count M_{r_i} of (7.1), that is, the count $M_{\mathrm{t}}(Nz_{r_i}, m_{r_i})$ of Lemma 4.5.

The *source tensor* $\mathcal{U}_i$ and the *canonical sector tensor* $\mathcal{V}_i$ of relation i are

$$\mathcal{U}_i = \begin{cases} \big(C(r_i, b_i) B_X(d_i)\big)^{\otimes Nw_i}, & i \in \mathcal{I}_c, \\ C(r_i, 4r_i - 1)^{\otimes Nz_{r_i}}, & i \in \mathcal{I}_t, \end{cases} \qquad \mathcal{V}_i = \begin{cases} \prod_{j=0}^{h_i} C(r_i, b_i + h_i - 2j)^{\otimes n_j^{(i)}}, & i \in \mathcal{I}_c, \\ C(r_i, r_i)^{\otimes (Nz_{r_i} - m_{r_i})} \big(C(r_i, r_i)^{(23)}\big)^{\otimes m_{r_i}}, & i \in \mathcal{I}_t. \end{cases}$$

By Lemmas 4.3 and 4.5, every sector of the type restriction of $\mathcal{U}_i$ is isomorphic to $\mathcal{V}_i$.

Words and sectors. Let T_i be the type restriction of Lemma 4.3 or Lemma 4.5, which acts on the second and third legs only, and index the tensor positions of $\mathcal{U}_i$ by θ. The M_i retained words, of type $(n_j^{(i)})_j$ for a split and with exactly m_{r_i} middle labels for a tripling, are numbered $g = 0, \dots, M_i - 1$ in lexicographic order, with letters ordered by filtration index for a split and as left, middle, right for a tripling; the coordinates whose word has number g form sector g. By the proofs of the two lemmas, every term that survives the type restriction has the same retained word on its second and third legs. Hence

$$\mathsf{T}_i(\mathcal{U}_i) = \sum_{g=0}^{M_i-1} T_{i,g}, \qquad T_{i,g} = \sum_{\alpha,\beta,\gamma} T_{i,g}[\alpha,\beta,\gamma]\, x_\alpha y_{g,\beta} z_{g,\gamma}, \tag{B.2}$$

where x_α runs over the coordinates of the shared first leg X_i, and $y_{g,\beta}$ and $z_{g,\gamma}$ over those of sector g, which span $Y_{i,g}$ and $Z_{i,g}$; this is the form to which Proposition 4.6 applies.

Coefficient arrays. The indices are multi-indices $\alpha = (\alpha_\theta)_\theta$, $\beta = (\beta_\theta)_\theta$ and $\gamma = (\gamma_\theta)_\theta$. For a split whose word g has letter j_θ at position θ, in the determinant bases and before any positions are reordered, α_θ indexes a monomial of V_{r_i}, and β_θ and γ_θ index basis vectors of filtration index j_θ in the input and output spaces, with ranges

$$0 \le \alpha_\theta < r_i, \qquad 0 \le \beta_\theta < b_i + h_i - 2j_\theta, \qquad 0 \le \gamma_\theta < r_i + b_i + h_i - 2j_\theta - 1, \tag{B.3}$$

and the coefficients are

$$T_{i,g}[\alpha,\beta,\gamma] = \prod_\theta \mathbf{1}_{\{\gamma_\theta = \alpha_\theta + \beta_\theta\}}. \tag{B.4}$$

Indeed, in a position with letter j the type restriction keeps exactly the diagonal block $C(r_i, b_i + h_i - 2j)$, and on this block multiplication by the monomial with index α sends monomial index β to $\beta + \alpha$ with coefficient one (Lemma 4.1(b),(c) and the computation that follows it in Section 4.1). For a tripling, (B.4) holds with $0 \le \alpha_\theta < r_i$ when β_θ and γ_θ are the original coordinates of $C(r_i, 4r_i - 1)$, restricted to the two intervals of the letter at position θ. The shifts of these interval coordinates and the reversal of the first index at middle positions are part of the reindexing R_i below.

B.2 The separated polynomial of one relation

Put $L_i = 2M_i - 1$ and $\zeta = \zeta_{L_i}$, and let $\mathsf{S}_i(t)$ be the substitutions of Proposition 4.6 for the M_i sectors in (B.2), with the phases of Step 1 and the weights of Step 2 of its proof combined. We suppress the index i on the new variables $X_{\alpha,g'}$, $Y_{g,\beta,e}$ and $Z_{g,\gamma,e'}$, where $g', e, e' \in \{0, \dots, M_i - 1\}$. In source copy ℓ, $0 \le \ell \le L_i - 1$, the substitutions are

$$x_\alpha \mapsto \sum_{g'} \zeta^{\ell g'} t^{g'^2} X_{\alpha,g'}, \quad y_{g,\beta} \mapsto \sum_e \zeta^{\ell(e-g)} t^{2ge-g^2} Y_{g,\beta,e}, \quad z_{g,\gamma} \mapsto \frac{1}{L_i} \sum_{e'} \zeta^{-\ell e'} t^{-2ge'} Z_{g,\gamma,e'}.$$

By Steps 1 and 2 of the proof of Proposition 4.6, $\mathsf{S}_i(t)$, applied after the type restrictions in the L_i source copies, transforms $L_i\mathcal{U}_i$ into exactly

$$\widetilde{T}_i(t) = \sum_{\substack{0 \le g,g',e,e' < M_i \\ e - e' + g' - g = 0}} t^{(g'-g)^2} \sum_{\alpha,\beta,\gamma} T_{i,g}[\alpha,\beta,\gamma]\, X_{\alpha,g'} Y_{g,\beta,e} Z_{g,\gamma,e'}, \tag{B.5}$$

where α, β, γ run over the ranges of word g; with (B.4), every coefficient is explicit, including those of the terms of positive degree. The exponents are δ^2 with $\delta = |g' - g| \le M_i - 1$, and we write $\widetilde{T}_i(t) = \sum_{\delta=0}^{M_i-1} t^{\delta^2} \widetilde{T}_{i,\delta^2}$, where the coefficient $\widetilde{T}_{i,\delta^2}$ of t^{δ^2} collects the terms with $|g' - g| = \delta$. The constant term consists of the terms with $g' = g$, for which $e = e'$, so $\widetilde{T}_{i,0} = \bigoplus_{g=0}^{M_i-1} T_{i,g} B_X(M_i)$, with variables $X_{\alpha,g}$, $Y_{g,\beta,e}$ and $Z_{g,\gamma,e}$.

B.3 Reindexing on the full target spaces

The target spaces of the separator are $\widetilde{X}_i = \bigoplus_{g'} X_i^{(g')}$, where $X_i^{(g')}$ is the copy of X_i with coordinates $X_{\alpha,g'}$, and $\widetilde{Y}_i = \bigoplus_g (Y_{i,g} \otimes \mathbb{C}^{M_i})$ and $\widetilde{Z}_i = \bigoplus_g (Z_{i,g} \otimes \mathbb{C}^{M_i})$, where the factor $\mathbb{C}^{M_i}$ carries the index e, respectively e'.

For each sector g, let $\mathsf{R}^X_{i,g}$, $\mathsf{R}^Y_{i,g}$ and $\mathsf{R}^Z_{i,g}$ be the bijections of coordinates from X_i, $Y_{i,g}$ and $Z_{i,g}$ to the three legs of $\mathcal{V}_i$ that identify $T_{i,g}$ with $\mathcal{V}_i$. For a split, they reorder the positions by increasing filtration index in word g, keeping the original order of positions with equal filtration index; after this reordering, (B.4) with the ranges (B.3) is the coefficient array of $\mathcal{V}_i$. For a tripling, they put all outer positions before all middle positions, again keeping the original order within each group; they subtract from β_θ and γ_θ the left endpoints of their intervals, and at middle positions they also reverse the first index, $\alpha_\theta \mapsto r_i - 1 - \alpha_\theta$. By the proof of Lemma 4.5, this identifies every outer block with $C(r_i, r_i)$ and every middle block with $C(r_i, r_i)^{(23)}$; left and right blocks give the same canonical block. Define the reindexing on the full target spaces by

$$\mathsf{R}_i = \Big(\bigoplus_{g'} \mathsf{R}^X_{i,g'},\ \bigoplus_g (\mathsf{R}^Y_{i,g} \otimes \mathrm{id}_{\mathbb{C}^{M_i}}),\ \bigoplus_g (\mathsf{R}^Z_{i,g} \otimes \mathrm{id}_{\mathbb{C}^{M_i}}) \Big). \tag{B.6}$$

The first component reorders the copy $X_i^{(g')}$ according to word g', and the other two reorder sector g according to word g. A monomial of positive degree has $g' \neq g$, so its legs are reordered according to different words and its image is not in a copy of $\mathcal{V}_i$; exactification removes such terms later. Nevertheless R_i is a bijection between the coordinates of the full spaces and does not depend on t, which is what Lemma 2.1 requires.

For an orientation $\pi \in \mathcal{A}_{\sigma_{r_i}}$, where $\sigma_{r_i} = (-1)^{a-r_i}$ is the sign of row r_i, the complete map of relation i is

$$\Psi_{i,\pi}(t) = \mathsf{R}_i^\pi \circ \mathsf{S}_i(t)^\pi \circ \Big(\bigoplus_{\ell=0}^{L_i-1} \mathsf{T}_i^\pi \Big). \tag{B.7}$$

It acts on $L_i \mathcal{U}_i^\pi$, and its separator uses as shared leg the global leg $\pi(1)$, which contains leg 1 of $\mathcal{U}_i$. By Lemma 2.1, R_i^π acts on $\widetilde{T}_i(t)^\pi$ coefficient by coefficient. In the constant term every monomial has $g' = g$, so R_i reindexes block g according to the single word g and maps $T_{i,g} B_X(M_i)$ onto $\mathcal{V}_i B_X(M_i)$. Hence

$$\Psi_{i,\pi}(t)(L_i \mathcal{U}_i^\pi) = \bigoplus_{g=0}^{M_i-1} \big(\mathcal{V}_i B_X(M_i)\big)^\pi + \sum_{\delta=1}^{M_i-1} t^{\delta^2} \big(\mathsf{R}_i(\widetilde{T}_{i,\delta^2})\big)^\pi, \tag{B.8}$$

and the first term is a full direct sum on all three legs.

Example B.1 (Two tensor positions)**.** Let i be a split of width $h_i = 1$ with type $n_0^{(i)} = n_1^{(i)} = 1$ (for instance $N = 1$ and $w_i = 2$, each child used once), and write $r = r_i$ and $b = b_i \geq 2$. Then $\mathcal{U}_i = (C(r,b) B_X(2))^{\otimes 2}$, $\mathcal{V}_i = C(r, b+1) C(r, b-1)$, and $M_i = 2$: sector 0 has the word $(0,1)$ and sector 1 the word $(1,0)$. By (B.3), the array of $T_{i,0}$ has ranges $\beta_1 < b+1$, $\gamma_1 < r+b$, $\beta_2 < b-1$, $\gamma_2 < r+b-2$, so it is the array of $\mathcal{V}_i$, and the array of $T_{i,1}$ is the same with the two positions exchanged. Hence $\mathsf{R}^X_{i,0}, \mathsf{R}^Y_{i,0}, \mathsf{R}^Z_{i,0}$ are identities, and $\mathsf{R}^X_{i,1}, \mathsf{R}^Y_{i,1}, \mathsf{R}^Z_{i,1}$ exchange the two positions.

The separator uses $L_i = 3$ copies, and $g', e, e' \in \{0,1\}$. The condition $e - e' + g' - g = 0$ leaves six index tuples (g, g', e, e'): the four with $g' = g$ and $e = e'$, which give $T_{i,0} B_X(2) \oplus T_{i,1} B_X(2)$, mapped by R_i onto $2\,\mathcal{V}_i B_X(2)$; and $(0,1,0,1)$ and $(1,0,1,0)$, which give the terms of degree one

$$t \sum_{\alpha,\beta,\gamma} T_{i,0}[\alpha,\beta,\gamma]\, X_{\alpha,1} Y_{0,\beta,0} Z_{0,\gamma,1} + t \sum_{\alpha,\beta,\gamma} T_{i,1}[\alpha,\beta,\gamma]\, X_{\alpha,0} Y_{1,\beta,1} Z_{1,\gamma,0}$$

of (B.5). In the first of them the first leg has label $g' = 1$ and the other legs have label $g = 0$, so R_i exchanges the two positions of α but not those of β and γ. The image pairs the first-leg factor of each position with the second- and third-leg factors of the other position; it is a well-defined term of degree one, but not a copy of $\mathcal{V}_i$, and exactification removes it.

B.4 Three orientations and all relations

Tensoring the maps $\Psi_{i,\pi}(t)$ over $\pi \in \mathcal{A}_\sigma$, where $\sigma = \sigma_{r_i}$, makes every global leg of the target a direct sum of blocks indexed by triples $(g_\pi)_{\pi\in\mathcal{A}_\sigma}$ of independently chosen labels. By (B.8), the constant term has M_i^3 such blocks, each carrying $\bigotimes_\pi (\mathcal{V}_i B_X(M_i))^\pi$. The proof of Corollary 4.8 identifies the source $\bigotimes_\pi \mathcal{U}_i^\pi$ and each of these blocks with the two sides of (4.7) or (4.8): it reorders tensor factors within each global leg, combines the three oriented dot products by (2.2), and, for a tripling, uses (4.6), including the diagonal isomorphism $S^+_{r,r} \cong S^-_{r,r}$ of (2.6). We apply each of these identifications as the same fixed coordinate map on every block of a global leg. It is therefore defined also on monomials of positive degree whose labels g_π differ, and it does not depend on t. With Lemma 2.1, each relation thus gives the degeneration of Corollary 4.8, of degree at most $3(M_i - 1)^2$, with every map defined on the full variable spaces.

The proof of Proposition 7.1 then tensors these maps over all $i \in \mathcal{J}$ with one parameter t, and identifies factors by the coordinate isomorphisms of (2.6) (at a row boundary, $S^\sigma_{r,b} \cong S^{-\sigma}_{b,r}$) and of (2.1), again applied identically on every block after all separators. Hence Lemma 2.1 applies to the complete composed map, which is the degeneration (7.4) of degree at most D.

C The general separator and exactification of one separator

This appendix proves the separator with a dot-product dimension K_0 and a phase slope c, and compares three ways of turning a single separator into a restriction: sparse interpolation, direct copying, and, for its tensor square, an average over a prime number of roots of unity. The construction of Section 7 uses none of this: it needs only the case $K_0 = M$, $c = 1$ (Proposition 4.6), and it exactifies the whole catalytic degeneration of Proposition 7.1 by Lemma 2.2.

C.1 The separator with general parameters

Let $T = \sum_{g=0}^{M-1} T_g$ be a tensor with M sectors sharing the first leg X, with coordinates x_α, $y_{g,\beta}$, $z_{g,\gamma}$ and coefficients $T_g[\alpha,\beta,\gamma]$ as in the proof of Proposition 4.6. For positive integers K_0 and c, put

$$L_0 = K_0 + c(M-1), \qquad \delta_* = \min\Big(M-1, \Big\lfloor \frac{K_0 - 1}{c} \Big\rfloor\Big). \tag{C.1}$$

For $K_0 = M$ and $c = 1$ these are $L_0 = 2M - 1$ and $\delta_* = M - 1$, and the next proposition is Proposition 4.6.

Proposition C.1 (General separator)**.** *Let T, K_0, c, L_0 and δ_* be as above. The substitutions in the proof, whose entries are Laurent polynomials in t, transform $L_0 T$ into a polynomial $\widetilde{T}(t)$ whose exponents lie in $\{c\delta^2 : 0 \le \delta \le \delta_*\}$ and whose constant term is $\widetilde{T}_0 = \bigoplus_g T_g B_X(K_0)$. In particular,*

$$L_0 T \trianglerighteq \bigoplus_{g=0}^{M-1} T_g B_X(K_0), \tag{C.2}$$

with degree at most $c\delta_*^2$. After a permutation of the legs, the same holds when the shared leg is the second or the third leg.

Proof. We follow the proof of Proposition 4.6 with $\zeta = \zeta_{L_0}$, source copies $\ell = 0, \dots, L_0 - 1$, and target variables $X_{\alpha,g'}$, $Y_{g,\beta,e}$, $Z_{g,\gamma,e'}$ with $0 \le g' \le M-1$ and $0 \le e, e' \le K_0 - 1$. In copy ℓ substitute

$$x_\alpha \mapsto \sum_{g'} \zeta^{\ell c g'} t^{c g'^2} X_{\alpha,g'}, \quad y_{g,\beta} \mapsto \sum_{e} \zeta^{\ell(e-cg)} t^{2ge - cg^2} Y_{g,\beta,e}, \quad z_{g,\gamma} \mapsto \frac{1}{L_0} \sum_{e'} \zeta^{-\ell e'} t^{-2ge'} Z_{g,\gamma,e'}.$$

Summing over ℓ, the term $x_\alpha y_{g,\beta} z_{g,\gamma}$ contributes to the coefficient of $X_{\alpha,g'} Y_{g,\beta,e} Z_{g,\gamma,e'}$ a power of t times the average over ℓ of $\zeta^{\ell(e-e'+c(g'-g))}$, which is 1 if L_0 divides $e - e' + c(g' - g)$ and 0 otherwise. Since $|e - e' + c(g' - g)| \le (K_0 - 1) + c(M-1) = L_0 - 1$, the surviving terms are exactly those with $e - e' = c(g - g')$. On them the exponent of t is $cg'^2 + 2g(e-e') - cg^2 = cg'^2 + 2cg(g-g') - cg^2 = c(g'-g)^2$. Hence the sum of the images of the L_0 copies is the polynomial

$$\widetilde{T}(t) = \sum_{\substack{g,g',e,e' \\ e-e'+c(g'-g)=0}} t^{c(g'-g)^2} \sum_{\alpha,\beta,\gamma} T_g[\alpha,\beta,\gamma]\, X_{\alpha,g'} Y_{g,\beta,e} Z_{g,\gamma,e'}.$$

The exponent vanishes exactly when $g' = g$, and then $e = e'$; as in Step 3 of the proof of Proposition 4.6, these terms form $\bigoplus_g T_g B_X(K_0)$. For every surviving term, $\delta = |g' - g|$ satisfies $\delta \le M - 1$ and $c\delta = |e - e'| \le K_0 - 1$, so $\delta \le \delta_*$. Permuting the legs permutes the three substitution maps. □

The two parameters trade source copies against sparsity of the exponent support: increasing c by one adds $M - 1$ source copies and never increases the number $\delta_* + 1$ of possible exponents, which drops below M once $c(M-1) > K_0 - 1$.

C.2 Sparse exactification and direct copying

With $k = 1$, Lemma 2.2 uses $D + 1$ evaluations for a polynomial of degree at most D. When the polynomial has few distinct exponents, fewer evaluations suffice.

Lemma C.2 (Sparse exactification)**.** *Suppose that a triple $\Psi(t)$ of linear maps with Laurent-polynomial entries transforms a tensor into $\widetilde{T}(t) = \sum_{\eta \in \mathcal{E}} t^\eta \widetilde{T}_\eta$, where $\mathcal{E}$ is a finite set of nonnegative integers that contains 0. Then the direct sum of $|\mathcal{E}|$ copies of that tensor restricts to $\widetilde{T}_0$, with rational interpolation coefficients.*

Proof. The polynomial $\psi(u) = \prod_{\eta \in \mathcal{E} \setminus \{0\}} (u - 2^\eta)/(1 - 2^\eta) = \sum_{\iota=0}^{|\mathcal{E}|-1} \psi_\iota u^\iota$ has rational coefficients, $\psi(1) = 1$, and $\psi(2^\eta) = 0$ for every $\eta \in \mathcal{E} \setminus \{0\}$. Hence $\sum_\iota \psi_\iota \widetilde{T}(2^\iota) = \sum_{\eta \in \mathcal{E}} \psi(2^\eta) \widetilde{T}_\eta = \widetilde{T}_0$. Each $\widetilde{T}(2^\iota)$ is the image of one copy of the source tensor under $\Psi(2^\iota)$, which is defined because $2^\iota \ne 0$. Sending the $|\mathcal{E}|$ copies to the common target variables and absorbing ψ_ι into one leg of copy ι gives the restriction. □

For a separator the source is $L_0 T$, and by Proposition C.1 the exponent support has $\delta_* + 1$ elements. Lemma C.2 therefore gives a restriction from $L_0(\delta_* + 1)$ copies of T to $\bigoplus_g T_g B_X(K_0)$.

Direct copying reaches the same target from MK_0 copies of T by a restriction, with no parameter t. Take one copy of T for each pair (g, e) with $0 \le g \le M - 1$ and $0 \le e \le K_0 - 1$, restrict it to sector g by setting the coordinates of the other sectors to zero, and map $x_\alpha \mapsto X_{\alpha,g}$, $y_{g,\beta} \mapsto Y_{g,\beta,e}$ and $z_{g,\gamma} \mapsto Z_{g,\gamma,e}$. For fixed g, the K_0 copies keep separate second and third legs but share the first leg, so together they form $T_g B_X(K_0)$; different values of g use different variables on all three legs. This gives $MK_0\, T \succeq \bigoplus_g T_g B_X(K_0)$.

Sparse exactification of one separator never uses fewer copies than direct copying:

$$L_0(\delta_* + 1) \geq MK_0. \tag{C.3}$$

If $\delta_* = M - 1$, this follows from $L_0 \geq K_0$. Otherwise $\delta_* = \lfloor (K_0 - 1)/c \rfloor$, so $\delta_* + 1 = \lceil K_0/c \rceil$ and $c(\delta_* + 1) \geq K_0$; hence $L_0(\delta_* + 1) = K_0(\delta_* + 1) + c(\delta_* + 1)(M - 1) \geq K_0 + K_0(M - 1) = MK_0$. The two-power refinement below shows that the tensor square of a separator can be made exact with fewer copies than direct copying needs.

In the main construction direct copying would be too expensive. With $K_0 = M_i$, it uses M_i^2 source copies in each of the three orientations of relation i, so M_i^6 in all, against M_i^3 output copies: a ratio M_i^3 instead of $(2M_i - 1)^3/M_i^3 < 8$. The copy overhead would be $3\sum_i \log M_i$, three times the contribution $\sum_i \log M_i$ of the separators to $\log B_1$, instead of less than $3I \log 2$ by (7.12).

C.3 A two-power refinement

For the tensor square of the separator polynomial $\widetilde{T}(t)$ of Proposition C.1, an average over a prime number of roots of unity removes all terms of positive degree. Let $\mathfrak{p}$ be a prime with $\mathfrak{p} > \delta_*$, $\mathfrak{p} \equiv 3 \pmod 4$ and $\mathfrak{p} \nmid c$; such primes exist, since there are infinitely many primes congruent to 3 modulo 4. Let $\zeta_\mathfrak{p}$ be a primitive $\mathfrak{p}$-th root of unity. Then

$$\frac{1}{\mathfrak{p}} \sum_{\iota=0}^{\mathfrak{p}-1} \widetilde{T}(\zeta_\mathfrak{p}^\iota)^{\otimes 2} = \widetilde{T}_0^{\otimes 2}. \tag{C.4}$$

Indeed, write $\widetilde{T}(t) = \sum_{\delta=0}^{\delta_*} t^{c\delta^2} \widetilde{T}_{c\delta^2}$, where some coefficients may vanish. Then $\widetilde{T}(t)^{\otimes 2}$ is the sum of the terms $t^{c(\delta_1^2+\delta_2^2)}\, \widetilde{T}_{c\delta_1^2} \otimes \widetilde{T}_{c\delta_2^2}$ with $0 \leq \delta_1, \delta_2 \leq \delta_*$, and the average in (C.4) keeps exactly those with $\mathfrak{p} \mid c(\delta_1^2 + \delta_2^2)$. Since $\mathfrak{p} \nmid c$, such a term has $\delta_1^2 + \delta_2^2 \equiv 0 \pmod{\mathfrak{p}}$. If $\mathfrak{p} \nmid \delta_1$, then $(\delta_2 \delta_1^{-1})^2 \equiv -1 \pmod{\mathfrak{p}}$, which is impossible because -1 is not a square modulo a prime congruent to 3 modulo 4. Hence $\mathfrak{p} \mid \delta_1$, and then $\mathfrak{p} \mid \delta_2$. As $0 \leq \delta_1, \delta_2 \leq \delta_* < \mathfrak{p}$, both vanish, and only $\widetilde{T}_0 \otimes \widetilde{T}_0$ remains. Each evaluation $\widetilde{T}(\zeta_\mathfrak{p}^\iota)^{\otimes 2}$ is the image of $(L_0 T)^{\otimes 2} = L_0^2 T^{\otimes 2}$ under the tensor square of the substitutions, so, as in the proof of Lemma C.2, $\mathfrak{p}\, L_0^2 T^{\otimes 2} \succeq (\bigoplus_g T_g B_X(K_0))^{\otimes 2}$.

For example, let $M = K_0 = 8$ and $c = 1$. Then $L_0 = 15$ and $\delta_* = 7$, and the smallest admissible prime is $\mathfrak{p} = 11$. The restriction above uses $11 \cdot 15^2 = 2475$ copies of $T^{\otimes 2}$. Direct copying applied to $T^{\otimes 2}$, which has M^2 sectors and needs a dot product of dimension K_0^2, uses $(MK_0)^2 = 8^4 = 4096$ copies. Lemma 2.2 with $k = 2$ and degree $c\delta_*^2 = 49$ would need $2 \cdot 49 + 1 = 99$ evaluations, that is, $99 \cdot 15^2 = 22275$ copies.

D Notation

This appendix lists the symbols that keep one meaning across sections, each with the place where it is defined. Notation local to Section 4, namely the binary forms, the filtration, words, types and the separator variables, is defined there and reused in Appendices B and C. Appendix A reuses the names of the walk and the counts for its binary certificate, as stated at its start. Some letters also have a local meaning. In sums $\sum_{j=0}^{h}$ over the children of a split, and as the filtration index (Sections 4 and 5, Appendix B), j is unrelated to the reuse count. In Section 4, Appendix C and the heuristic of Section 1.2, M is the sector count of a single relation. The lifts $L_{c,h}(i)$ of Section 4 are unrelated to L_i and L. In Section 4, c counts coefficients, and in Appendix C and Remark 4.7 it is a phase slope; neither is related to the catalyst multiplicities $c_{\sigma,r,b}$. In Appendix C, Remark 4.7 and Lemma 9.1(b), K_0 is the dimension of a dot product, unrelated to the rounding scale K. The degree bound D and the relative entropy $D(\cdot\|\cdot)$ are distinguished by their arguments.

Section 9 uses local notation. The rational variant has the Lagrange basis polynomials $\varpi_{n,\ell}(\xi)$ of the nodes $1, \ldots, n$, with coefficients $\varpi_{n,\ell,\gamma}$, and $[\xi^\gamma]$ takes the coefficient of ξ^γ (Section 9.1); its counts L, D, E,

r_*, m_*, A and B are those of Section 7. Theorem 9.2 uses the capped allowance $\varepsilon' = \min(\varepsilon, 1/2)$, the leaf threshold n_* of Lemma 2.4 and the rank constant Ξ. The fraktur letters $\mathfrak{T}, \mathfrak{F}, \mathfrak{W}, \mathfrak{H}, \mathfrak{Q}, \mathfrak{S}$ are size bounds used only in Theorem 9.3 and Remark 9.8, where b_ε is the bit length of ε, $\mathfrak{b}$ a bit length and $\mathfrak{b}_0$ a bound on the bit lengths of factors. For a fixed rational seed (Section 9.4) with R products, $p_{\mathrm{seed}} = \log_{d_0} R$ and c_{seed} is the constant of Lemma 9.4; the coefficient matrices are $\mathsf{U}, \mathsf{V}, \mathsf{W}$, with common denominator c_{den}, cleared-coefficient bound c_{coef} and bit lengths $c_{\mathrm{in}}, c_{\mathrm{out}}$; ℓ is the recursion depth, b_{in} the bit bound of the input entries, u the unit roundoff, ϑ an error target, and Υ, χ enter the forward-error bound. In Corollary 9.7 and Remark 9.8, $1/v(n)$ is the allowance of the seed selected at input side n.

Section 11.2 also uses local notation: the double factorial $(2a-3)!! = \prod_{r=2}^{a-1}(2r-1)$, a potential ψ on the states of a row, the ratio $\eta_a = F(a,a)/(2a-1)$, the least source side $A_{\min}(\varepsilon)$ at allowance ε, and the number R of terms of a decomposition (Remark 11.5).

Tensors and operations

ω	exponent of matrix multiplication (§1)
$\mathrm{R}(T)$	rank of the tensor T (§2.1)
t	degeneration parameter (§2.3)
$T \trianglerighteq T'$	restriction: linear maps on the three legs turn T into T' (§2.1)
$T \trianglerighteq T'$	degeneration: substitutions depending on the parameter t turn T into a polynomial in t with constant term T' (§2.3)
mT	direct sum of m copies of T, with independent variables on all three legs (§2.1)
T^π	leg permutation: leg i of T moves to position $\pi(i)$, for $\pi \in S_3$ (§2.1)
$\mathcal{A}_+$, $\mathcal{A}_-$, (23)	even and odd permutations in S_3; the transposition of legs 2 and 3 (§2.1)
ζ_m	primitive m-th root of unity (§2.1)
M_d	matrix multiplication tensor of side d (§2.2)
$B_X(d)$, $B_Y(d)$, $B_Z(d)$	dot-product tensors of dimension d; the subscript names the leg with a single variable (§2.2)
$C(r,b)$	convolution tensor: multiplication of polynomials with r and b coefficients (2.3)
$S^\sigma_{r,b}$	symmetrized convolution tensor $\bigotimes_{\pi\in\mathcal{A}_\sigma} C(r,b)^\pi$, with sign $\sigma \in \{+,-\}$ (2.5)
$Q_{r,b}$	sixfold product $S^+_{r,b}S^-_{r,b} = \bigotimes_{\pi\in S_3} C(r,b)^\pi$ (2.7)

Profile

P	a profile: positive symmetric function satisfying the normalization, concavity and tripling axioms (Definition 3.1)
F, s_m	least profile; $s_m = \prod_{h=1}^{m-1}\left(1+\frac{1}{3h}\right)$ (3.4)
τ_a	profile exponent $3\log(2a-1)/\log F(a,a)$ (§1.2)

Row walk and integer counts

$\mathbf{e}_x$	standard basis vector of state x (5.3)
H, $D(\cdot\|\cdot)$	Shannon entropy and relative entropy, with natural logarithms (§6.4)
a	target index, an integer $a \geq 2$ (§1.1)
r, b	row and column: the two lengths of $C(r,b)$ and $S^\sigma_{r,b}$; the walk uses the rows $2 \leq r \leq a$ (§5.1)
x, y	states $0, \ldots, 3r$ of row r; state x is column $b = r-1+x$ (§5.1)
σ_r	sign $(-1)^{a-r}$ of row r (§5.1)
$f_r(x)$	$\frac{3(r-1)}{2} + x$, so that $F(r, r-1+x) = s_r f_r(x)$ (5.1)
$h_{r,x}$, $d_{r,x}$	width $\min(x, 3r-x)$ and auxiliary dimension $h_{r,x}+1$ of the split at state x (5.2)
$\mathcal{Y}_{r,x}$	children $\{x-h_{r,x}, x-h_{r,x}+2, \ldots, x+h_{r,x}\}$ of the split at (r,x) (5.2)
$q_{r,x,y}$	probability that the split at state x moves to child y (5.2)
x^{in}_r	injection state of row r: $x^{\mathrm{in}}_a = 1$, and $x^{\mathrm{in}}_r = 2$ for $r < a$ (§5.1)

$\lambda_{r,x}$, μ_r	occupation: expected number of splits at state x of row r; expected number of triplings in row r (Definition 5.1)
Γ_a	$105 + 180(a-2)$, bound for $\sum_{r,x} \lambda_{r,x}$ and for $\sum_{r,x} \lambda_{r,x} \log d_{r,x}$ (Proposition 5.3)
K	rounding scale, a positive integer (§6.2)
$n_{r,x,y}$, $w_{r,x}$	number of splits at (r,x) with child y; number $w_{r,x} = \sum_y n_{r,x,y}$ of splits at (r,x) (§6.2)
z_r	number of triplings in row r (§6.2)
$\bar{w}$, $\bar{z}$	totals $\sum_{r,x} w_{r,x}$ and $\sum_r z_r$ (§6.2)
W	$\sum_{r,x} w_{r,x} \log d_{r,x}$ (§6.5)
$\hat{q}_{r,x}$	empirical distribution of children of the split at (r,x): $\hat{q}_{r,x,y} = n_{r,x,y}/w_{r,x}$ (6.7)
J_a	$\sum_{r,x} d_{r,x}$, the number of transition coordinates (6.8)
$[\sigma; r, b]$, $[\mathsf{M}_b]$	formal symbol for $S^{\sigma}_{r,b}$ in the ledger; boundary symbol, the class of $[\sigma; 1, b]$ (6.3)
$c_{\sigma,r,b}$, $\mathcal{C}$, $\mathcal{C}_N$	catalyst multiplicity of $S^{\sigma}_{r,b}$; catalyst $\mathcal{C} = \bigotimes_{\sigma,r,b} (S^{\sigma}_{r,b})^{\otimes c_{\sigma,r,b}}$; $\mathcal{C}_N = \mathcal{C}^{\otimes N}$ (6.4)
G, $\mathcal{L}$	rounded potential and rounding loss (6.7); $G = K \log F(a,a) - \mathcal{L}$ by Proposition 6.3

Finite construction

ε	positive rational exponent allowance (Theorem 1.1)
d_0	seed side: the side of a rational seed (§1.1); given by a formula in Theorem 9.2
N, k, j	type scale; interpolation power; reuse count, the number of uses of one copy of the catalyst (§7.1)
$\mathcal{I}_c$, $\mathcal{I}_t$	active splits: pairs (r,x) with $w_{r,x} > 0$; active triplings: rows r with $z_r > 0$ (§7.1)
$\mathcal{J}$, I	the relations $\mathcal{J} = \mathcal{I}_c \sqcup \mathcal{I}_t$, indexed by i; $I = \lvert\mathcal{J}\rvert$ (§7.1)
d_i, w_i, $n_{i,y}$, $\hat{q}_i$	$d_{r,x}$, $w_{r,x}$, $n_{r,x,y}$ and $\hat{q}_{r,x}$ for a split $i = (r,x)$ (§7.1)
M_i, m_r	sector count of relation i; middle count of the tripling in row r (7.1)
$M_{\mathrm{t}}(n,m)$	number $\binom{n}{m} 2^{n-m}$ of sectors of the tripling of $C(r, 4r-1)^{\otimes n}$ with middle count m (Lemma 4.5)
M, L_i, L	$M = \prod_{i \in \mathcal{J}} M_i$, so that the catalytic degeneration has M^3 output copies; $L_i = 2M_i - 1$, the source copies of the separator of relation i; $L = \prod_i L_i^3$ (7.2)
D	$3 \sum_i (M_i - 1)^2$, degree bound of the catalytic degeneration (7.3)
E	$kD + 1$, the number of interpolation points (§7.3)
A_1, B_1	one-step source side $\prod_{i \in \mathcal{I}_c} d_i^{N w_i}$ and target side $2^{NK} M$ (7.3)
Δ_N	$\log(B_1/A_1)$ (7.8)
κ_N	rank bound for the catalyst $\mathcal{C}_N$ (7.5)
r_*, m_*, A, B	relative scheme $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$ with source side A and target side B (Definition 2.3); in the construction, $A = A_1^{kj}$ and $B = B_1^{kj}$ (7.6)
p	pooled exponent $\log(r_*/m_*)/\log(B/A)$ (Definition 2.3); its value for the construction is (7.9)
Φ, $\overline{\Phi}$	fixed-batch (unpooled) exponent bound (10.1); sufficient upper expression (10.5), built from logarithms of integers, which bounds Φ from above when $\underline{\Delta}_N > 0$ (Lemma 10.2)
H_i	entropy charge of relation i: $w_i H(\hat{q}_i)$ for a split, $z_r \log 3$ for the tripling in row r (7.10)
$\bar{H}_N$, $\mathcal{T}_N$	largest scaled entropy charge $N \max_{i \in \mathcal{J}} H_i$; type penalty (7.10)
ℓ_a	$\lceil \log_2 a \rceil$ (Theorem 1.2)
Λ	order of the cyclotomic field $\mathbb{Q}(\zeta_\Lambda)$ that contains all coefficients of the cyclotomic construction (Theorem 8.4)

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